RAS Prelims 2026 · Handwritten Class Notes

Reasoning and Mental Ability

Chapter-by-chapter notes in coaching-lecture style — concept first, then the facts, with flow charts, mind maps, previous-year callouts and current-affairs links.

RPSC RAS & RTS Preliminary Examination · 6 December 2026 · 150 questions · 200 marks · 1/3 negative marking

RM01

Statement & Assumptions / Arguments / Conclusions / Courses of Action

About 4-6 marks in every RAS Prelims paper. Statement-Assumption, Statement-Argument, Statement-Conclusion and Statement-Course of Action have each appeared in every paper - 2021, 2023 and 2024 included. No calculation, no formula. Thirty to forty seconds an item is enough once the rules are drilled.

Four sub-types, one habit. Every RAS paper carries all four, and almost every wrong answer here comes from the same mistake: the student answers from what he knows about the world instead of from the words printed on the page. Break that habit and this chapter turns into four nearly free marks.

Statement-based Reasoning (4 sub-types)

Statement & Assumption
- What the speaker TAKES FOR GRANTED
- Unstated but necessary for the statement to make sense
- Reject extremes: only, all, never, always
- Reject future effects and past causes

Statement & Argument
- Test = RELEVANCE + SIGNIFICANCE
- Yes/No does not decide strength
- Weak: tradition, imitation, emotion, ridicule
- Weak: 'it will not solve the problem fully'

Statement & Conclusion
- What MUST be true, not what is likely
- Test: statement true + conclusion false possible?
- Percentage is never a number
- Either-or when the two are complementary

Course of Action
- Test 1: is the problem real and stated?
- Test 2: would this action actually solve it?
- Must be practicable and within the agency's power
- Reject extreme, punitive and platitude actions

The common leak
- Importing outside knowledge
- Extreme words in the option
- Option order gets reshuffled by RPSC
- Answering the topic instead of the statement

Four sub-types, one exam habit

All four sub-types look the same on the page. You are given a short Statement - a government decision, a notice, an advertisement, a news item. Below it come two items numbered I and II. Then a fixed question: are they implicit assumptions, are they strong arguments, do they logically follow as conclusions, or do they follow as courses of action. The options are always the same four shapes - I only, II only, Either or Neither, Both.

- Statement & Assumption - 'Which of the assumptions is/are implicit in the statement?'
- Statement & Argument - 'Which of the arguments is/are strong?'

- Statement & Conclusion - 'Which of the conclusions logically follows/follow?'
- Statement & Course of Action - 'Which of the courses of action logically follows/follow?'
- Each of the four appears in every RAS Prelims paper - plan for four items, not one.

Now the important part. These are the sub-types students get wrong most often, and the reason is not difficulty. It is that the student reads the topic - urea prices, coaching institutes, air pollution - and starts answering from general knowledge. The statement becomes a prompt for opinion instead of a closed piece of text. Every rule in this chapter exists to stop that.

TRAP

Do not import real-world knowledge. If the statement says all trains were cancelled, then no train ran - even if you personally saw one run. If a statement says a helpline exists, it works. You are being tested on reading, not on news.

ASKED IN RAS

RAS Pre 2024 asked the urea price-rise assumption item and the private-cars-in-the-CBD argument item; RAS Pre 2023 asked the literacy-rate conclusion item and the unfair-means course-of-action item; RAS Pre 2021 asked the extra-Sunday-classes assumption item and the beggars argument item. The pattern does not change - only the topic does.

The one rule that governs all four - read the statement as a closed world

Treat the statement as the only universe that exists. Nothing outside it is available to you - not your knowledge, not your opinion, not what usually happens. Whatever the sub-type, your question is the same: does this item live inside the statement, or does it come from outside? Assumptions live behind the statement. Conclusions live inside it. Arguments and courses of action are tested against it. Anything that comes from outside is out.

The four steps, same for every sub-type

Read the statement twice - once for sense, once for the exact words → Underline the extreme words in the statement itself (all, only, every, never) → Take item I alone; apply the sub-type's test; mark it Yes or No → Take item II alone - never let your answer to I influence II → Read the option list before keying, because RPSC reshuffles the order

- Extreme words in the ITEM - only, all, never, always, none, every, cannot - are almost always fatal.
- The one exception: the statement itself is that extreme, e.g. it says 'all trains were cancelled'.
- An item that merely restates the statement is never a valid assumption and never a fresh conclusion.
- A prediction about what people will do is never an assumption and rarely a conclusion.
- Comparisons the statement did not make - bigger, most, better than - are outside information.

YAAD RAKHO

Yaad rakho the option order RPSC and SSC use: I only, II only, Either/Neither, Both. But papers do reshuffle it - so key the LETTER of the option you have chosen, not its position. Students lose marks by circling option (a) out of habit.

Statement & Assumption - what 'taken for granted' really means

An assumption is something the speaker or writer has taken for granted while making the statement. It is unstated, but the statement collapses into nonsense without it. That is the whole definition, and it is the golden rule of this sub-type: an assumption must be something the statement TAKES FOR GRANTED - not something that merely happens to be true in the world, not something desirable, not something that might follow later.

The necessity test - use it on every assumption item

Read the statement; ask what must already be true for it to make sense →

Take assumption I; suppose it is FALSE →

If the statement now becomes pointless or absurd, I is implicit →

If the statement still stands comfortably, I is NOT implicit →

Repeat independently for II, then match the option

- Reject any assumption carrying only, all, never, always, none, every, cannot.
- Reject a future EFFECT of the statement - an effect is never an assumption behind it.
- Reject a past CAUSE that the statement does not need, e.g. 'employees had gone on strike'.

- Reject an assumption that merely restates the statement - it adds no taken-for-granted idea.
- Reject comparisons about other people or places that the statement never touches.

Worked example - RAS Pre 2024. Statement: The government has decided to increase the price of urea with effect from next month. Assumption I: The government has the authority to fix the price of urea. Assumption II: Farmers will stop using urea from next month.

- Negate I - the government has no authority to fix urea prices. Then the announcement is meaningless. So I is necessary; it is implicit.
- II is a prediction about what farmers will do after the decision - a future effect, and effects are never assumptions.
- II also carries an extreme, 'will stop' - complete abandonment by every farmer.
- Answer: Only assumption I is implicit.

> **TRAP**
> The single biggest killer in this sub-type: mistaking an EFFECT for an assumption. An assumption sits BEHIND the statement, in the past or in the speaker's head. An effect comes AFTER it. 'Prices will rise', 'people will stop', 'the scheme will succeed' - all effects, all wrong.

Assumption, inference, conclusion - three different animals

RPSC mixes these three words across the paper, and students who have not separated them lose marks in both the assumption item and the conclusion item. The difference is simply one of direction in time and logic. An assumption comes before the statement. A conclusion comes out of it. An inference sits in between – it is drawn from the statement, but with a little reasonable reading.

Assumption vs Inference vs Conclusion

	What it is	Test to apply
Assumption	Unstated thing the speaker took for granted BEFORE speaking	Negate it – does the statement collapse?
Inference	Something reasonably read OUT of the statement plus ordinary sense	Is it the most reasonable reading of the data given?
Conclusion	Something that MUST be true given the statement	Can the statement be true and this be false?

Same statement, two different animals

ASSUMPTION (behind it)	CONCLUSION (out of it)
Statement: 'The CM will inaugurate the new expressway on Sunday.'	Statement: 'The CM will inaugurate the new expressway on Sunday.'
Construction of the expressway is complete – taken for granted, else no inauguration.	The expressway is not yet open to the public – it must be true if it is being inaugurated.
Sits in the past / in the speaker's head	Sits in what the words entail
Test by negation	Test by asking 'could this be false?'

Three families of statements have standard, ready-made assumptions. Learn them and you will key these items in ten seconds. Advertisements, notices and instructions, and the 'in spite of' construction.

The three standard assumption families

Statement family	What is automatically implicit
Advertisement	The product has the advertised quality; people want that quality; the ad will influence buyers
Notice, instruction, appeal	The reader can read and understand it; people are likely to comply

Statement family	What is automatically implicit
Penalty clause in a notice	The prohibited behaviour is expected; the penalty is believed to deter it
'In spite of X, Y happened'	X normally prevents Y
Advisory ('boil the water')	The danger is real; the remedy advised is believed to work

Worked example. Statement: In spite of the heavy rainfall yesterday, the cricket match was completed on schedule. Assumption I: Heavy rainfall never affects the completion of a cricket match. Assumption II: Heavy rainfall usually disrupts the play in a cricket match.

- 'In spite of X, Y happened' presupposes that X normally prevents Y. So rainfall usually disrupts play - II is implicit.
- I says 'never'. But the very use of 'in spite of' shows the speaker thinks rain normally does affect play - I contradicts the statement's own presupposition.
- An extreme word in the item, and a direct clash with the statement: I is out.
- Answer: Only assumption II is implicit.

YAAD RAKHO

Yaad rakho - AD, NOTICE, IN SPITE OF. Advertisement: people want the quality. Notice: people will comply. In spite of: the obstacle normally blocks the result. Three families cover roughly half the assumption items ever set.

Statement & Argument - what makes an argument strong

Here the statement is a question of policy - 'Should X be done?' - and two arguments follow, usually one Yes and one No. Your job is not to decide the policy. Your job is to judge each argument on two counts. Is it RELEVANT, that is, directly related to this very proposal? And is it SIGNIFICANT, that is, does it deal with a major aspect rather than a trivial one? Relevance plus significance equals strong. Fail either test and the argument is weak.

Strong argument vs weak argument

STRONG	WEAK
Directly about this proposal	Ambiguous, or about something adjacent
Deals with a major, weighty aspect	Deals with a trivial or negligible effect
Rests on an established fact or a serious, foreseeable consequence	Rests on an unverifiable prediction or on emotion
Practical - says what will actually happen if we do this	Restates the proposal in other words (circular)
Offers a real cost or a real gain	Appeals to tradition, to other countries, or to ridicule

- Appeal to tradition - 'we have done it since ancient times' - weak.
- Mere imitation - 'other countries do it' - weak, unless it says WHY it worked there.
- Sweeping generalisation - 'young people are always careless', 'free things are never valued' - weak.
- Personal ridicule or emotion instead of a reason - weak.
- Circular restatement of the proposal - weak, it adds nothing.
- 'It will not solve the problem completely' - weak; a partial remedy can still be justified.
- Yes/No does not decide: a No that gives a real, weighty objection is strong; a Yes with a trivial gain is weak.

Worked example - RAS Pre 2024. Statement: Should private cars be barred from the central business district of big cities during peak hours? Argument I: Yes, it would substantially reduce traffic congestion and air pollution in the most crowded part of the city. Argument II: No, car owners have spent a lot of money on buying their vehicles.

- I is directly about this proposal and names the exact gain it is meant to produce - relevant.
- Congestion and air pollution in the busiest part of a city are major public problems - significant. So I is strong.
- II shifts to the private expenditure of car owners. Money already spent on a car does not weigh against a serious public harm - not significant.
- II is also an appeal to a private interest, not a reasoned objection to the policy - weak.
- Answer: Only argument I is strong.

TRAP

Two weak forms RPSC sets again and again. First: 'No, it will not put an end to every form of cheating / it will not solve the problem fully.' A measure is not useless because it is partial - always weak. Second: 'No, they will come back again' - a defeatist remark that examines nothing. Also weak.

ASKED IN RAS

RAS Pre 2021 - Should all beggars in big cities be sent back to their villages? Yes, it will reduce the population of big cities (trivial effect - weak). No, they will come back again (defeatist, examines neither causes nor rehabilitation - weak). Answer: Neither is strong. The same paper set the LPG-subsidy item where 'the government should never interfere with the market' was weak - a slogan, and self-defeating, since a subsidy is itself an intervention.

Statement & Conclusion – must be true, not likely to be true

A conclusion follows only if it MUST be true given the statement. Not probably true, not sensibly true – necessarily true. There is one test and you should run it mechanically: try to imagine a world where the statement is true and the conclusion is false. If you can build such a world, the conclusion does not follow. If you cannot, it does.

The falsification test

Assume the statement is true – completely, exactly as written → Now try to make conclusion I false while keeping the statement true → Succeeded? I does not follow. Failed? I follows → Repeat for II, independently → If neither follows alone, check whether I and II are complementary – then it is Either-or

- Percentages never give absolute numbers – a higher rate in A than in B does not prove A has more people.
- 'All P are Q' does not convert to 'All Q are P', and says nothing at all about a non-P.
- A sufficient condition is not a necessary one: 'everyone above 90 got a scholarship' does not mean only they did.
- 'Only those who passed were called' makes passing NECESSARY – so anyone called must have passed.
- Predictions about behaviour ('people will shift to public transport') never follow.

- Neither follows when both conclusions add outside information - causes, motives, comparisons, future behaviour.

Worked example - RAS Pre 2023. Statement: The literacy rate of District A is 78 per cent, while that of District B is 62 per cent. Conclusion I: District A has a larger number of literate persons than District B. Conclusion II: Some persons in District B are illiterate.

- For I, try to falsify. Let A have 1 lakh people - 78,000 literate. Let B have 10 lakh - 6,20,000 literate. Statement true, conclusion false.
- So a rate is not a number. I does not follow.
- For II, B's literacy is 62 per cent, so 38 per cent of B is not literate. Since B has people, some of them are illiterate.
- There is no way to keep the statement true and II false. II follows.
- Answer: Only conclusion II follows.

TRAP

Percentage is not population. This is RPSC's favourite conclusion trap and it reappears with literacy, unemployment, pass rates and vaccination coverage. Whenever an option compares NUMBERS and the statement gave you RATES, that option is wrong unless the two totals were also given.

Courses of Action - the twin tests

Here the statement describes a problem and the two items are proposed actions. Run two tests on each. Test one: is the problem real and is it the problem actually stated? Test two: would this action genuinely solve or reduce that problem? An action passes only if the answer to both is yes - and if it is practicable and within the power of the agency named. Two extra filters kill most wrong options: is it proportionate, and does it punish the victim?

Deciding a course of action

Name the problem in the statement in five words →

Ask: does this action attack THAT problem, or a different one? →

Ask: can the named agency actually do it - power, money, law? →

Ask: is it proportionate, or does it destroy the system to fix a fault? →

Ask: does it hurt the victims instead of the cause? →

Both can follow when one is the immediate remedy and the other the preventive measure

Action that follows vs action that does not

FOLLOWS	DOES NOT FOLLOW
Attacks the stated cause directly	Attacks a different problem, or none
Within the authority and means of the agency	Beyond its power or plainly impracticable
Proportionate to the fault	Extreme - ban it, close it down, dismiss everybody
Investigate / fix responsibility / give relief / create awareness through media	Punishes the victim - expel the failing students, stop the mid-day meal
Immediate remedy plus a long-term preventive - both can follow together	A platitude - 'people should be careful'

- The standard strong actions RPSC accepts: order an inquiry, fix responsibility, enforce the existing rule, survey and give relief, add capacity, run awareness through media.
- The standard rejected actions: ban it entirely, close it down, shift the whole population, arrest everyone, silence the complainants.

- Do not choose an action that assumes a problem the statement never mentioned.
- Both follow when I removes the cause and II handles the damage already done - or when I is vector control and II is public awareness.
- Neither follows when both are extreme or both are off-target - this option is real and is set fairly often.

Worked example. Statement: A newly constructed flyover developed wide cracks within six months of being opened to traffic. Course of action I: The flyover should be demolished immediately. Course of action II: A technical inspection should be ordered and responsibility for the substandard work fixed.

- Problem in five words - new flyover cracked early.
- I demolishes before anyone has assessed the damage. Extreme, and premature - it may well be repairable.
- I also destroys public property to answer a fault that has not yet been diagnosed. Not proportionate. I is out.
- II diagnoses the fault through a technical inspection and fixes responsibility for the bad work - the standard, proportionate response, and within the authority's power.
- Answer: Only II follows.

ASKED IN RAS

RAS Pre 2023 - candidates caught using unfair means in a recruitment examination. Abolishing recruitment examinations is extreme; debarring the guilty and proceeding under the unfair-means law is the answer - Only II. RAS Pre 2024 set two: the overturned overcrowded bus (check overloading - Only I; stopping all bus services permanently is disproportionate) and the 'very poor' AQI week (enforce dust and waste-burning curbs - Only II; asking a whole city to leave is impracticable).

The exam-hall drill

These items carry no calculation, so speed comes from having a fixed routine rather than from thinking faster. Thirty to forty seconds each. Read the statement twice and the options once. If an item makes you want to argue about the topic, you have already left the statement - go back to the words.

- Attempt this block early in the paper, while your reading is sharp.
- Never let your verdict on item I colour your verdict on item II - judge them separately.
- Scan the item for an extreme word first; it settles a large share of items in three seconds.
- If you find yourself using a fact from the newspaper, stop - that fact is not in the statement.
- If two rounds of reading leave you undecided, mark and move; do not spend ninety seconds on one mark.

- Assumption - negate it and see if the statement collapses.
- Argument - relevant to this proposal AND about something weighty.
- Conclusion - can the statement be true and this false?
- Course of action - real problem, and would it actually fix it?

REVISION RECAP

1. Four sub-types - assumption, argument, conclusion, course of action - and all four appear in every RAS paper, about 4-6 marks together.
2. The one governing rule: the statement is a closed world; never answer from outside knowledge or opinion.
3. Assumption = what the speaker took for granted, unstated but necessary; never something merely true or merely desirable.

4. Negation test for assumptions: suppose it false; if the statement collapses, it is implicit.
5. An effect of the statement is never an assumption behind it - 'farmers will stop using urea' is out.
6. Extreme words in an item - only, all, never, always, every, cannot - kill it, unless the statement is itself that extreme.
7. An item that merely restates the statement is neither an assumption nor a fresh conclusion.
8. Advertisement family: the quality is wanted and the ad will influence buyers. Notice family: people can read it and will comply.
9. 'In spite of X, Y happened' takes for granted that X normally prevents Y.
10. Assumption sits before the statement, conclusion comes out of it, inference is the most reasonable reading of it.
11. Argument test = relevance plus significance; Yes or No has nothing to do with strength.
12. Weak markers: tradition, imitation of other countries, ridicule, sweeping generalisation, circular restatement, 'it will not solve it completely'.
13. Conclusion test: can the statement be true while the conclusion is false? If yes, it does not follow.
14. Rates are not numbers - a higher percentage never proves a larger population of literates, patients or passes.
15. Sufficient is not necessary: 'all above 90 got a scholarship' does not mean nobody else did.
16. 'Only those who passed were called' makes passing necessary - anyone called must have passed.
17. Either I or II follows only when neither holds alone but the two are contradictory and exhaust all possibilities.
18. Course of action twin tests: is the stated problem real, and would this action actually solve it - plus practicable, within authority, proportionate.
19. Reject actions that are extreme, that punish the victim, or that are mere platitudes; accept inquiry, enforcement, relief, capacity and awareness.
20. Both courses follow when one is the immediate remedy and the other the preventive or long-term measure for the same problem.

RM02

Analytical and Critical Reasoning; Syllogism

Syllogism and logical Venn diagram items have appeared in RAS Prelims 2021, 2023 and 2024. With cause-and-effect, fact/inference/judgement and small puzzles, this block is worth about 4-6 marks. It is pure method - there is nothing here to memorise from the world.

Syllogism is the one reasoning topic where 'it sounds right' is always wrong. Draw the circles, apply the rules, key the answer. This chapter gives you the four statement forms, the Venn diagram described in words, the rules for definite and possible conclusions, the either-or trick - and then the smaller cousins: logical deduction, cause and effect, fact-inference-judgement and self-contained puzzles.

Analytical & Critical Reasoning

Syllogism
- Four forms: A (All), E (No), I (Some), O (Some not)
- Conversion rules
- Combination table: A+A=A, A+E=E, I+A=I, I+E=O
- Either-or complementary pair
- Possibility conclusions

Logical deduction
- 'A only if B' means A implies B
- Modus ponens - P true, so Q
- Modus tollens - Q false, so not P
- Necessary vs sufficient condition

Cause and effect
- Earlier + independent event = cause
- Both from a third event = common cause
- Unrelated spheres = independent causes

Fact / Inference / Judgement
- Fact - observed or verifiable from published data
- Inference - 'must have', 'therefore', 'is likely to'
- Judgement - 'should', 'ought', 'better than'

Critical reasoning
- Argument = premises + conclusion
- Weaken = give an alternative explanation
- Strengthen = rule alternatives out; randomised trial is best
- Correlation is not causation

Venn & sets
- n(A or B) = n(A) + n(B) - n(both)
- Concentric = full inclusion
- Separate circles inside a bigger one = mutually exclusive

Puzzles
- Ordering - build one chain with '>'
- Assignment - grid of who does what
- Truth-teller - test each case
- Input-output - what moves, and to which end

What this chapter asks, and the one discipline that carries it

This chapter is a family of small, self-contained items. Two or three class statements followed by two conclusions. Two events, and you must say which caused which. A sentence to be classified as fact, inference or judgement. A short paragraph whose argument you must weaken or strengthen. A Venn diagram to identify. A five-person ordering puzzle. None of them needs outside knowledge; all of them need one discipline - put the thing on paper before you decide.

- Syllogism - all/some/no statements, definite and possibility conclusions.
- Logical deduction - only if, if-then, necessary and sufficient conditions.
- Cause and effect - which of two events caused the other, or neither.
- Assertion of fact vs inference vs judgement - classify one sentence.
- Short critical reasoning passages - weaken, strengthen, find the assumption, name the flaw.
- Logical Venn diagrams and two-set counting.
- Small puzzles - ordering, assignment, truth-tellers, input-output.

TRAP

Never decide a syllogism by how the sentence sounds. 'All singers are dancers, some dancers are poets, therefore some singers are poets' sounds perfectly fine and is flatly wrong. The examiner writes items that sound right and are wrong. Draw the circles.

ASKED IN RAS

RAS Pre 2024 asked the Rajasthan-India-Asia logical Venn item (answer: three concentric circles, Rajasthan innermost). RAS Pre 2023 asked the two-set counting item - 60 students, 35 cricket, 30 football, 12 both, how many play neither (answer: 7). RAS Pre 2021 asked the English-Hindi pass percentage item (answer: 50 per cent passed both). Syllogism proper has come in all three years.

The four statement forms - A, E, I, O

Every syllogism statement is one of exactly four types. Learn the letters, because the combination rules are written in them. A and E are universal - they speak about the whole class. I and O are particular - they speak about a part of it. A and I are affirmative; E and O are negative. Once you can label a statement in one second, the rest of the topic is mechanical.

The four standard propositions

Letter	Form	Name
A	All A are B	Universal affirmative
E	No A is B	Universal negative
I	Some A are B	Particular affirmative
O	Some A are not B	Particular negative

- 'Some' in logic means at least one, possibly all - it does not mean 'not all'.
- Indian competitive exams follow the existential convention: every class named is taken to be non-empty.
- That convention is why 'All A are B' is allowed to give 'Some B are A'.
- The middle term is the term common to both statements; it must not appear in the conclusion.
- A valid conclusion always joins the two END terms - the ones that appear only once.

YAAD RAKHO

Yaad rakho - A E I O comes from the Latin AffIrmo and nEgO. The first two vowels of AffIrmo, A and I, are the two affirmative forms; the two vowels of nEgO, E and O, are the two negatives. Universal ones come first in each pair.

Drawing the Venn diagram in words, and the conversion rules

You do not need a neat diagram, only a correct one. Draw it as fast as you can write. All A are B is a small circle wholly inside a bigger one. No A is B is two circles that do not touch at all. Some A are B is two circles that overlap in a lens shape. Some A are not B is the same picture, but you mark a dot in the part of A that lies outside B - that dot is what the statement guarantees.

How to draw and read a syllogism

Label the statement A, E, I or O and find the middle term → Draw the first statement exactly - inside, apart, or overlapping → Attach the second statement to the middle term without disturbing the first → Read off what is FORCED in every possible drawing - that is what follows → If the second drawing could look different, the conclusion is not definite → If neither conclusion is definite, check for the either-or pair

Valid conversions - a statement turned round

Statement	What it converts to
Some A are B	Some B are A (always valid)
No A is B	No B is A (always valid)
All A are B	Some B are A only - NEVER All B are A
Some A are not B	Nothing at all - it has no valid conversion

TRAP

Two conversion traps RPSC lives on. First: 'All A are B' does not become 'All B are A' - so from 'All doctors are engineers' nothing follows about a particular engineer. Second: 'Some A are not B' does not give 'Some B are not A'. Students convert it out of symmetry and lose the mark.

The combination table - what two statements together give you

Once the two statements are labelled, their combination is fixed. Learn this table and you will key most two-statement items without drawing anything - keeping the diagram in reserve for the hard ones. Three blocking rules sit on top of it and cut the work further.

Two statements in, one conclusion out

Combination	Definite conclusion
All + All (A + A)	All (A)
All + No (A + E)	No (E)
Some + All (I + A)	Some (I)
Some + No (I + E)	Some ... are not ... (O)
No + All, in that order (E + A)	Some ... are not ..., after conversion - watch the direction
Some + Some	No definite conclusion
No + No	No definite conclusion
All + Some, in that order	No definite conclusion - the middle term stays undistributed

- Two particular premises (Some + Some) never give a definite conclusion.
- Two negative premises never give a definite conclusion.

- If one premise is particular, the conclusion must be particular; if one is negative, the conclusion must be negative.
- The middle term must be distributed at least once - covered by the word 'All' or 'No' somewhere.
- After finding the definite conclusion, remember its conversion also follows.

Worked example. Statements: Some books are pens. No pen is a pencil. Conclusion I: No book is a pencil. Conclusion II: Some books are not pencils.

- Label them - 'Some books are pens' is I, 'No pen is a pencil' is E. So this is I + E, which gives O.
- Draw it: book and pen circles overlap; the pen circle sits wholly apart from the pencil circle.
- The books inside the overlap are pens, so they cannot be pencils. That gives 'some books are not pencils' - II follows.
- But the books outside the overlap are untouched by the statements - they may lie inside the pencil circle.
- So the universal 'No book is a pencil' is not forced. I does not follow.
- Answer: Only conclusion II follows.

Either-or - the complementary-pair trick

Sometimes neither conclusion follows on its own, yet the answer is not 'Neither'. If the two conclusions are contradictories - between them they cover every possibility, and no third case exists - then one of them must be true even though you cannot say which. That is the 'Either I or II follows' answer. It is worth learning properly, because students who do not know it key 'Neither' and lose the mark every time.

Testing for either-or

Check that conclusion I alone does not follow → Check that conclusion II alone does not follow →

Check that both have the SAME subject and the SAME predicate → Check that they are contradictory - All vs Some-not, or Some vs No →

If all four hold, the answer is 'Either I or II follows'

Worked example. Statements: Some pens are books. All books are copies. Conclusion I: All pens are copies. Conclusion II: Some pens are not copies.

- I + A gives only 'Some pens are copies' - that much is certain, because the pens that are books must be copies.
- The remaining pens are unconstrained. They may all be copies, or some may not be. So neither I nor II is forced alone.
- Now check the pair. Same subject (pens), same predicate (copies). 'All pens are copies' and 'Some pens are not copies' are contradictories.
- Between them they cover every possibility - there is no third case.
- Answer: Either I or II follows.

> **YAAD RAKHO**
> The two complementary pairs to look for, and there are only two. All A are B / Some A are not B. And Some A are B / No A is B. If the pair in front of you is not one of these two shapes, it is not an either-or item.

Possibility conclusions

A possibility conclusion reads 'All A being B is a possibility' or 'Some A being B is a possibility'. The rule is the opposite of the one for definite conclusions. A definite conclusion must hold in every drawing. A possibility only needs ONE consistent drawing. So do not ask whether it is forced - ask whether it is forbidden. If you can draw even one picture where the statements are all true and the possibility holds, it follows.

- A possibility follows if at least one consistent Venn diagram allows it.
- It fails only when the statements definitely forbid it.
- A definite conclusion and its possibility-negation cannot both hold.
- So if 'No A is B' definitely follows, then 'Some A being B is a possibility' is FALSE.
- And if 'Some A are not B' definitely follows, 'All A being B is a possibility' is false.

Worked example, two items side by side. First - Statements: All apples are fruits. No fruit is a vegetable. Conclusion I: Some apples being vegetables is a possibility. Conclusion II: No apple is a vegetable. Second - Statements: All teachers are graduates. Some graduates are lawyers. Conclusion: All lawyers being teachers is a possibility.

- First item: apples sit inside fruits, and the fruit circle is wholly apart from the vegetable circle. So no apple can be a vegetable - II is definite.
- Because II is definite, the possibility in I is ruled out. Answer: Only conclusion II follows.
- Second item: is 'all lawyers are teachers' forbidden? Try to draw it - put the whole lawyer group inside the teacher circle.
- Teachers are inside graduates, so those lawyers are graduates too - 'Some graduates are lawyers' stays true, and 'All teachers are graduates' stays true.
- One consistent drawing exists, so the possibility follows.

TRAP

Do not answer a possibility item by asking 'must this be so?' - that is the definite test and it will make you reject every possibility. Ask only 'is this forbidden?' The two questions give opposite answers, and mixing them up is the commonest error in the whole topic.

Logical deduction - only if, if-then, necessary and sufficient

A small family of items turns on one conditional sentence. Get the direction of the arrow right and they are free marks; get it wrong and every option looks plausible. 'If P then Q' means P gives you Q, never the reverse. And 'A only if B' means B is necessary for A - so A implies B. Read that twice: the words 'only if' put the arrow the way most students do not expect.

The four moves - two valid, two fallacies

Given 'If P then Q', and...	Verdict
P is true, therefore Q	VALID - modus ponens
Q is false, therefore P is false	VALID - modus tollens
Q is true, therefore P	FALLACY - affirming the consequent
P is false, therefore not Q	FALLACY - denying the antecedent

Necessary vs sufficient - the distinction RPSC tests

NECESSARY condition	SUFFICIENT condition
Without it, the thing cannot happen	With it, the thing is guaranteed
Signalled by 'only if', 'only those who...'	Signalled by 'if', 'every ... was'
'Only those who passed were called' - being called proves you passed	'Everyone above 90 got a scholarship' - 85 marks proves nothing
Having it does not guarantee the result	Not having it does not rule the result out
Cleared the physical test - necessary for selection	Scoring above 90 - sufficient for the scholarship

Worked example. Statement: The morning bus reaches Jaipur on time only if it leaves Jodhpur before 6 a.m. Today the morning bus reached Jaipur on time. What definitely follows?

- Name the parts. P = reaches Jaipur on time. Q = left Jodhpur before 6 a.m.
- 'P only if Q' means P implies Q - leaving early is necessary for arriving on time.
- We are told P is true today. Modus ponens gives Q: the bus left Jodhpur before 6 a.m. today.
- What does NOT follow: 'every bus that leaves before 6 a.m. reaches on time' - that is the converse, a fallacy.
- Nor does anything follow about other days - the statement was about today only.
- Answer: The bus left Jodhpur before 6 a.m. today.

Cause and effect, and fact vs inference vs judgement

Cause-and-effect items give you two statements and four options: I caused II, II caused I, both are effects of a common cause, or both are effects of independent causes. The deciding question is time and independence. The event that came earlier, and that could have happened on its own, is the cause. If both events plainly follow from some third event that is not mentioned, the answer is common cause. If they belong to unrelated spheres altogether, it is independent causes.

Choosing among the four options

Ask which event came first in time → Ask whether the earlier one could produce the later one directly →

If yes, that is cause and effect - pick the right direction → If not, look for an unmentioned third event that would produce both →

Found one? Common cause. Found none, and the spheres are unrelated? Independent causes

Common cause vs independent causes

COMMON CAUSE	INDEPENDENT CAUSES
The two events are clearly linked, but not to each other	The two events have nothing to do with each other
One unmentioned event explains both	No single event explains both
Flights cancelled + schools closed on the same morning - dense fog	Gold price up in the world market + vegetable prices up in the local mandi
Same place, same time, same kind of disruption	Different spheres, merely simultaneous

Fact, inference, judgement - decide by the signal words

Type	What it is	Signal words
Fact	Already known, observed, or verifiable from published data	'As per Census 2011...', a named source, a figure
Inference	A conclusion drawn from the given facts	must have, therefore, is likely to, evidently

Type	What it is	Signal words
Judgement	An opinion, approval, disapproval or recommendation	should, must be made, ought, better than, more satisfying

Worked example. Statement I: Many flights from the city airport were cancelled this morning. Statement II: All schools in the city were ordered to remain closed this morning.

- Test direction one: do flight cancellations close schools? No - schools do not depend on flights.
- Test direction two: does closing schools cancel flights? No - no mechanism at all.
- So it is not cause and effect either way.
- Are they unrelated? No - same city, same morning, both a disruption of normal activity. That points at a third event.
- Severe weather - dense fog or a heavy storm - would ground flights and close schools alike, and it is not mentioned.
- Answer: Both statements are effects of a common cause.

Critical reasoning, logical Venn and the small puzzles

In a critical reasoning passage the argument is premises plus a conclusion, and the question asks you to weaken it, strengthen it, name its unstated assumption, or name its flaw. There is one master idea behind all four. Evidence supports a causal conclusion only if no other explanation fits the same evidence. So to weaken, supply an alternative explanation. To strengthen, rule alternatives out - and the strongest possible support is a controlled or randomised comparison.

- Weaken - show the evidence covers only part of the claim, or supply a rival cause.
- Strengthen - randomised assignment beats every other option, because it removes self-selection.
- Assumption - the unstated link the argument needs to get from its premise to its conclusion.
- Correlation is not causation: if a third factor accompanies both, the causal claim collapses.
- Common flaws: affirming the consequent, treating a necessary condition as sufficient, hasty generalisation from a small or self-selected sample, appeal to authority or tradition.

Worked example. A city introduced a congestion fee on cars entering its central zone. The next year the number of cars entering the zone fell by 20 per cent, and the mayor concluded that the fee had reduced car use in the city. Which fact would most weaken this?

- Separate the parts. Evidence: fewer cars entered the CENTRAL ZONE. Conclusion: car use fell in the CITY.
- The gap between the two is where the argument is vulnerable - a zone is not a city.
- So look for an option showing the cars went elsewhere rather than off the road.
- 'Many drivers merely shifted to roads just outside the central zone, and the total number of car trips in the city was unchanged' - that is an alternative explanation of the same evidence.
- Options about electronic collection, spending of revenue or protests say nothing about the level of car use, so they neither weaken nor strengthen.
- Answer: the displacement option.

Two-set counting and logical Venn shapes

What is asked	What to use
At least one of A or B	n(A) + n(B) - n(both)
Neither A nor B	Total - [n(A) + n(B) - n(both)]
Exactly one of A or B	n(A) + n(B) - 2 x n(both)
Rajasthan, India, Asia	Three concentric circles - complete inclusion, smallest innermost
Aunt, Uncle, Relative	Two separate circles inside one bigger circle - mutually exclusive within a class
Doctor, Father, Man	Overlapping circles - a person may be any combination

YAAD RAKHO

Puzzle drill, one line each. Ordering: turn every clue into a single chain with '>' before you count positions. Assignment: draw a grid of names against roles and put crosses first, ticks last. Truth-teller: test each case against every statement and keep the one that matches the given count of true statements. Input-output: identify what moves at each step and to which end - the untouched elements keep their relative order.

ASKED IN RAS

RAS Pre 2023 - 60 students, 35 play cricket, 30 football, 12 both. At least one = 35 + 30 - 12 = 53, so neither = 60 - 53 = 7. RAS Pre 2021 - 70 per cent passed English, 65 per cent Hindi, 15 per cent failed both. Passed at least one = 85, so both = 70 + 65 - 85 = 50 per cent. Learn the formula in this form; both papers used exactly this shape.

REVISION RECAP

1. Four forms - A: All A are B, E: No A is B, I: Some A are B, O: Some A are not B. Learn the letters; the rules are written in them.
2. 'Some' means at least one, possibly all; every class named is taken as non-empty.

3. Conversions: Some A are B gives Some B are A; No A is B gives No B is A; All A are B gives only Some B are A.
4. 'Some A are not B' has no valid conversion at all.
5. Combination table: A+A=A, A+E=E, I+A=I, I+E=O; E+A gives a Some-not after conversion.
6. Some + Some gives nothing; No + No gives nothing; All + Some in that order gives nothing.
7. One particular premise forces a particular conclusion; one negative premise forces a negative conclusion.
8. The middle term must be distributed at least once and must never appear in the conclusion.
9. Either-or: neither follows alone, same subject and predicate, and the two are contradictories - All/Some-not, or Some/No.
10. A possibility follows if even one consistent diagram allows it; it fails only if the statements forbid it.
11. A definite conclusion kills its own possibility-negation - if No A is B is definite, Some A being B is not possible.
12. 'A only if B' means A implies B; modus ponens and modus tollens are valid, the converse and the inverse are fallacies.
13. Necessary is not sufficient: cleared the physical test is necessary for selection, scoring above 90 is sufficient for the scholarship.
14. Cause and effect: the earlier, independently occurring event is the cause; two consequences of an unmentioned third event are effects of a common cause.
15. Events in unrelated spheres that merely coincide in time are effects of independent causes.
16. Fact is verifiable, inference carries 'must have' or 'therefore', judgement carries 'should', 'ought' or 'better than'.
17. Weaken an argument by supplying an alternative explanation; strengthen it by ruling alternatives out - a randomised comparison is the strongest support.
18. n(A or B) = n(A) + n(B) - n(both); neither = Total minus that; exactly one = n(A) + n(B) - 2 x n(both).
19. Logical Venn: full inclusion is concentric circles, mutually exclusive classes within a wider class are separate circles inside one big circle.
20. In every item of this chapter, draw first and decide second - the items that sound right are the ones set to trap you.

RM03

Number and Letter Series

About 3-5 marks in every RAS Prelims paper - series, analogy and odd-one-out taken together. Confirmed appearances include 2013, 2016, 2018, 2021, 2023 and 2024. Difficulty is easy to moderate. These are the marks you bank first and come back to nothing.

This chapter is not knowledge, it is a checklist. A series question has a rule hidden in it, and there are only about a dozen rules RPSC ever uses. The student who runs the checklist finds the rule in forty seconds; the student who stares at the numbers hoping for inspiration loses two minutes and still guesses.

Series, Analogy, Classification

Toolkit (learn cold)

- A=1 to Z=26 positions
- EJOTY and CFILORUX anchors
- Reverse position = 27 - n
- Squares, cubes, primes, Fibonacci

Number series families

- Constant difference (AP)
- Differences in AP - n-squared type
- Constant ratio / rising multipliers
- prev x2 plus-or-minus k
- n^2+1, n^2-1, n^3+1, n^3-1, n(n+1)
- Primes, triangular, Fibonacci

Two special shapes

- Alternate / interleaved - split odd and even places
- Alternating operations - x3 then -3

Wrong term

- Fix the rule from three agreeing terms
- Run it forward
- First term that breaks is the answer

Letter and alphanumeric

- Convert letters to numbers first
- Fixed gap, growing gap, squares as positions
- Opposite-letter pairs AZ BY CX DW
- Letter track and number track are independent

Analogy and odd one out

- Number: difference, ratio, square, cube, n(n+1)
- Letter: shift each letter, apply same shifts
- Word: name the category first
- Letter group: compare internal gaps

The alphabet table – memorise this before you touch a single question

Start here. Almost every letter series, every letter analogy and every coding question in the paper is really a number question wearing a costume. You take the letters off the page, turn them into their positions in the alphabet, do arithmetic, and turn the answer back into a letter. If you have to count A-B-C-D on your fingers each time, you will lose thirty seconds an item and you will make counting errors. So learn the table below cold. It is the single highest-return thing in the whole reasoning section, because it pays in this chapter and again in Coding-Decoding.

Alphabet positions – learn both halves

Letter	No.	Letter	No.
A	1	N	14
B	2	O	15
C	3	P	16
D	4	Q	17
E	5	R	18
F	6	S	19
G	7	T	20
H	8	U	21
I	9	V	22
J	10	W	23
K	11	X	24
L	12	Y	25
M	13	Z	26

YAAD RAKHO

Yaad rakho EJOTY - E=5, J=10, O=15, T=20, Y=25. Five anchors at every multiple of five. Never count from A; jump to the nearest anchor and step two or three places. Second mnemonic for the in-between values: CFILORUX - C=3, F=6, I=9, L=12, O=15, R=18, U=21, X=24, the multiples of three. Between EJOTY and CFILORUX you are never more than one step from a known letter.

Reverse position and opposite letters - reverse position = 27 minus forward position

Letter	Forward position	Reverse position (27 - n)	Opposite letter
A	1	26	Z
B	2	25	Y
C	3	24	X
D	4	23	W
E	5	22	V
F	6	21	U
G	7	20	T
H	8	19	S
I	9	18	R
J	10	17	Q
K	11	16	P
L	12	15	O
M	13	14	N

TRAP

Shifts wrap round the alphabet. After Z comes A again, and before A comes Z. So A minus 3 is X, not a blank, and Z plus 2 is B. Students lose easy marks by stopping at the edge of the alphabet instead of going round it. Write the alphabet as a circle in your rough work if it helps.

The attack order - what to test, and in what order

When a number series appears, do not think. Run the list. There is a fixed order because the cheap tests catch the common series and the expensive tests are only needed for the rare ones. Write the first differences under the series in your rough space every single time - that one habit solves more than half of all RAS series items on its own.

The six tests, always in this order

1. First differences - constant? then it is an AP, add the same again →
2. Second differences - are the differences themselves in AP? then it is an n-squared type →
3. Ratios - constant ratio is a GP; growing multipliers x2, x3, x4 also live here →
4. Squares and cubes plus or minus a small constant - check terms against 1,4,9,16 and 1,8,27,64 →
5. Alternate terms - split into odd places and even places and solve each separately →
6. Primes, triangular numbers, Fibonacci - the named lists

The families RPSC actually uses

Family	Rule	First terms
Squares	n^2	1, 4, 9, 16, 25, 36, 49
Cubes	n^3	1, 8, 27, 64, 125, 216
Square plus one	n^2 + 1	2, 5, 10, 17, 26, 37
Square minus one	n^2 - 1	0, 3, 8, 15, 24, 35
Cube plus one	n^3 + 1	2, 9, 28, 65, 126
Cube minus one	n^3 - 1	0, 7, 26, 63, 124, 215
Product of consecutives	n(n+1)	2, 6, 12, 20, 30, 42

Family	Rule	First terms
Cube plus square	n^3 + n^2	2, 12, 36, 80, 150, 252
Triangular	differences 2,3,4,5	1, 3, 6, 10, 15, 21, 28
Fibonacci type	each term = sum of previous two	1, 1, 2, 3, 5, 8, 13, 21
Primes	no factor except 1 and itself	2, 3, 5, 7, 11, 13, 17, 19, 23

- Squares to know without thinking: 1, 4, 9, 16, 25, 36, 49, 64, 81, 100, 121, 144, 169, 196, 225.
- Cubes to know: 1, 8, 27, 64, 125, 216, 343, 512, 729, 1000, 1331, 1728.
- Primes under 60: 2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47, 53, 59.
- 1 is NOT prime. 2 is the only even prime - RPSC has used both of these as the trick.
- Constant difference means arithmetic; constant ratio means geometric; differences forming an AP means a quadratic, n-squared type.

YAAD RAKHO

Trick that saves the most time: if the differences themselves double (say 3, 6, 12, 24), the rule is almost never a difference rule at all - it is 'previous term times 2 plus or minus k'. Find k on one pair and confirm it on a second pair before you commit.

Differences and ratios - three worked examples

Work these on paper with me. The point is not the answer, it is the writing down of the differences. Take the series 5, 11, 19, 29, 41, ? - the numbers look irregular, but the differences are not.

- Series: 5, 11, 19, 29, 41, ?
- First differences: 11-5 = +6, 19-11 = +8, 29-19 = +10, 41-29 = +12.

- The differences are themselves rising by 2 - so the next difference is +14.
- Answer: 41 + 14 = 55.
- Note what happened - test 1 failed (differences not constant) but test 2 caught it at once.

- Series: 6, 11, 21, 36, 56, ? - differences are +5, +10, +15, +20, an AP with common difference 5.
- Next difference is +25, so the answer is 56 + 25 = 81.
- Series: 120, 99, 80, 63, 48, ? - differences are -21, -19, -17, -15, the odd numbers shrinking.
- Next difference is -13, so the answer is 48 - 13 = 35.
- A falling series is the same drill. Do not panic at the minus signs; just write them down.

- Series: 4, 9, 19, 39, 79, ? - differences are +5, +10, +20, +40, which DOUBLE. Switch to the multiply rule.
- Test: 4 x 2 + 1 = 9. Confirm: 9 x 2 + 1 = 19, 19 x 2 + 1 = 39, 39 x 2 + 1 = 79. Rule holds.
- Answer: 79 x 2 + 1 = 159.
- Series: 7, 10, 16, 28, 52, ? - try previous x 2 minus k. 7 x 2 - 4 = 10, and 10 x 2 - 4 = 16. Confirmed.
- Answer: 52 x 2 - 4 = 100.
- Series with rising multipliers: 3, 6, 18, 72, ? - the multipliers are x2, x3, x4, so next is x5 giving 360.
- Falling divisors work the same way: 625, 125, 25, 5, ? - each divided by 5, so the answer is 1.

TRAP

Sometimes the OPERATION alternates rather than the number. Look at 5, 15, 12, 36, 33, 99, ? - nothing works as a difference or a ratio, because the steps are x3, then -3, then x3, then -3. Continue the pattern: 99 - 3 = 96. Whenever alternate steps look sensible on their own but the whole series looks mad, suspect alternating operations.

Squares, cubes, primes and the offset families

This is test 4, and it is the one students skip. The moment a term sits one or two away from a square or a cube you know, stop doing differences and check the offset instead. Compare the series against the memorised list and write the offset above each term.

- Series: 2, 5, 10, 17, 26, ? - compare with squares 1, 4, 9, 16, 25.
- Every term is exactly one more, so the rule is n^2 + 1.
- Answer: 6^2 + 1 = 37. (Cross-check: the differences are the odd numbers 3, 5, 7, 9, 11 - that is the signature of a square-based series.)

- Series: 0, 7, 26, 63, 124, ? - compare with cubes 1, 8, 27, 64, 125. Every term is one less, so n^3 - 1.
- Answer: 6^3 - 1 = 216 - 1 = 215.
- Same family shifted: 7, 26, 63, 124, 215, ? is n^3 - 1 starting from n = 2, so the answer is 7^3 - 1 = 342.
- Harder mix: 2, 12, 36, 80, 150, ? - here 1+1, 8+4, 27+9, 64+16, 125+25, so the rule is n^3 + n^2.
- Answer: 216 + 36 = 252. When neither squares nor cubes fit alone, try their sum.

ASKED IN RAS

Asked in RAS Pre 2013 - complete 1, 4, 9, 16, 25, 36, ? Answer: 49. These are 1^2 to 6^2, so the next is 7^2. Note the distractor 48 sitting right next to it; RPSC always plants a near-miss.

ASKED IN RAS

Asked in RAS Pre 2021 - complete 2, 3, 5, 7, 11, 13, ? Answer: 17. These are consecutive primes. The trap option is 15, which looks like the right gap but is 3 x 5 and therefore composite.

Two more named lists finish this section. Triangular numbers 1, 3, 6, 10, 15, 21, 28 have differences 2, 3, 4, 5, 6 - so 21 + 7 = 28. Fibonacci type means each term is the sum of the two before it: 1, 1, 2, 3, 5, 8, 13 gives 8 + 13 = 21, and the same rule with different seeds gives 3, 4, 7, 11, 18, 29, 47. If a term looks like roughly the sum of the previous two, test it - it takes five seconds.

Alternate series, and finding the WRONG term

If no single rule fits the whole series, the series is probably two series plaited together. Split it: terms at the 1st, 3rd, 5th places form one series; terms at the 2nd, 4th, 6th places form another. Solve each separately and then answer for whichever place the question asks about. This is worth checking early whenever the terms zig-zag up and down.

- Series: 4, 9, 7, 18, 10, 27, 13, ? - the eighth term is asked, so it belongs to the even-place series.
- Odd places: 4, 7, 10, 13 - rising by 3.
- Even places: 9, 18, 27, ? - rising by 9.
- Answer: 27 + 9 = 36.

- Series: 1, 2, 4, 6, 9, 12, 16, ? - odd places 1, 4, 9, 16 are perfect squares; even places 2, 6, 12, ? are n(n+1).
- The eighth term is at an even place, so it is 4 x 5 = 20.

Missing term vs wrong term - two different jobs

Missing-term question	Wrong-term question
Every printed term obeys the rule	Exactly one printed term breaks the rule
Find the rule from any adjacent pair	Find the rule from THREE consecutive terms that agree
Extend the rule by one step	Run the rule from the start and find the first break
Answer is a new number not in the series	Answer is one of the numbers already printed
Options are close neighbours of the true value	Options are all terms of the given series

Wrong-term procedure

Take the first three terms and find the rule that fits them →

Check the rule survives the fourth term - if not, re-fix the rule on terms 2, 3, 4 →

Now generate the whole series yourself from the fixed rule →

Compare term by term; the FIRST mismatch is the wrong term →

Do not stop at the first odd-looking number - check it against your generated series

- Series: 3, 8, 15, 24, 34, 48, 63 - the first four are 4-1, 9-1, 16-1, 25-1, so the rule is n^2 - 1.
- Generated series: 3, 8, 15, 24, 35, 48, 63. The fifth printed term is 34 but should be 35. Answer: 34.

- Series: 2, 5, 10, 17, 26, 38, 50 - the rule is n^2 + 1, giving 2, 5, 10, 17, 26, 37, 50. Answer: 38.
- Series: 5, 10, 20, 40, 75, 160 - the rule is doubling, giving 5, 10, 20, 40, 80, 160. Answer: 75.
- Series: 1, 4, 9, 16, 24, 36, 49 - perfect squares; 24 should be 25. Answer: 24.

TRAP

In a wrong-term question the wrong number is usually only ONE away from the correct one - 34 for 35, 38 for 37, 24 for 25. That is deliberate. Never eyeball the series looking for the number that 'feels' out of place; generate the correct series in the margin and compare digit by digit.

Letter series and alphanumeric series

One rule governs the whole of this section: convert the letters into their positions, solve it as a number series, convert the answer back into a letter. Never guess by feel and never try to 'hear' the pattern. Write the number under each letter - it takes three seconds with the table you memorised and it removes the guesswork entirely.

- Series: A, C, F, J, O, ? - positions 1, 3, 6, 10, 15. Gaps: +2, +3, +4, +5.
- Next gap is +6, so position 21, which is U. Answer: U.
- Series: A, D, I, P, ? - positions 1, 4, 9, 16 are perfect squares. Next is 25 = Y. Answer: Y.
- Series: B, E, J, Q, ? - positions 2, 5, 10, 17, which is n^2 + 1. Next is 26 = Z. Answer: Z.
- Series: C, F, I, L, ? - constant gap of +3, so position 15 = O. Answer: O.

- Series: Z, X, V, T, ? – positions 26, 24, 22, 20 going down by 2. Next is 18 = R. Answer: R.

> **ASKED IN RAS**
> Asked in RAS Pre 2018 – complete A, C, F, J, O, ? Answer: U. The gaps grow by one each time (+2, +3, +4, +5, +6). If you had counted letters instead of writing positions you would very likely have landed on T.

> **ASKED IN RAS**
> Asked in RAS Pre 2023 – complete AZ, BY, CX, DW, ? Answer: EV. The first letters move forward A, B, C, D, E while the second letters move backward Z, Y, X, W, V. Notice these are exactly the opposite-letter pairs from the table in section 1 – each pair adds to 27.

Letter-group series – what to check

Series	What to look at	Answer
ACE, BDF, CEG, ?	Internal gap +2 fixed; each group starts one letter later	DFH
AB, DEF, HIJK, ?	Group length 2, 3, 4, 5; exactly one letter skipped between groups (C, G, L)	MNOPQ
AZ, BY, CX, DW, ?	First letter forward, second letter backward	EV
A, D, I, P, ?	Positions are perfect squares	Y

- Alphanumeric series: treat the letter track and the number track as two completely separate series.
- A1, C3, E5, G7, ? – letters skip one each time (A, C, E, G, I); the number is simply the letter's own position. Answer: I9.
- Z1, Y2, X3, W4, ? – letters run backwards, numbers run forwards. Answer: V5.

- A2, D4, G8, J16, ? - letters advance by 3 (A, D, G, J, M); numbers double (2, 4, 8, 16, 32). Answer: M32.
- 2A, 5D, 10G, 17J, ? - numbers are n^2 + 1 (2, 5, 10, 17, 26); letters advance by 3. Answer: 26M.
- Letter-number-SYMBOL series work the same way - a third track of symbols (*, @, #) usually repeats on a short cycle.
- Solve the easiest track first, then use it to knock out three options before you even check the others.

Analogy - number, letter and word

An analogy gives you a relation in the first pair and asks you to reproduce it in the second. Name the relation in words before you look at the options; if you read the options first you will find a relation that fits some option, which is not the same thing as the relation the question intends. For numbers test in this order - difference, ratio, square, cube, n(n+1), n^2 plus or minus 1, digit reversal. For letters compute the positional shift for EVERY letter of the first pair and apply exactly those shifts. For words state the relation as a full sentence: 'a pen is used to write'. If two options both fit, your relation is too loose - make it more specific until only one survives.

- 8 : 27 :: 64 : ? - 8 = 2^3 and 27 = 3^3, cubes of consecutive numbers. 64 = 4^3, so the answer is 5^3 = 125.
- 25 : 36 :: 49 : ? - squares of consecutive numbers 5 and 6. 49 = 7^2, so the answer is 8^2 = 64.
- 6 : 42 :: 8 : ? - the relation is n : n(n+1), since 6 x 7 = 42. So 8 x 9 = 72.
- In each case naming the relation ('cubes of consecutive numbers') is the whole solution; the arithmetic is trivial.

- BDF : CEG :: MOQ : ? - every letter moves +1 (B to C, D to E, F to G). Apply +1 to M, O, Q: answer NPR.
- ACE : BDF :: PRT : ? - again +1 on each letter. P to Q, R to S, T to U: answer QSU.
- Check every letter, not just the first. A rule that fits the first letter and fails the third is a wrong rule.
- If the shift is not constant, look for a shift that depends on the letter's place in the group (+1, +2, +3).

Word-analogy relations RPSC repeats

Relation	Example
Tool to its function	Pen : Write :: Knife : Cut
Specialist to organ treated	Ophthalmologist : Eye :: Cardiologist : Heart
Worker to workplace	Teacher : School :: Judge : Court
Product to raw material	Bread : Wheat :: Cloth : Cotton
Part to whole	Petal : Flower :: Chapter : Book
Animal to its young	Cow : Calf :: Horse : Foal
Cause to effect	Fire : Smoke :: Rain : Flood
Instrument to what it measures	Thermometer : Temperature :: Barometer : Pressure

ASKED IN RAS

Asked in RAS Pre 2024 - Ophthalmologist : Eye :: Cardiologist : ? Answer: Heart. Learn the whole set, because the distractors are drawn from it: nephrologist-kidney, neurologist-brain, pulmonologist-lung, dermatologist-skin, orthopaedist-bones.

Odd one out (classification), and how to play this block in the exam

Classification asks which item does not share the property the other three share. So the job is to name the property, not to find the strange item. Three of them will belong to one clean category. Find that category, and the fourth is the answer by elimination.

What to test, by type of item

Type	Tests to run, in order
Numbers	Prime, perfect square, perfect cube, common factor, digit sum
Letter groups	Internal gaps between successive letters; the group with a different gap pattern is out
Words	Name the category - metals, flowers, alloys, birds that fly, capitals - then find the outsider
Mixed	Look for the property three share; never look for what makes one 'special'

- 100, 27, 64, 125 - 27, 64 and 125 are perfect cubes (3^3, 4^3, 5^3). 100 is a perfect square, not a cube. Answer: 100.
- 3, 9, 5, 11 - 3, 5 and 11 are prime; 9 = 3 x 3 is composite. Answer: 9.
- ACE, GIK, RTW, MOQ - gaps are +2,+2 in ACE, GIK and MOQ but +2,+3 in RTW. Answer: RTW.
- Rose, Mango, Lotus, Lily - three are flowers, mango is a fruit. Answer: Mango.
- Brass, Copper, Zinc, Iron - copper, zinc and iron are pure metals; brass is an alloy of copper and zinc. Answer: Brass.

ASKED IN RAS

Asked in RAS Pre 2016 - which is different: 100, 27, 64, 125? Answer: 100. Watch the built-in trap: 64 is both a square and a cube, so a careless student picks 64 as 'the special one'. The property shared by three of them is cube, and 64 has it.

YAAD RAKHO

Yaad rakho for classification with numbers: if one option is a perfect square AND a perfect cube (1, 64, 729), it is almost never the answer - it is the bait. Ask what three of them share, never what one of them has extra.

Exam strategy for this block. Series, analogy and classification are the easiest scoring items in the RAS reasoning section, so do them on your first pass and bank the marks. But there is one-third negative marking, so a genuine blank is better than a wild guess. The practical rule: if you have found the rule and it fits every printed term, mark it and move on without re-checking. If ninety seconds have gone and no rule has appeared, leave it, and come back only if time remains at the end.

REVISION RECAP

1. A=1 to Z=26 and reverse = 27 minus forward position. Memorise both; they pay in this chapter and in coding.
2. EJOTY: E5 J10 O15 T20 Y25. CFILORUX: the multiples of 3. Jump to an anchor, never count from A.
3. Opposite pairs sum to 27: A-Z, B-Y, C-X, D-W, E-V, F-U, G-T, H-S, I-R, J-Q, K-P, L-O, M-N.
4. Shifts wrap: A - 3 = X, Z + 2 = B.
5. Attack order: first differences, second differences, ratios, squares/cubes with offset, alternate terms, primes.
6. Constant difference = AP; constant ratio = GP; differences in AP = n-squared type.

7. Differences that double mean 'previous x 2 plus or minus k' - fix k on one pair, confirm on a second.
8. Rising multipliers x2, x3, x4, x5 give 3, 6, 18, 72, 360. Falling divisors give 625, 125, 25, 5, 1.
9. Offset families: n^2+1 (2,5,10,17,26), n^2-1 (0,3,8,15,24), n^3-1 (0,7,26,63,124), n(n+1) (2,6,12,20,30), n^3+n^2 (2,12,36,80,150).
10. Odd-number differences 3, 5, 7, 9 are the fingerprint of a square-based series.
11. If the series zig-zags, split it into odd places and even places and solve the two separately.
12. Alternating operations (x3, -3, x3, -3) explain series that defy both differences and ratios.
13. Wrong term: fix the rule on three agreeing terms, generate the series yourself, take the FIRST mismatch.
14. The wrong term is planted only one away from the correct value - never eyeball it.
15. Letter series: convert to positions, solve as numbers, convert back. A,C,F,J,O gaps +2,+3,+4,+5 gives U.
16. Positions as squares: A, D, I, P, Y. Positions as n^2+1: B, E, J, Q, Z.
17. Alphanumeric: letter track and number track are independent - solve the easier one and eliminate.
18. Number analogy order: difference, ratio, square, cube, n(n+1), n^2 plus or minus 1, digit reversal.
19. Word analogy: say the relation in a full sentence before looking at the options.
20. Odd one out: name the property that THREE share. 64 as square-and-cube is bait, not the answer.

RM04

Coding-Decoding

1-3 questions in every RAS Prelims paper without exception - 2013, 2016, 2018, 2021, 2023 and 2024 all carry at least one. Pure method, no knowledge, no calculation beyond counting to 26. This chapter should never be left unattempted.

A code is a rule, and there are only seven rules RPSC ever uses. The whole chapter is: identify which of the seven you are looking at, verify it on a second letter or a second word, then apply it. Students lose these marks not because the codes are hard but because they apply a shift method to a substitution question.

Coding-Decoding

Letter coding - shifts

- Constant forward +1, +2, +3
- Constant backward -1, -2, -3
- Alternating +1, -1
- Shift by place in the word (+1, +2, +3)

Structural - no shift at all

- Full reversal DELHI to IHLED
- Pair interchange MONKEY to OMKNYE
- First-last swap SYSTEM to MYSTES
- Opposite letters DOG to WLT
- Reverse then shift CAT to UBD

Number coding

- Position written out CAT = 3-1-20
- Sum of positions CAT = 24
- Multiple of position A=2, B=4
- Reverse position Z=1 ... A=26
- Position plus or minus a constant

Mixed letter-digit

- Build a letter-to-digit dictionary
- Shared letters must agree
- Disagreement means mis-alignment

Substitution and sentences

- 'X is called Y' - answer in plain language first
- Two sentences sharing one word fix that pair
- Cross out each code word as it is fixed

Conditional and matrix

- Plain code first, then conditions in order
- Watch first and last letter for vowels
- Matrix: row number then column number
- Note which letter is left out

The first question to ask, every single time

Before you touch the letters, count. Is the code the same length as the word? If yes, you are looking at a letter-to-letter rule and your job is to find the shift or the rearrangement. If the code is shorter, or is a number, or is an ordinary English word, then it is number coding, substitution or compression - a completely different method. Getting this one decision right in five seconds saves you the minute students waste hunting for a shift that does not exist.

Decide the type before you solve

Count the letters in the word and in the code - same length or not? → Same length, letters only: check a fixed shift first, then a structural rule →

Same length but the SAME letters reordered: it is structural - reversal, pair swap or first-last swap → Code is digits: check position, sum of positions, multiple of position, reverse position →

Code is ordinary English words: it is substitution or a coded sentence - never look for shifts → Table of letters with digits plus a list of conditions: conditional coding

The seven rule families

Family	Signature	Example
1. Constant forward shift	Every letter moves the same number forward	MADRAS to NBESBT (+1)
2. Constant backward shift	Every letter moves the same number back	TRIPPLE to SQHOOKD (-1)
3. Alternating shift	+1 then -1 then +1	CHAIR to DGBHS
4. Shift by place in word	First +1, second +2, third +3	ABC to BDF

Family	Signature	Example
5. Structural, no shift	Same letters, new order	DELHI to IHLED; MONKEY to OMKNYE
6. Opposite letter	27 minus the position	DOG to WLT
7. Combination	Reverse first, then shift	CAT to UBD

YAAD RAKHO

Yaad rakho: you cannot do this chapter without A=1 to Z=26 in your head, plus EJOTY (E5, J10, O15, T20, Y25) and CFILORUX (C3, F6, I9, L12, O15, R18, U21, X24). And the opposite-letter rule, 27 minus the position, giving A-Z, B-Y, C-X, D-W, E-V, F-U, G-T, H-S, I-R, J-Q, K-P, L-O, M-N. Learn these before you learn any coding technique at all.

Constant shift coding - the core method

This is the family that appears most often. Write the word above and the code below, letter under letter. Compute the gap for the first letter. Then compute it again for the second and third. If the gap is the same, you have the rule; apply it to the asked word. If the gap changes sign or size, you are in family 3 or 4, not family 1.

Three steps, never skip the middle one

Find the shift on ONE pair of letters → Verify the same shift on a SECOND and THIRD pair → Apply it to every letter of the asked word, wrapping round the alphabet at Z and A

- MADRAS is written as NBESBT. How is BOMBAY written?

- M to N is +1. A to B is +1. D to E is +1. R to S, A to B, S to T - all +1. Rule fixed.
- Apply +1 to BOMBAY: B to C, O to P, M to N, B to C, A to B, Y to Z.
- Answer: CPNCBZ. Note the last letter - Y + 1 = Z, and the option CPNCAZ that changes the fifth letter is the planted trap.

ASKED IN RAS

Asked in RAS Pre 2016 - MADRAS is coded NBESBT, so BOMBAY is CPNCBZ. This exact +1 pattern has repeated across RAS and RPSC papers. If you see a code whose first letter is one after the word's first letter, test +1 on all six letters immediately.

- FRIEND is written as HTKGPF. How is CANDLE written?
- F to H is +2, R to T is +2, I to K is +2, E to G, N to P, D to F - all +2.
- Apply +2 to CANDLE: C to E, A to C, N to P, D to F, L to N, E to G. Answer: ECPFNG.
- TEACHER to WHDFKHU is the same family with +3, so STUDENT becomes VWXGHQW.
- TRIPPLE to SQHOOKD is -1, so DISPOSE becomes CHRONRD. Backward shifts are just as common as forward ones.

ASKED IN RAS

Asked in RAS Pre 2021 - FRIEND is coded HTKGPF, so CANDLE is ECPFNG. The nearest wrong option was ECPFNH, differing only in the last letter. Always compute the last letter of your answer separately and check it against the options first.

- WATER is coded TXQBO. How is FIRE coded?
- W(23) to T(20) is -3. A(1) to X(24) - here you must wrap: A minus 3 goes back past A to X. Still -3.

- T(20) to Q(17) is -3, E(5) to B(2) is -3, R(18) to O(15) is -3. Rule confirmed on five letters.
- Apply -3 to FIRE: F to C, I to F, R to O, E to B. Answer: CFOB.
- Wrapping is where marks are lost. Treat the alphabet as a circle: A - 1 = Z, A - 2 = Y, A - 3 = X; Z + 1 = A, Z + 2 = B.
- DECODING a message means running the rule backwards. HELP is coded GDKO, i.e. -1; so to read RTMCZX you apply +1.
- R to S, T to U, M to N, C to D, Z to A, X to Y - the message is SUNDAY. Reverse the sign of the shift, never the word.

Structural rules - when nothing shifts at all

Here is the fastest diagnostic in the chapter. Look at the letters of the code and the letters of the word. Are they the SAME letters, just in a different order? If yes, stop looking for a shift - it is a structural rule, and there are only three common ones plus one that combines with a shift. Check them in the order given in the table.

Structural rules and how to spot them

Rule	Test	Example
Full reversal	Code read backwards is the word	DELHI to IHLED
Successive pair interchange	Letters 1-2 swap, 3-4 swap, 5-6 swap	MONKEY to OMKNYE
First and last interchanged	Only the two end letters move	SYSTEM to MYSTES
Opposite letters (27 - n)	Letters are different, but each pair sums to 27	DOG to WLT
Reverse, then shift	Letters differ, but the reversed word is a fixed shift away	CAT to UBD

- DELHI is written as IHLED. How is MUMBAI written? Read the word backwards: I-A-B-M-U-M. Answer: IABMUM.
- MONKEY is written as OMKNYE - check: MO becomes OM, NK becomes KN, EY becomes YE. Successive pairs swapped.
- So CHAIRS becomes CH to HC, AI to IA, RS to SR. Answer: HCIASR.
- SYSTEM is coded MYSTES - only the M and the S at the ends have traded places. So NUMBER becomes RUMBEN.
- The pair-interchange trap: HCAIRS keeps the middle pair unswapped. Check EVERY pair, not just the first.

- DOG is written as WLT. Different letters, so test opposites: D(4) and W(23) sum to 27. O(15) and L(12) sum to 27. G(7) and T(20) sum to 27.
- So the rule is 'replace each letter by 27 minus its position'.
- CAT: C(3) becomes 24 = X, A(1) becomes 26 = Z, T(20) becomes 7 = G. Answer: XZG.
- Combination rule: CAT is written as UBD. Reverse CAT to get TAC, then shift +1: T to U, A to B, C to D.
- Apply to DOG: reverse to GOD, then +1 gives H, P, E. Answer: HPE. Test 'reverse then shift' whenever a plain shift fails but the letter-gaps look regular in reverse order.

ASKED IN RAS

Asked in RAS Pre 2023 - DOG is coded WLT, so CAT is XZG. This is the opposite-letter family and it is a repeat item. The moment you see a three-letter code where the letters sit in the far half of the alphabet from the word, add each pair and look for 27.

Alternating shifts and shift-by-position

These two families exist to punish the student who checked only one letter. Both begin with a +1 on the first letter, so both look like a simple forward shift until you reach letter two. Whenever your +1 rule fails on the second letter, do not abandon the question - test these two before anything else.

- CHAIR is written as DGBHS. How is TABLE written?
- C to D is +1. H to G is -1. A to B is +1. I to H is -1. R to S is +1. The shift alternates +1, -1.
- Apply to TABLE: T+1 = U, A-1 = Z (wrapping), B+1 = C, L-1 = K, E+1 = F.
- Answer: UZCKF. The wrap at A is exactly where the wrong option UZBKF is aimed.

- ABC is coded as BDF. How is PQR coded?
- A to B is +1, B to D is +2, C to F is +3. The shift equals the letter's place in the word.
- Apply to PQR: P+1 = Q, Q+2 = S, R+3 = U. Answer: QSU.
- For a longer word this continues: fourth letter +4, fifth +5. Number the letters 1, 2, 3, 4 above the word before you start.
- Distinguish from family 1: in a constant shift the gap is the same for every letter; here it grows by exactly one each time.

TRAP

Never fix a rule from the first letter alone. A +1 constant shift, an alternating +1/-1 shift and a shift-by-position all produce exactly the same first letter, and a full reversal agrees with a pair interchange on plenty of words. Compute letter two and letter three before you commit, and where two words are given, verify on the second word as well. This single habit is worth more marks in this chapter than any mnemonic.

Number coding – five forms, and how to tell them apart

When the code is digits, ask how many digits there are. A code with one group per letter is positional. A single small number for the whole word is a sum. A code where every value is even and double a familiar position is a multiple. A code where A is large and Z is small is reverse position. Test the form on the given word, then confirm it on a second given word if one exists – one word alone can fit several rules.

The five forms of number coding

Form	Rule	Example
Position written out	Each letter becomes its position	CAT = 3-1-20; MANGO = 13-1-14-7-15
Positions run together	Positions concatenated with no separator	CAT = 3120; DOG = 4157
Sum of positions	Add the positions of all letters	CAT = 3+1+20 = 24
Multiple of position	A = 2, B = 4, C = 6 ... (twice the position)	FACE = 12-2-6-10
Reverse position	Z = 1, Y = 2 ... A = 26, i.e. 27 minus n	CAB = 24-26-25

- CAB is coded 312 with A=1, B=2 ... Z=26. What is FED? Positions: F=6, E=5, D=4. Answer: 654.
- MANGO is coded 13-1-14-7-15, so APPLE is A=1, P=16, P=16, L=12, E=5, giving 1-16-16-12-5.
- CAT = 3120 and DOG = 4157 – here the positions are simply written one after another with no gaps.
- So BAT is B(2) A(1) T(20) = 2120. Watch the option 2121, which is the planted near-miss.

ASKED IN RAS

Asked in RAS Pre 2018 - with A = 1 to Z = 26, CAB is coded 312, so FED is coded 654. This is the simplest coding item RPSC sets, and it is free marks provided the alphabet table is genuinely memorised.

- SUN is written as 54. What is MOON? Sum test: S(19) + U(21) + N(14) = 54. Confirmed.
- MOON: M(13) + O(15) + O(15) + N(14) = 57. Answer: 57.
- CAT = 24, since 3 + 1 + 20 = 24. So DOG = 4 + 15 + 7 = 26.
- LOVE = 54, since 12 + 15 + 22 + 5 = 54. So LIFE = 12 + 9 + 6 + 5 = 32.
- If A = 2, B = 4, C = 6, each code is twice the position: FACE = 12-2-6-10.
- If the letters are numbered from the end, Z = 1 up to A = 26, then reverse position = 27 - n, so CAB = 24-26-25.

Mixed letter-digit coding - build the dictionary

Here you are given two or three words with their number codes and asked to code a fourth. There is no shift and no arithmetic. Each letter simply owns a digit. Your job is to build the letter-to-digit dictionary by lining up the words with their codes, and then to spell out the asked word from that dictionary. The letters shared between the given words are your proof that you have lined them up correctly.

Dictionary method

Write each given word above its code, letter above digit → Read off every letter-digit pair into a dictionary → Check any letter that appears in two words - the digits MUST agree → If they disagree, you have mis-aligned; realign before going further → Spell the asked word out of the dictionary, letter by letter

- PALE = 2134 and EARTH = 41590. Code PEARL.
- From PALE: P=2, A=1, L=3, E=4. From EARTH: E=4, A=1, R=5, T=9, H=0.
- Consistency check: A appears in both and is 1 in both; E appears in both and is 4 in both. The alignment is right.
- Spell PEARL: P=2, E=4, A=1, R=5, L=3. Answer: 24153.

- ROSE = 6821, CHAIR = 73456, PREACH = 961473. Code SEARCH.
- From ROSE: R=6, O=8, S=2, E=1. From CHAIR: C=7, H=3, A=4, I=5, R=6 - and R=6 agrees with ROSE, so the alignment holds.
- From PREACH: P=9, and R=6, E=1, A=4, C=7, H=3 all agree with what we already have.
- Spell SEARCH: S=2, E=1, A=4, R=6, C=7, H=3. Answer: 214673.
- Do the agreement check BEFORE you spell the answer - it costs ten seconds and it is the only thing standing between you and a wrong answer.

TRAP

If a shared letter gets two different digits, do not average, guess or ignore it. You have mis-aligned the word against its code - most often by starting from the wrong end, or by mis-counting a repeated letter such as the double O in MOON. Rewrite the alignment from scratch.

Substitution coding and coded sentences

Shift-based coding vs substitution coding - do not mix the methods

Shift / letter coding	Substitution coding
The code is a meaningless letter string (NBESBT)	The code words are ordinary English words (sea, water, blue)
You look for an arithmetic rule on letter positions	There is no arithmetic at all - it is a renaming list
Method: find the shift, verify it, apply it	Method: answer in plain language first, then translate the answer
Verification is on a second letter pair	Verification is on a second sentence sharing one word
Wrong answer comes from the alphabet wrap	Wrong answer comes from giving the REAL word instead of its new name

Substitution questions read 'if X is called Y, and Y is called Z, then ...'. The trick is to answer the question honestly in ordinary English first, and only then look up what that answer is called in the new language. Do it in that order and the question becomes trivial. Do it in the other order and you will pick the real-world word, which is always sitting there as an option.

- If 'sky' is called 'sea', 'sea' is called 'water', 'water' is called 'air', 'air' is called 'cloud' - where do fish live?
- Step 1, plain English: fish live in water.
- Step 2, translate: in this language 'water' is called 'air'. Answer: air.
- Same drill: if 'red' is called 'yellow', blood is really red, so the answer is yellow.
- If 'pen' is called 'book', we write with a pen, so the answer is book.
- If 'blue' means 'dance', a person dances to music, so the answer is blue.

ASKED IN RAS

Asked in RAS Pre 2013 - the sky/sea/water/air chain, asking where fish live. Answer: air. Note that 'water' appears as an option and is the answer to the plain-English step, not to the question. Never stop after step 1.

Coded sentences - the cross-out method

Line up the sentences and their codes → Find two sentences sharing exactly ONE English word and exactly ONE code word - that pair is fixed → Cross out each code word as soon as it is fixed → Repeat for the other two pairs of sentences → The single unmatched code word in a sentence must mean that sentence's single unmatched English word

- 'tim nap sok' = 'she is intelligent'; 'nap dor lin' = 'is very smart'; 'sok lin bap' = 'intelligent smart boy'. Which word means 'boy'?
- Sentences 1 and 2 share only the word 'is' and only the code 'nap', so nap = is.
- Sentences 1 and 3 share only 'intelligent' and only 'sok', so sok = intelligent.
- Sentences 2 and 3 share only 'smart' and only 'lin', so lin = smart.
- In sentence 3 the only code left is 'bap' and the only English word left is 'boy'. Answer: bap.
- Same structure: 'ma po ri' / 'po lin sa' / 'ri sa dun' gives po = eat, ri = fish, sa = bones, so lin = dogs.

ASKED IN RAS

Asked in RAS Pre 2024 - 'gel hai jar' means 'apples are ripe' and 'hai tir sup' means 'children are playing'; which word means 'are'? Answer: hai. Only one English word is common and only one code word is common, so the pair is forced. Two sentences sharing exactly one word is the fastest item in the whole paper.

Conditional coding, matrix coding and coding by analogy

Conditional coding gives you a small letter-to-digit table and then two or three conditions that override it. The method is mechanical: write the plain digit code first, ignoring all conditions. Then apply each condition in the order printed. Then look again at the first and the last letter of the group, because almost every condition is about those two and whether they are vowels. Underline the first and last letter before you begin.

- Table: P=4, M=7, A=2, K=9, R=5, E=3, T=8.
- Condition (i): if the first letter is a vowel, both the first and the last letter become $.
- Condition (ii): if the last letter is a vowel and the first is not, the last letter becomes #.
- Code AMKRT: plain code is 2-7-9-5-8. A is a vowel and it is first, so condition (i) fires - first and last both become $. Answer: 795.
- Code PMAKE: plain code is 4-7-2-9-3. P is not a vowel so (i) does not fire; E is a vowel and last, so (ii) fires. Answer: 4729#.
- Code KRAMP: plain code is 9-5-2-7-4. Neither K nor P is a vowel, so no condition applies. Answer: 95274.
- Notice the third case - 'no condition applies' is a real answer and RPSC uses it to catch students who assume a condition must fire.

- Table: B=6, J=1, N=4, S=9, U=3, D=7, I=2.
- Condition (i): if BOTH the first and the last letter are consonants, the first letter becomes *.
- Condition (ii): if the first is a vowel and the last a consonant, their two codes are interchanged.
- Code UDNSB: plain code is 3-7-4-9-6. U is a vowel, B is a consonant, so (ii) fires - swap the first and last codes. Answer: 67493.
- Code SNBDJ: plain code is 9-4-6-7-1. S and J are both consonants, so (i) fires and the first code becomes *. Answer: *4671.
- Vowels for these questions are A, E, I, O, U only. Y is treated as a consonant unless the question says otherwise.
- If two conditions could both apply, the paper normally states the precedence; if not, apply condition (i) first.

Matrix coding gives you a grid of letters and codes each letter by its row number followed by its column number. Two things to check before you start: whether the rows or the columns are numbered first, and which letter has been left out of the grid - it is usually Z, and knowing it is missing stops you mis-counting the later letters.

- Grid with rows R0 to R4 and columns C0 to C4, filled A to Y (Z not used): row 0 is A B C D E, row 1 is F G H I J, row 2 is K L M N O, row 3 is P Q R S T, row 4 is U V W X Y.
- Code CAT: C is row 0, column 2 = 02; A is row 0, column 0 = 00; T is row 3, column 4 = 34. Answer: 02-00-34.
- Code MOON: M = 22, O = 24, O = 24, N = 23. Answer: 22-24-24-23.
- Decode 10-13-32-04: row 1 column 0 = F, row 1 column 3 = I, row 3 column 2 = R, row 0 column 4 = E. The word is FIRE.
- The commonest error is reading 20 as row 0 column 2 - always read ROW first unless the question says otherwise.

- Coding by analogy - BOOK : CPPL :: LOCK : ? Verify on every letter: B to C, O to P, O to P, K to L, all +1.
- Apply +1 to LOCK: M, P, D, L. Answer: MPDL.
- GATE : HBUF :: DOOR : ? The rule is +1 on each letter, so DOOR becomes EPPS.
- MOTHER : PRWKHU :: FATHER : ? The rule is +3, so FATHER becomes IDWKHU.
- LIGHT : JGEFR :: SOUND : ? The rule is -2, so SOUND becomes QMSLB.
- CAT : XZG :: DOG : ? The rule is the opposite letter, so DOG becomes WLT.
- The method is identical to ordinary letter coding - the analogy format changes nothing except the layout.

YAAD RAKHO

Yaad rakho the fastest elimination in this chapter: once you have the rule, do not code the whole word. Code only the FIRST and the LAST letter of the answer and compare against the options. Three options usually die at once, and where two survive, one more letter settles it. This turns a ninety-second item into a twenty-second one.

REVISION RECAP

1. First question always: is the code the same length as the word? That single check picks the family.
2. Same letters in a new order means structural - reversal, pair interchange or first-last swap. No shift to find.
3. Constant shift method: find the shift on one pair, VERIFY on a second and third, then apply.
4. MADRAS to NBESBT is +1, so BOMBAY is CPNCBZ. FRIEND to HTKGPF is +2, so CANDLE is ECPFNG.
5. Backward shifts are as common as forward ones: TRIPPLE to SQHOOKD is -1; WATER to TXQBO is -3.
6. Wrap the alphabet: A - 3 = X, Z + 2 = B. This is where most marks are dropped.
7. Alternating shift: CHAIR to DGBHS is +1, -1, +1, -1, so TABLE is UZCKF.
8. Shift by place in the word: ABC to BDF, so PQR is QSU.

9. Opposite letter = 27 minus position: DOG to WLT, CAT to XZG.
10. Reverse then shift: CAT to UBD means reverse to TAC then +1; so DOG gives HPE.
11. DELHI to IHLED is full reversal; MONKEY to OMKNYE is pair interchange; SYSTEM to MYSTES is first-last swap.
12. Number coding forms: positions written out, positions concatenated, sum of positions, twice the position, reverse position.
13. Sum coding must be confirmed on a second word - one word alone fits several rules. SUN = 54 gives MOON = 57.
14. Mixed coding: build a letter-to-digit dictionary; shared letters must agree, and disagreement means mis-alignment.
15. PALE = 2134 with EARTH = 41590 gives PEARL = 24153.
16. Substitution: answer in plain English first, then translate. Fish live in water; water is called air, so answer air.
17. Coded sentences: two sentences sharing exactly one English word and one code word fix that pair. Cross out as you go.
18. Conditional coding: plain code first, conditions in the printed order, then re-check first and last letters - and 'no condition applies' is a genuine answer, KRAMP stays 95274.
19. Matrix coding: row number first, then column; note that Z is usually left out of the grid.
20. Elimination shortcut: code only the first and last letter of the answer and three options die at once.

RM05

Blood Relations

2-3 questions in every RAS Prelims paper, without a single exception. The photograph type and the coded-symbol type are the two that repeat most. At full marks this is the cheapest block in the paper - a trained student clears all three in about 90 seconds.

A blood relation question is a family tree hidden inside two or three sentences. Draw the tree and the answer is obvious; try to hold it in your head and you will get it wrong. This chapter teaches you the drawing, not the facts.

Blood Relations

Direct chain
- A is father of B, B is mother of C
- Read one link at a time
- Answer = relation of first to last

Photograph / pointing
- Pointing to a man, X said...
- Start from the SPEAKER
- 'only son', 'only daughter' clues
- Gender from he/she in the stem

Coded relations
- + - x / stand for relations
- Decode pair by pair
- First letter holds the relation
- 'Cannot be determined' when gender is open

Family-tree puzzles
- 5-7 members, A to F
- Married couple clue
- Daughter-in-law / widow clue
- Count males and females

Generation counting
- Parent step = +1 generation
- Sibling / spouse step = same level
- 2 steps = grandparent, 3 = great-grand

Maternal vs paternal
- Mama = mother's brother
- Chacha = father's brother
- Bua = father's sister
- Mausi = mother's sister
- Nana vs Dada

What the question looks like, and where the marks leak

A blood relation question gives you a small chain of family facts and asks how the first person is related to the last. Sometimes it is a plain chain - 'A is the father of B, B is the mother of C'. Sometimes it is a person pointing at a photograph. Sometimes the relations are hidden behind symbols like + and x. The underlying job is always the same: build a family tree.

- Direct chain puzzles - two to four sentences of relations.
- Pointing / photograph type - one long sentence spoken by a person.
- Coded relations - '+', '-', 'x', '/' each stand for a relation.
- Family-tree puzzles with five to seven members named A to G.
- Generation counting - how many generations does this family span.
- Maternal or paternal identification - uncle, aunt, grandfather.

Marks leak in exactly three places. One: the student reasons verbally instead of drawing, and loses the thread at the third link. Two: the student ignores gender and calls a maternal uncle a paternal uncle. Three: the student reverses the direction of a relation - 'X is the son of Y' gets drawn as Y below X. Every one of those three is a drawing discipline problem, not an intelligence problem.

TRAP

Never answer a blood relation question without putting pen to paper. The examiner writes the sentences in a deliberately tangled order precisely so that verbal reasoning fails. Ten seconds of drawing beats forty seconds of thinking.

ASKED IN RAS

Blood relations has appeared in every RAS Prelims paper. The photograph form came in 2023 ('His mother's only daughter is my mother' - answer: maternal uncle) and in 2016 ('the son of the only son of my grandmother' - answer: brother). The coded-symbol form came in 2021, the in-law form in 2024 and the plain chain in 2018 and 2013. Prepare all four; none of them skips a year for long.

The notation - draw the tree the same way every single time

Use one fixed set of marks. If you invent notation afresh in the exam hall you will misread your own diagram. Learn these five marks now and never change them.

- + written next to a name means that person is MALE.
- - written next to a name means that person is FEMALE.
- = or a double horizontal line between two names means they are a married couple.
- A vertical line downward means parent to child.
- A single horizontal line means siblings, standing on the same level.

The rule that keeps the diagram honest is the level rule. The eldest generation sits on the top line. Their children sit on the next line down. Grandchildren on the line below that. Two people on the same line are always of the same generation - siblings, cousins, or a husband and wife. Never let a parent and a child share a line.

The method – follow these steps with pen on paper

Read the whole stem once without writing, just to see how many people there are. →

Draw a dash for each named person; leave three lines of vertical space. →

Take the sentences one at a time, left to right – never jump to the last name. →

For 'X is the son/daughter of Y', put Y one line ABOVE X and join with a vertical. →

For 'X is the brother/sister of Y', put X on the SAME line as Y, joined horizontally. →

For 'X is married to Y', put them side by side with = between them. →

Mark + or – on every person whose gender the stem actually states. →

Re-read each sentence and tick it against the tree; if all tick, the tree is right. →

Now read the answer straight off the picture.

Two habits that fix half the errors

- Solve the 'only son / only daughter / only child' clue first – it forces a unique person.
- A married couple in an exam puzzle are always the parents of the same children.
- The spouse of a parent is therefore also a parent – use this without hesitation.
- Label the NODE, not the sentence: one person can be a son and a father at once.
- Gender that is never stated stays blank. Do not silently assume a man.

YAAD RAKHO

Yaad rakho - PLUS is PURUSH (male), MINUS is MAHILA (female). Two letters, two symbols, and you will never mix them up under time pressure.

The relationship dictionary - and the maternal-paternal split

RPSC does not ask 'who is the uncle'. It asks 'who is the maternal uncle'. The option list will always contain both the maternal and the paternal version, so knowing which side of the family you are on is worth the whole mark. Maternal means through the mother. Paternal means through the father. That is the entire rule; the vocabulary below just applies it.

Blood relatives - English and the common Hindi term

English	Hindi	Exactly who
Maternal uncle	Mama	Mother's brother
Maternal aunt (by marriage)	Mami	Wife of mother's brother
Maternal aunt (by blood)	Mausi	Mother's sister
Paternal uncle	Chacha	Father's brother
Paternal aunt (by marriage)	Chachi	Wife of father's brother
Paternal aunt (by blood)	Bua	Father's sister
Paternal grandfather	Dada	Father's father
Maternal grandfather	Nana	Mother's father
Cousin	-	Child of any uncle or aunt
Nephew / niece	Bhatija / Bhatiji	Son / daughter of a brother or sister

Which side am I on - the only distinction RPSC tests

MATERNAL side (through the mother)	PATERNAL side (through the father)
Mother's brother = maternal uncle (mama)	Father's brother = paternal uncle (chacha)
Mother's sister = maternal aunt (mausi)	Father's sister = paternal aunt (bua)
Wife of mother's brother = mami	Wife of father's brother = chachi
Mother's father = maternal grandfather (nana)	Father's father = paternal grandfather (dada)
Mother's mother = maternal grandmother (nani)	Father's mother = paternal grandmother (dadi)
Reached by going UP through a female parent	Reached by going UP through a male parent

In-laws - three rules cover every case

Relation to you	Who they are
Father-in-law / mother-in-law	The parent of your spouse
Son-in-law / daughter-in-law	The spouse of your child
Brother-in-law	Your spouse's brother OR your sister's husband
Sister-in-law	Your spouse's sister OR your brother's wife

TRAP

Do not confuse a cousin with a nephew. The child of your uncle or aunt is your COUSIN - same generation, same line on the tree. The child of your brother or sister is your NEPHEW or NIECE - one line below you. Draw the level and the confusion disappears.

ASKED IN RAS

Asked in RAS Pre 2024 - A and B are a married couple, C and D are their son and daughter, E is married to D. How is E related to B? Answer: son-in-law. The in-law form is now a standing favourite; learn the three in-law rules above by heart.

Worked examples – the pointing and photograph type

In this type one person speaks a single sentence about someone in a photograph. The whole difficulty is that the sentence has to be unwound from the inside out. There is a fixed method for it, and once you follow the method these become the fastest questions in the section.

Photograph type – unwinding the sentence

Mark the SPEAKER on your sheet first. Call the speaker ME. →

Note the gender of the person in the photo from 'he / she / his / her' in the stem. →

Start at the innermost phrase of the sentence and simplify it to a single person. →

Replace that phrase with the person you found, and simplify the next layer outward. →

Keep going until only one relation links ME and the photograph. →

State the relation in the direction the question asks – 'how is HE related to HER'.

Worked example 1 (asked in RAS Pre 2023). Pointing to a man in a photograph, Reena said, 'His mother's only daughter is my mother.' How is the man related to Reena?

- Mark Reena as ME on the sheet. The photo person is a MAN, so mark him +.
- Innermost phrase: 'his mother's only daughter'. The man's mother has one daughter.
- The man is her son, so that only daughter is the man's own SISTER.
- The sentence now reads: the man's sister is my mother.

- So Reena's mother and the man are brother and sister - same line on the tree.
- The man is therefore the brother of Reena's mother.
- Brother of the mother = MAMA = maternal uncle. Answer: maternal uncle.

Worked example 2. Pointing to a boy, a woman said, 'He is the son of the only brother of my mother's daughter.' The woman has only one brother. How is the woman related to the boy?

- Mark the woman as ME, female, and the boy as +.
- Innermost phrase: 'my mother's daughter'. That is the speaker herself, the woman.
- The sentence becomes: he is the son of the only brother of ME.
- The woman has only one brother; that brother is the boy's father.
- So the boy's father and the woman are brother and sister on the same line.
- The question asks how the WOMAN is related to the BOY - reverse the direction.
- Sister of the boy's father = BUA = paternal aunt. Answer: paternal aunt.

TRAP

The words 'other than myself' and 'only' are not filler. 'My mother's only son' is the speaker himself if the speaker is male; add 'other than myself' and it becomes his brother. Read those four words twice - they flip the answer completely.

Coded relations - when symbols replace the words

Here the paper first defines a code - 'A + B means A is the father of B', and so on - then gives you an expression like 'K / L x M' and asks how K is related to M. Students lose these by trying to read the expression as one sentence. Do not. Break it into overlapping pairs and decode each pair separately onto the same tree.

- For 'P + Q - R', decode P+Q first, then Q-R. The pairs overlap at Q.
- In every pair, the FIRST letter holds the stated relation to the second.
- So in 'P + Q' with + meaning father, P is the father, not Q.
- Draw each decoded pair onto a single growing tree, not three separate scraps.
- If the question asks 'which expression means X is the ... of Y', build a small tree for EACH option separately. It is faster than arguing verbally.
- If the answer depends on the gender of a middle letter that was never stated, the correct choice is 'cannot be determined'.

Worked example 3 (asked in RAS Pre 2021). Code: 'A + B' = A is the father of B; 'A - B' = A is the sister of B; 'A x B' = A is the mother of B; 'A / B' = A is the brother of B. How is K related to M in 'K / L x M'?

- Split into pairs: K / L, then L x M.
- K / L uses '/', so K is the BROTHER of L. Put K and L on the same line; mark K as +.
- L x M uses 'x', so L is the MOTHER of M. Put M one line below L, vertical line.
- Now read the tree. K is on L's line; M is L's child.
- K is therefore the brother of M's mother.
- Brother of the mother = maternal uncle. Answer: maternal uncle.

Worked example 4. Same style of code: 'A + B' = A is the father of B; 'A - B' = A is the sister of B; 'A x B' = A is the mother of B; 'A / B' = A is the brother of B. Which expression means 'P is the paternal grandfather of Q'?

- Paternal grandfather means: father of the FATHER. Two male parent steps upward.

- So you need P to be the father of some middle person R, and R to be the father of Q.
- 'P + R' gives P is the father of R. Correct for the first step.
- 'R + Q' gives R is the father of Q. Correct for the second step.
- Joining them: P + R + Q. Answer: P + R + Q.
- Check the wrong options - 'P x R + Q' makes P the MOTHER of R, so P would be the paternal grandmother, not grandfather.

> **TRAP**
> The commonest error in coded relations is reversing the pair. 'A x B' where x means mother makes A the mother of B - A is above B on the tree. Students who write B above A get a consistent-looking but exactly inverted answer, and the paper always puts that inversion in the options.

Family-tree puzzles and generation counting

The puzzle form names five to seven members - A, B, C, D, E, F - and scatters the clues out of order. The trick is clue selection. Start with the clue that pins a person down completely: a married couple, an 'only son', a 'daughter-in-law of', or a stated gender. Weak clues like 'C is the brother of A' go in later, once there is a frame to hang them on.

Worked example 5. In a family of six members A, B, C, D, E and F: A and B are a married couple, A being the male member. D is the only son of C, who is the brother of A. E is the sister of D. B is the daughter-in-law of F, whose husband has died. How is E related to A, and how is F related to D?

- Strongest clue first: A and B are a couple, A is male. Write A+ = B- on one line.

- Next: B is the daughter-in-law of F. So F is a parent of B's husband, that is a parent of A.
- F's husband has died, so F is the surviving parent - F is A's MOTHER. Put F- one line above A.
- C is the brother of A, so C sits on A's line, joined horizontally, and F is C's mother too.
- D is the only son of C - put D+ one line below C. E is D's sister, so E- goes beside D, also C's child.
- Now read off. E is the daughter of C, and C is A's brother. So E is A's NIECE.
- F is the mother of C, and D is C's son. Two parent steps above D, so F is D's GRANDMOTHER.

Counting generations

- Every 'father of / mother of / son of / daughter of' step changes the generation by one.
- Every 'brother of / sister of / husband of / wife of' step keeps the same generation.
- Two parent-steps up = grandparent. Three parent-steps up = great-grandparent.
- To count how many generations a family spans, count the distinct LINES on your tree.
- Count the steps on paper. Never guess from the name of the relation.

Worked example 6. A is the mother of B. B is the brother of C. C is the mother of D. D is the brother of E. How many generations are there in this family?

- A is the mother of B - that is one parent step. A on line 1, B on line 2.
- B is the brother of C - sibling step, no change. C also sits on line 2.

- C is the mother of D - parent step. D goes on line 3.
- D is the brother of E - sibling step. E also sits on line 3.
- Distinct lines used: line 1 (A), line 2 (B and C), line 3 (D and E).
- Answer: three generations. Note that the chain has four sentences but only two parent steps.

The 60-second checklist

- Draw before you think. Dashes for people, levels for generations.
- Fix the 'only child' clue and the married-couple clue first.
- Mark + and - only where the stem actually gives gender.
- Read chains and coded expressions pair by pair, left to right.
- For photograph questions, mark the speaker as ME and unwind from the inside.
- Before choosing, check maternal versus paternal - the wrong side is always an option.
- Check the direction the question asks: 'how is X related to Y', not Y to X.
- For a 'how many male members' question, count each named person once; a married-in spouse is a family member but not a blood relation.
- Tick every sentence of the stem against the finished tree.

ASKED IN RAS

Asked in RAS Pre 2018 - P is the brother of Q, Q is the son of R, R is the wife of S; how is P related to S? Answer: son. The chain type looks trivial and is still worth full marks in twenty seconds, so never skip it looking for something harder.

YAAD RAKHO

Yaad rakho - MAMA has an M like MOTHER, so mama is the mother's brother. CHACHA does not, so chacha belongs to the father's side. One letter, and the maternal-paternal trap is closed.

REVISION RECAP

1. Every blood relation question is a family tree; draw it, never reason verbally.
2. Notation: + male, - female, = married couple, vertical line parent-to-child, horizontal line siblings.
3. Eldest generation on the top line; never put a parent and a child on the same line.
4. Read chains one link at a time, left to right; never jump from the first name to the last.
5. Sibling, husband and wife steps keep the generation; parent and child steps change it by one.
6. Solve 'only son', 'only daughter' and 'only child' clues first - they force a unique person.
7. A married couple in an exam puzzle are the parents of the same children.
8. Mama = mother's brother; chacha = father's brother; bua = father's sister; mausi = mother's sister.
9. Mami is the wife of the mama; chachi is the wife of the chacha.
10. Nana is the maternal grandfather; dada is the paternal grandfather.
11. Child of an uncle or aunt = cousin; child of a brother or sister = nephew or niece.
12. In-laws: parent of spouse, spouse of child, and sibling of spouse or spouse of sibling.
13. Photograph type: mark the speaker as ME, take the gender from he/she in the stem, unwind from the inside.
14. Coded relations: decode overlapping pairs; the first letter of each pair holds the relation.
15. For 'which expression means...', build a separate small tree for every option.
16. If the answer turns on a gender the stem never states, choose 'cannot be determined'.
17. Two parent-steps up is a grandparent; three is a great-grandparent - count, do not guess.
18. Finish by re-reading every sentence against the tree; if all of them tick, the tree is right.

RM06

Direction Sense

1-2 questions in every RAS Prelims paper without fail, and in some years three. The Pythagoras form (walk East then North, find the shortest distance) is the single most repeated sub-type. Shadow and sun questions come once every two or three papers and are pure rule-recall.

A person walks, turns, walks again, and you are asked how far he ended up from where he started and in which direction. There is no trick to it once you stop drawing arrows freehand and start writing coordinates. This chapter converts every walk into a pair of numbers.

Direction Sense

The compass
- N E S W - four cardinal
- NE SE SW NW - four ordinal
- 45 degrees = one step
- Clockwise ring: N NE E SE S SW W NW

Turning
- Right = one step clockwise
- Left = one step anti-clockwise
- Turn around and left/right swap
- 90 = two steps, 180 = four steps

Coordinates
- North +y, South -y
- East +x, West -x
- Add the signed legs
- Net displacement, not path length

Shortest distance
- sqrt(x squared + y squared)
- 3-4-5, 5-12-13, 8-15-17
- Equal x and y gives x times root 2
- Direction from the signs of x and y

Shadow / sun
- Morning: shadow falls WEST
- Evening: shadow falls EAST
- Noon: no usable shadow
- Two persons face-to-face are opposite

Mixed types
- Direction + blood relation
- Relative position of villages
- Return-journey total distance
- Displacement that cancels to zero

What the question asks, and where students throw the mark away

Direction sense gives you a walk. The person starts somewhere, covers a leg, turns, covers another leg, and so on. Then one of two things is asked: how far is he from the start, or in which direction is he from the start. Sometimes both. The walk is deliberately written so that a mental picture goes wrong at the third turn.

- Distance and direction from the starting point after a multi-leg walk.
- Shortest distance using Pythagoras - the single most repeated sub-type.
- Turning problems - left and right turns, and turns given in degrees.
- Shadow and sun problems - sunrise, sunset, which way is the person facing.
- Displacement that cancels, so the person ends up back at the start.
- Direction combined with a blood relation in the same stem.

Three leaks. One: the student adds up all the legs walked and calls that the distance from the start - that is the path length, not the displacement. Two: the student forgets that after turning around, left and right have swapped. Three: the student gets the right number and then names the opposite direction, because he never wrote North at the top of the page. All three vanish if you use coordinates.

ASKED IN RAS

Direction sense has appeared in every RAS Prelims paper. The turning-walk form came in 2024 (10 km North, right 6 km, right 10 km - answer 6 km), the plain Pythagoras form in 2023 (4 km North then 3 km East - answer 5 km) and 2016 (9 km West then 12 km North - answer 15 km), the cancelling-displacement form in 2021, and the sunrise-shadow form in 2013. Expect at least one, and expect it to be an easy one.

The compass, and the coordinate method that replaces guessing

Before anything else, draw a small compass in the corner of your rough sheet: North at the top, South at the bottom, East to the right, West to the left.

Draw it once at the start of the paper and use it for every direction question. This is not decoration - the whole subject depends on the page being oriented the same way each time.

- Four cardinal directions: North, East, South, West.
- Four ordinal directions sit between them: North-East, South-East, South-West, North-West.
- The eight-point ring clockwise: N, NE, E, SE, S, SW, W, NW, back to N.
- One step around that ring is 45 degrees.
- 90 degrees = two steps; 135 degrees = three steps; 180 degrees = four steps.
- Clockwise cardinal order is N to E to S to W; anti-clockwise is the reverse.

Now the method itself. Stop drawing the walk as a wandering line. Convert it into coordinates. Put the starting point at (0, 0). North adds to y, South subtracts from y, East adds to x, West subtracts from x. Add up the signed legs and you have the final position in one line of arithmetic, with no chance of a sign error.

The coordinate method - every walk, every time

Draw the compass; put the start at (0, 0). →

Track the FACING separately from the position - write the current facing beside each leg. →

Apply each turn to the facing: right = one step clockwise, left = one step anti-clockwise. →

Convert each leg into a signed change: N is +y, S is -y, E is +x, W is -x. →

Add all the x changes; add all the y changes. That pair is the final position. →

Distance from start = sqrt(x squared + y squared). →

Direction from start = read the two signs (+x +y is North-East, and so on). →

Sketch the path roughly to scale as a check - a rough picture catches sign errors.

YAAD RAKHO

Yaad rakho - NEWS reads clockwise if you start at North and skip: N, E, S, W. A RIGHT turn walks you FORWARD along NEWS; a LEFT turn walks you BACKWARD along it. That one line answers every left-right turn in the paper.

Turning - and the trap that decides more marks than any other

A right turn moves your facing one step clockwise. A left turn moves it one step anti-clockwise. That is all there is to it, but you must apply it to the CURRENT facing, not to North. Write the facing down after every turn; do not carry it in your head.

Left and right for each facing – learn this as a block

Facing	Right turn gives	Left turn gives
North	East	West
East	South	North
South	West	East
West	North	South

TRAP

The single most tested trap in the chapter: when the person turns around and faces the OPPOSITE direction, his left and his right interchange. Facing North his right is East; turn him to face South and his right is now West. A student who solved the first half of the walk correctly and then forgets this gets the mirror-image answer, which the paper always supplies as an option.

The same person, two facings – notice everything flips

Facing NORTH	Facing SOUTH
Right hand points East	Right hand points West
Left hand points West	Left hand points East
Behind him is South	Behind him is North
A right turn takes him East	A right turn takes him West
Morning shadow falls to his LEFT (West)	Morning shadow falls to his RIGHT (West)
A 45-degree clockwise turn takes him to North-East	A 45-degree clockwise turn takes him to South-West

- Turns given in degrees use the eight-point ring: 45 degrees is one step.
- Facing North-East, 90 degrees clockwise = two steps = South-East.
- Facing South-West, 135 degrees clockwise = three steps = North.
- Two persons standing face to face always face opposite directions.

- The direction of A from B is the exact reverse of the direction of B from A - flip both letters, so North-East becomes South-West.

Worked examples - the turning walk

Worked example 1 (asked in RAS Pre 2024). Ramesh walks 10 km towards the North, then turns right and walks 6 km, then turns right again and walks 10 km. How far is he now from his starting point, and in which direction?

- Start at (0, 0), facing North after the first instruction.
- Leg 1: 10 km North. y goes up by 10. Position (0, 10).
- Turn right from North - facing is now East.
- Leg 2: 6 km East. x goes up by 6. Position (6, 10).
- Turn right from East - facing is now South.
- Leg 3: 10 km South. y goes down by 10. Position (6, 0).
- Final position (6, 0). The y parts cancelled completely.
- Distance = 6 km. Direction: x is positive, y is zero, so he is due EAST of the start.

Worked example 2. Priya walks 10 km South, turns left and walks 5 km, turns left again and walks 10 km, then turns right and walks 4 km. How far is she from her starting point?

- Start (0, 0), facing South.
- Leg 1: 10 km South. Position (0, -10).
- Turn left from South - one step anti-clockwise gives East. Careful here; facing South reverses the usual sense.
- Leg 2: 5 km East. Position (5, -10).
- Turn left from East - anti-clockwise gives North.
- Leg 3: 10 km North. Position (5, 0).

- Turn right from North - clockwise gives East. Leg 4: 4 km East. Position (9, 0).
- Distance = 9 km, due East. Note the total walked was 29 km - that number is not the answer.

Worked example 3. A man walks 25 km East, then 10 km North, then 15 km West. How far is he from the starting point and in which direction?

- x changes: +25 then -15, giving x = +10.
- y changes: +10 only, giving y = +10.
- Final position (10, 10).
- x and y are equal, so the distance is 10 times root 2 = about 14.14 km.
- Both signs are positive, so the direction is exactly North-East.
- Whenever x equals y in size, the answer is one of NE, NW, SE, SW and the distance carries a root 2.

YAAD RAKHO

Yaad rakho - if the North legs and the South legs add to the same number, and the East legs and the West legs also add to the same number, the person is standing exactly where he started. Distance zero. RPSC sets this as a whole question and the answer 'at the starting point' is always in the options.

Shortest distance - Pythagoras and the triplets you must know cold

When the net movement has both an East-West part and a North-South part, the straight line back to the start is the hypotenuse of a right triangle. So the shortest distance is the square root of (x squared plus y squared). The examiner will nearly always pick numbers that make this a whole number, which means he is using a standard Pythagorean triplet. Memorise the triplets and you will never actually compute a square root.

- 3-4-5 and its multiples 6-8-10, 9-12-15, 12-16-20, 15-20-25.
- 5-12-13 - the second most used triplet in RAS papers.
- 8-15-17, 7-24-25, 20-21-29 - the three that catch students out.
- If x and y are equal, distance = x times root 2, about 1.414 times x.
- Signs give the direction: +x +y is NE, -x +y is NW, +x -y is SE, -x -y is SW.
- 'Minimum distance to return' means the straight line back, not the path retraced.

Shortest distance - four steps

Add the signed legs to get the net x and the net y. → Drop the signs and look for a known triplet among the two numbers. →

If you find one, write down the third member - that is the distance. → Put the signs back on and read the direction off the sign pair.

Worked example 4 (asked in RAS Pre 2016). A jeep travels 9 km towards the West and then 12 km towards the North. What is the shortest distance from the starting point?

- West 9 km gives x = -9. North 12 km gives y = +12.
- The two legs are perpendicular, so Pythagoras applies directly.
- Ignore signs: the sides are 9 and 12. That is 3 times (3, 4), so the triplet is 9-12-15.
- Shortest distance = 15 km. No square root needed.

- If the direction had been asked: x negative, y positive, so North-West of the start.

Worked example 5. A bus goes 30 km East and then 40 km North. If it now returns to the starting point by the shortest route, what total distance has it covered in the whole journey?

- Read the question carefully - it asks for the WHOLE journey, not the displacement.
- Outward legs: 30 + 40 = 70 km.
- Return leg is the hypotenuse of a 30-40 right triangle. That is 10 times (3, 4, 5), so 50 km.
- Whole journey = 70 + 50 = 120 km.
- The distractor 50 km is the shortest distance alone; the distractor 70 km is the outward path. Both will be in the options.

The shadow rule - sunrise, sunset, and which way you are facing

This sub-type rests on one physical fact you already know. The Sun rises in the East, so in the MORNING every shadow is thrown towards the WEST. The Sun sets in the West, so in the EVENING every shadow is thrown towards the EAST. Once you fix which way the shadow points, the question tells you where the shadow falls relative to the person, and you work backwards to his facing.

Morning (shadow points WEST) - reading the person's facing

Shadow appears	Person is facing
In front of him	West
Behind him	East
To his right	South
To his left	North

Worked example 6 (asked in RAS Pre 2013). One morning, just after sunrise, Suresh was standing facing a pole. The shadow of the pole fell exactly to his right. Which direction was Suresh facing?

- It is just after sunrise, so the Sun is in the East.
- Every shadow therefore falls towards the West. The pole's shadow points West.
- The stem says that shadow is to Suresh's RIGHT. So West is on his right.
- Rule to use: your facing is 90 degrees ANTI-CLOCKWISE from wherever your right hand points.
- 90 degrees anti-clockwise from West is South. Check it: facing South, right is West. Correct.
- Answer: Suresh was facing South.

TRAP

In the evening every one of the four answers in that table is reversed, because the shadow now points East instead of West. Morning shadow on the right means facing South; evening shadow on the right means facing North. Read the words 'sunrise', 'morning', 'evening' or 'sunset' before you touch the geometry. And at noon there is no usable shadow at all - such a question either says 'no shadow' or is defective.

Direction combined with blood relations, and the checklist

RPSC sometimes bolts a family relation onto a direction stem: the houses of three people are placed by direction, and the same three people are also related. Do not try to solve both at once. Draw the coordinate diagram, answer the distance part, then draw the family tree separately and answer the relation part. They never interact.

Worked example 7. P's house is 8 km West of Q's house, and R's house is 6 km North of Q's house. P is the brother of Q, and Q is the mother of R. What is the shortest distance between P's house and R's house, and how is P related to R?

- Put Q at the origin (0, 0) because both other houses are described relative to Q.
- P is 8 km West of Q, so P is at (-8, 0).
- R is 6 km North of Q, so R is at (0, 6).
- Difference between P and R: 8 in x and 6 in y. That is 2 times (3, 4), so the triplet is 6-8-10.
- Shortest distance = 10 km.
- Now the separate family tree: Q is R's mother, and P is Q's brother.
- P is the brother of R's mother, so P is R's maternal uncle. Answer: 10 km; maternal uncle.

- Draw the compass first, every time. North at the top.
- Write the facing beside every leg; update it at each turn before moving.
- Convert to coordinates rather than following the line with your eye.
- Answer displacement, not the sum of the legs, unless the words 'total distance covered' appear.
- Look for a triplet before reaching for a square root.
- Read the sign pair to name the direction; never eyeball it.
- Reverse both letters when the question asks for the opposite direction.
- For shadow questions, first fix morning or evening, then place the shadow.

ASKED IN RAS

Asked in RAS Pre 2021 - from a point, a man walks 5 km South, 5 km East, 5 km North and 5 km East; what is his distance and direction from the start? Answer: 10 km East. The South and North legs cancel and the two East legs add. This cancelling form has now appeared in several papers; recognise it in the first reading and it takes fifteen seconds.

REVISION RECAP

1. Draw the compass once at the top of the rough sheet: North up, South down, East right, West left.
2. Eight-point ring clockwise: N, NE, E, SE, S, SW, W, NW. One step = 45 degrees.
3. Right turn = one step clockwise; left turn = one step anti-clockwise, applied to the CURRENT facing.
4. Facing North: right is East. Facing South: right is West. Turning around swaps left and right.
5. 90 degrees is two steps on the ring, 135 degrees is three, 180 degrees is four.
6. Coordinates: North +y, South -y, East +x, West -x. Add the signed legs.
7. Distance from the start is the net displacement, never the sum of the legs walked.
8. If the N-S legs cancel and the E-W legs cancel, the person is back at the start.
9. Shortest distance = sqrt(x squared + y squared) when both components exist.
10. Triplets to know: 3-4-5, 6-8-10, 9-12-15, 5-12-13, 8-15-17, 7-24-25, 20-21-29.
11. Equal x and y gives distance x times root 2 and an exactly diagonal direction.
12. Sign pairs: +x +y is NE, -x +y is NW, +x -y is SE, -x -y is SW.
13. Direction of A from B is the exact reverse of the direction of B from A.
14. Morning shadows fall West because the Sun is in the East; evening shadows fall East.
15. Morning: shadow on the right means facing South; in the evening that answer reverses.
16. At noon the shadow is negligible - the answer is usually 'no shadow'.
17. Two persons facing each other face exactly opposite directions.
18. For mixed stems, solve the coordinate diagram and the family tree separately.

RM07

Ranking, Seating Arrangement and Logical Venn Diagrams

The biggest reasoning chapter - 3-5 questions in every RAS Prelims paper. Seating arrangement usually comes as a small set of two or three questions on one common stem, which means one good diagram pays for three marks. Ranking and numeric Venn are single, quick questions and should never be lost.

Three different question families share one chapter because they share one habit: you fix a reference point first, then place everything else against it. Ranking is one formula. Seating is a set of dashes on paper. Venn is two formulas and five standard pictures. None of it needs cleverness, only order.

Ranking, Seating and Venn

Ranking
- Bottom = Total - Top + 1
- Same person: Total = a + b - 1
- Between = Total - left rank - right rank
- Shifting and interchange

Comparison
- One single inequality chain
- Tallest at one end
- Count from the correct end

Linear seating
- Fix an END or the MIDDLE first
- Facing North: their right = your right
- Facing South: everything flips
- Third to the left = two people between

Circular / square
- Facing centre: clockwise = LEFT
- Facing outward: clockwise = RIGHT
- Opposite seat is n/2 away
- Square: diagonal = two corners

Double row
- Row 1 North, row 2 South
- Mirror-image left-right senses
- X's right faces Y's left

Floors / days
- Floor 1 at the bottom
- 'Above' = larger number
- Days as slots 1 to n
- Test both cases of a gap clue

Logical Venn
- All A are B = A inside B
- Some A are B = overlap
- No A is B = separate circles
- Nested vs co-ordinate patterns

Numeric Venn
- Two-set: A + B - both
- Three-set inclusion-exclusion
- Exactly one, exactly two
- Neither = Total - at least one

Three question families, one working habit

This chapter carries three separate things. Ranking asks where a person stands in a row or a merit list. Seating arrangement gives you a row, a circle, a square, a set of floors or a set of days, plus five or six clues, and asks you to place everyone. Logical Venn asks either which picture shows the relation between three classes, or a numeric set problem. What ties them together is the working habit: never start placing until you have found the one clue that fixes a reference point.

- Order and ranking - single-ended and double-ended, shifting, interchange.
- Comparison and ordering puzzles - tallest, heaviest, third from the top.
- Linear seating - single row facing North or facing South.
- Double-row seating - two parallel rows facing each other.
- Circular seating - facing the centre or facing outward; square and rectangular tables.
- Floor puzzles and day or month scheduling puzzles.
- Logical Venn - which diagram represents Doctors, Men and Human beings.
- Numeric Venn - two-set and three-set set problems.

Where marks leak. In ranking, the student forgets the plus one or subtracts it in the wrong place. In seating, the student starts from a weak clue like 'X is not next to Y' and builds three branches instead of one. In circular seating, the student uses left and right as if the persons were facing outward when they are facing the centre. In numeric Venn, the student treats the pairwise figure as if it excluded the people who take all three. Each is a single specific slip, and each is fixable in a minute.

ASKED IN RAS

Seating arrangement has appeared in every RAS Prelims paper, usually as a linked set of two or three questions. Ranking came in 2018 (rank from the bottom in a class of 40 - answer 29th) and 2023 (10th from the front, 15th from the back - answer 24). Circular seating came in 2021 and linear seating in 2024. Logical Venn class-relationship questions came in 2016, 2021, 2023 and 2024. This chapter alone can be worth four marks.

Ranking – three formulas and the plus-one

All of ranking rests on one idea: when you count a person from both ends, that person gets counted twice. Every plus one and minus one in the formulas below is just correcting for that double count. Learn the three formulas as they stand and stop deriving them in the hall.

- Rank from the bottom = Total – rank from the top + 1.
- When both ranks belong to the SAME person: Total = rank from one end + rank from the other end – 1.
- When the two ranks belong to DIFFERENT persons who do not overlap: number between them = Total – rank of the first from the left – rank of the second from the right.
- Moving k places to the right: new position from the left = old position + k.
- After two persons interchange, each occupies the other's ORIGINAL seat – convert the new rank back to the old person's position first.

Ranking – what to write down

Read whether the two given ranks belong to one person or to two different persons. → If one person: add the two ranks and subtract 1 to get the total. →

If two persons: convert both ranks to the SAME end – usually both from the left. → For a shift, apply the shift to the position from the left, then convert back. →

For an interchange, remember the mover has taken over somebody else's old seat. → Sanity-check: no position may exceed the total, and none may be less than 1.

Worked example 1. In a row of children, Nita is 7th from the left end and Radha is 9th from the right end. When they interchange their places, Nita becomes 11th from the left end. How many children are there in the row?

- After the exchange, Nita is sitting in Radha's ORIGINAL seat. That is the key sentence.
- That seat is 11th from the left, because that is where Nita now is.
- The same seat was Radha's, and Radha was 9th from the right.
- So one single seat is 11th from the left and 9th from the right.
- Both ranks now belong to the same seat, so Total = 11 + 9 - 1 = 19.
- Answer: 19 children. Note that Nita's original 7th from the left was never needed - it is a deliberate distractor.

TRAP

Do not use Total = a + b - 1 when the two ranks belong to two different people. That formula is only for one person counted from both ends. Different people need the subtraction formula, and if they overlap in the row the answer changes again. Read the names before you pick the formula.

Comparison and ordering puzzles

Here you get four or five separate comparisons - A is taller than B, D is taller than C - and must find who is tallest or who is third. The mistake is to keep the pairs as pairs. Merge them into one single chain and the answer is just a matter of counting along it.

- Write every clue in the same direction. Convert 'X is shorter than Y' into 'Y > X'.
- Link the pairs into one chain: A > B > C and so on. Do not keep separate fragments.

- The tallest or heaviest sits at one end of the chain, the shortest or lightest at the other.
- Count from the end the question names – 'third heaviest' counts from the heavy end.
- If two people cannot be linked by any clue, the answer is 'cannot be determined'.

Worked example 2. Five boxes P, Q, R, S and T have different weights. P is heavier than S, R is heavier than P, Q is heavier than R, and T is lighter than S. Which box is the third heaviest?

- P is heavier than S gives P > S.
- R is heavier than P gives R > P, so the chain grows to R > P > S.
- Q is heavier than R gives Q > R, so Q > R > P > S.
- T is lighter than S gives S > T, so the full chain is Q > R > P > S > T.
- Count from the heaviest end: Q is 1st, R is 2nd, P is 3rd.
- Answer: P is the third heaviest.

Linear seating – fix a reference point before anything else

This is the heart of the chapter and it has a fixed procedure. Draw as many blank dashes as there are seats. Then hunt for the clue with the fewest possibilities – a clue that names an END seat or the exact MIDDLE seat. That clue is your reference point. Every other clue gets attached to it. If you start from a floating clue like 'D sits third to the right of A' you will be carrying four cases instead of one.

Seating arrangement – the fixed procedure

Count the seats and draw exactly that many dashes in a line. →

Write down which way the persons face; if they face South, note the flip at once. →

Scan every clue and pick the one that fixes an END or the MIDDLE seat. Place it. →

Take the next strongest clue – the one that ties two named persons at a fixed gap. →

Slide that block of two along the dashes and keep only the positions that fit. →

Add the remaining positive clues one at a time; negative clues ('not next to') go LAST. →

When one arrangement survives, re-read every clue against it before answering. →

If two arrangements survive, you have misread a clue – go back, do not guess.

- For a row facing NORTH, the persons' right is towards the right of the page as you read it.
- For a row facing SOUTH, everything reverses: the persons' right is towards the LEFT of the page.
- 'X sits third to the left of Y' means exactly TWO persons sit between X and Y.
- 'n persons sit between X and Y' means the gap between their seats is n + 1.
- 'Immediately to the right of' means the very next seat on that person's right side.
- Write the row once in printed order and flip mentally; do not redraw it upside down.

Worked example 3. Eight students A, B, C, D, E, F, G and H sit in a straight row facing North. C sits third to the right of H. G is not an immediate neighbour of E. H is not an immediate neighbour of F. Only one person sits between G and H. D sits at the extreme left end. C is not an immediate neighbour of G. Two persons sit between B and F. G sits third to the left of A. Who sits exactly between H and F, and which pair holds the two ends?

- Start with the end clue: D is at the extreme left. Seat 1 = D. That is the reference point.
- Next take the linked pair: only one person sits between G and H, so G and H are two seats apart.
- G sits third to the left of A, so A sits three seats to the right of G. Try G at seat 2: then A is at seat 5.
- With G at seat 2, the gap clue puts H at seat 4 (H at seat 0 is impossible).
- C sits third to the right of H, so C is at seat 7.
- Two persons sit between B and F, a gap of three seats. The free seats are 3, 6 and 8. B at 3 and F at 6 fits.
- Seat 8 is left for E. Row from the left: D, G, B, H, A, F, C, E.
- Check the negatives: G at 2 is not beside E at 8; H at 4 is not beside F at 6; C at 7 is not beside G. All hold.
- H is at seat 4 and F at seat 6, so the person exactly between them is at seat 5 - A.
- The two ends are seat 1 and seat 8 - D and E.

TRAP

When a row faces South, the phrase 'immediately to the right of P' means the seat to the LEFT of P on your page. Students solve the whole arrangement correctly and then read the final answer off in printed order. Write a small arrow above the row showing which way the persons face, and read the answer off that arrow, not off the page.

Circular, square and rectangular tables

A circle has no fixed left end, so you must create one. Pick any person, draw him at the top of your circle, and place everybody else relative to him. Then the only thing that can go wrong is the direction of left and right - and that depends entirely on whether the group is facing the centre or facing outward.

Facing the centre versus facing outward - learn this pair together

ALL FACING THE CENTRE	ALL FACING OUTWARD
The clockwise neighbour is on a person's LEFT	The clockwise neighbour is on a person's RIGHT
The anti-clockwise neighbour is on the RIGHT	The anti-clockwise neighbour is on the LEFT
'Second to the left' = two seats clockwise	'Second to the left' = two seats anti-clockwise
'Third to the right' = three seats anti-clockwise	'Third to the right' = three seats clockwise
This is the default - assume it unless the stem says otherwise	Applies only when the stem explicitly says facing outward
The seat exactly opposite is n/2 seats away	The seat exactly opposite is still n/2 seats away

- Fix one person as the reference; a circle has relative order only, never absolute position.
- In a circle of n persons, the seat exactly opposite is n/2 seats away - defined only when n is even.
- Persons between X and Y counted clockwise = clockwise gap - 1; the other way it is n - clockwise gap - 1.
- The opposite-seat rule does NOT change with facing; only left and right change.

- At a SQUARE table with one person per corner, 'diagonally opposite' means two corners away, which is also second to the left and second to the right.
- At a rectangular table with two persons on each long side and one on each short side, treat the six seats as one ring and use the circle rules.

Worked example 4 (asked in RAS Pre 2021). Eight friends P, Q, R, S, T, U, V and W sit around a circular table facing the centre. S sits immediately to the right of P. W sits second to the right of Q. V sits third to the left of W. T sits fourth to the left of U. T sits third to the left of S. Who sits exactly opposite W, and who sits third to the right of Q?

- All face the centre, so translate first: 'to the right' means anti-clockwise, 'to the left' means clockwise.
- Fix S at the top of the circle as the reference point.
- S is immediately to the right of P, so S is one seat anti-clockwise from P; P sits immediately clockwise of S.
- T sits third to the left of S means T is three seats clockwise from S.
- T sits fourth to the left of U places U four seats clockwise from T, which is seven clockwise from S.
- Applying the Q and W clues the same way, exactly one seating survives.
- Clockwise order: S, P, W, T, Q, V, R, U, and back to S.
- Opposite W: with eight persons the opposite seat is four away. Counting four from W gives R.
- Third to the right of Q: right means anti-clockwise. Step anti-clockwise from Q - first T, second W, third P.
- Answers: R sits opposite W; P sits third to the right of Q.

Double rows, floors, days and months

A double-row puzzle has five persons in row 1 facing North and five in row 2 facing South, sitting directly opposite each other. Because the two rows face each other, their left-right senses are mirror images. Write both rows in printed left-to-right order, one above the other, and put a small arrow on each row showing the facing. Then never redraw.

- Row 1 faces North: their right is the printed right. Row 2 faces South: their right is the printed LEFT.
- The person at printed position k in row 1 faces the person at printed position k in row 2.
- If X faces Y, then the person on X's right faces the person on Y's LEFT - this is the double-row rule.
- Solve the row that has the strongest clue first, then transfer through the 'X faces Y' clue.
- Always name the final answer using the facing of the row the question asks about.

Worked example 5. Ten persons sit in two parallel rows of five. A, B, C, D and E sit in row 1 facing North; P, Q, R, S and T sit in row 2 facing South; each row-1 member faces exactly one row-2 member. A sits immediately to the right of C. Three persons sit between C and D. Only one person sits between A and B. Only one person sits between T and R. C faces S. R sits third to the left of Q. Who faces B, and who sits at the extreme right end of the second row?

- Row 1 faces North, so for row 1 the printed right is their right. Three persons between C and D means C and D are four seats apart, i.e. seats 1 and 5.
- A sits immediately to the right of C. If C is at printed seat 1, A is at printed seat 2.
- Only one person sits between A and B, a gap of two seats, so B is at printed seat 4. E takes seat 3.
- Row 1 printed left to right: C, A, E, B, D.

- Row 2 faces South, so for row 2 their left is the printed right. R sits third to the left of Q places R three printed seats to the RIGHT of Q.
- C faces S, and C is at printed seat 1, so S is at printed seat 1 of row 2.
- R = Q + 3 in printed terms, and seat 1 is taken by S, so Q must be at printed 2 and R at printed 5.
- Only one person sits between T and R, a gap of two seats, so T is at printed 3. P takes the last free seat, printed 4.
- Row 2 printed left to right: S, Q, T, P, R.
- B is at printed seat 4 of row 1, so the person facing him is at printed seat 4 of row 2 - P.
- Extreme right end of the SECOND row: that row faces South, so their right is the printed left - seat 1, which is S.

- FLOOR puzzles: always draw floor 1 at the bottom and the top floor at the top. 'Above' means a larger floor number.
- 'Only k persons live between X and Y' fixes the gap but not who is higher - test both cases before rejecting either.
- DAY and MONTH puzzles: number the days or months as slots 1 to n, then convert every 'before' and 'after' clue into a difference of slot numbers.
- 'R visits two days after P' means slot of R = slot of P + 2, not 'somewhere later'.
- Verify the finished arrangement by reading every clue once more; if a second valid arrangement exists, you have mis-read a clue.

YAAD RAKHO

Yaad rakho for circles - CLOCK LEFT. When everyone faces the CentrE, CLockwise is LEft. Flip both words for facing outward. Two words, and the biggest circular-seating error is closed.

Logical Venn diagrams - picking the right picture

This form gives you three class names - say Doctors, Men and Human beings - and four candidate pictures. You choose the one that shows the real relation. There are only three drawing rules, and only about five pictures RPSC ever uses, so this is close to free marks once you have seen the standard patterns.

- If EVERY A is a B, draw the circle A wholly inside the circle B.
- If SOME A are B, draw two circles that partly overlap.
- If NO A is a B, draw two completely separate circles.
- Ask the two questions in order: can something be both? and must everything of one be the other?
- Test each option against a real example before choosing - one counter-example kills a picture.

The five standard patterns RPSC reuses

Pattern	Example	Picture
Fully nested	Jaipur, Rajasthan, India	Smallest inside middle inside largest
Co-ordinate inside a whole	Table, Chair, Furniture	Two separate circles both inside a bigger one
One nested, one separate	Pigeon, Bird, Dog	Pigeon inside Bird; Dog a separate circle
Two overlapping inside a whole	Doctors, Men, Human beings	Doctors and Men overlap, both inside Human beings
Two separate, both overlapping a third	Boys, Girls, Students	Boys and Girls disjoint, Students cuts across each

ASKED IN RAS

Asked in RAS Pre 2023 - the relation between Doctors, Men and Human beings. Answer: Doctors and Men are two intersecting circles, both lying wholly inside a larger circle for Human beings, because some doctors are women. The same form came in 2024 (Jaipur, Rajasthan, India - fully nested), in 2021 (Boys, Girls, Students) and in 2016. Memorise the five patterns above and this question is worth a mark in ten seconds.

Numeric Venn - the two-set and three-set formulas

The numeric form gives you how many people do each of two or three things and asks how many do both, or none, or exactly one. It is pure inclusion-exclusion. You add the singles, then take out what you double-counted, then put back what you removed twice. Write the formula on the sheet before you substitute a single number.

- TWO SETS: n(A or B) = n(A) + n(B) - n(A and B).
- Those in NEITHER = Total - n(A or B).
- THREE SETS: n(A or B or C) = n(A) + n(B) + n(C) - n(A and B) - n(B and C) - n(A and C) + n(A and B and C).
- Exactly ONE of three = sum of the singles - 2 times the sum of the pairwise + 3 times all three.
- Exactly TWO of three = sum of the pairwise - 3 times all three.
- If the paper says everybody does at least one thing, then n(A or B) equals the total - use that to find the overlap.

Worked example 6 (asked in RAS Pre 2016). In a class of 60 students, 35 play cricket, 25 play hockey and 10 play both games. How many students play neither game?

- Write the two-set formula first: n(A or B) = n(A) + n(B) - n(A and B).
- Substitute: 35 + 25 - 10 = 50 students play at least one game.
- The 10 who play both were counted once in the 35 and once in the 25, so removing 10 once is exactly right.
- Neither = Total - at least one = 60 - 50 = 10.
- Answer: 10 students play neither game.

Worked example 7 (asked in RAS Pre 2024). Out of 200 students, 100 study Mathematics, 90 Physics and 80 Chemistry. 40 study both Mathematics and Physics, 35 both Physics and Chemistry, 30 both Mathematics and Chemistry, and 20 study all three. How many study none of the three?

- Write the three-set formula before substituting anything.
- Sum of singles: 100 + 90 + 80 = 270.
- Sum of pairwise: 40 + 35 + 30 = 105. Subtract it: 270 - 105 = 165.
- Add back the all-three figure once: 165 + 20 = 185 study at least one subject.
- None = 200 - 185 = 15.
- Answer: 15 students. If the question had asked for exactly one instead: 270 - 2(105) + 3(20) = 270 - 210 + 60 = 120.

TRAP

In set questions the pairwise figures normally INCLUDE the people who do all three. '40 study both Mathematics and Physics' means 40 in total, of whom 20 also do Chemistry. Only if the paper actually prints the word 'only' does the figure exclude them. Read for that one word before you substitute - the whole answer turns on it.

REVISION RECAP

1. Rank from the bottom = Total - rank from the top + 1; the +1 corrects the double count.

2. Both ranks of the SAME person: Total = rank from one end + rank from the other end - 1.
3. Different persons: convert both ranks to the same end before subtracting; comparison puzzles merge into one inequality chain.
4. After an interchange, the mover occupies the other person's original seat - use that seat's two ranks.
5. Seating: draw the dashes, then start from the clue that fixes an END or the MIDDLE seat.
6. Negative clues such as 'not an immediate neighbour' are used last, to eliminate, never to start.
7. 'Third to the left of Y' means exactly two persons between; 'n persons between' means a gap of n+1.
8. Row facing South: the persons' right is the printed left. Mark the facing with an arrow.
9. Double row: row 1 North and row 2 South face each other, so their left-right senses are mirrored.
10. In a double row, if X faces Y then the person on X's right faces the person on Y's left.
11. Circle facing the CENTRE: clockwise is LEFT, anti-clockwise is RIGHT. Facing outward reverses both.
12. In a circle of n persons the opposite seat is n/2 away, whichever way they face; at a square table diagonally opposite is two corners.
13. Floors: number 1 at the bottom; a gap clue does not say which of the two is higher, so test both.
14. Day and month puzzles: number the slots 1 to n and turn every before/after clue into a slot difference.
15. Logical Venn: all A are B means A inside B; some means overlap; no A is B means separate circles.
16. Two sets: n(A or B) = n(A) + n(B) - n(A and B); neither = Total - n(A or B).
17. Three sets: add the singles, subtract all three pairwise, add back the all-three figure.
18. Pairwise figures include the all-three people unless the paper prints the word 'only'.

RM08

Shapes, Figure Counting, Mirror and Water Images, Embedded Figures

About 2-3 questions in every RAS Prelims paper. Counting triangles or squares comes almost every year (2023 and 2024 both carried one), mirror images came in 2023 and 2024, dice in 2024. The bank carries 45 questions on this chapter - do all of them, because speed here buys you time for the puzzle questions.

This is the picture chapter of the reasoning paper - count the triangles, name the mirror image, find the opposite face of the dice. There is nothing to memorise about the world here; there is a method for each type, and the method is short. Learn nine methods and the whole chapter becomes free marks.

Figures, Images and Counting

Counting triangles

- Count in layers: 1 region, then 2, then 3, then 4
- Rectangle + both diagonals = 8
- Rectangle + diagonals + one mid-line = 12
- Triangle + three medians = 16
- n cevians from one vertex = C(n+2, 2)

Counting squares & rectangles

- n x n grid squares = n(n+1)(2n+1)/6
- 3x3 = 14, 4x4 = 30, 8x8 = 204
- Rectangles = C(m+1,2) x C(n+1,2)
- Chessboard rectangles = 1296

Mirror image

- Reflection about a VERTICAL axis
- Unchanged: A H I M O T U V W X Y
- Letter order always reverses
- Clock: true time = 11:60 minus image

Water image

- Reflection about a HORIZONTAL axis
- Unchanged: B C D E H I K O X
- Both mirror and water: H I O X, digits 0 and 8
- Clock: 18:30 minus the given time

Paper folding & punching

- n folds = 2^n layers
- One punch = 2^n holes
- k punches = k x 2^n holes
- Punch on a fold line gives fewer

Cube & dice

- Standard dice: opposite faces add to 7
- A face touches 4, is opposite the 5th
- Two views, same top: four sides in a cycle
- Painted cube: 8, 12(n-2), 6(n-2)^2, (n-2)^3

Polygons & embedded figures

- Diagonals = n(n-3)/2
- Interior sum = (n-2) x 180
- n = 360/(180 - interior angle)
- Trapezium is the usual odd one out

What the paper actually asks - and where the marks leak

Everything in this chapter is a picture question. But the picture is always one of a small number of standard shapes, and RPSC describes it in plain words - 'a rectangle with both its diagonals drawn', 'two positions of the same dice'. So read the description, draw it yourself in the margin in five seconds, and then apply the method. Never try to hold the figure in your head.

- Counting - triangles, squares, rectangles inside a described figure.
- Mirror image - of a letter, a word, a number or a clock face.
- Water image - the same job, but the reflection is top-to-bottom.
- Paper folding and punching - how many holes after unfolding.
- Dice - which face is opposite which, from one or two positions.
- Painted cube - how many small cubes have 3, 2, 1 or 0 painted faces.
- Embedded or hidden figures, and odd-one-out among shapes.

Marks leak in two places only. One: the student counts the smallest triangles, finds four, and ticks 4 - forgetting that two small triangles joined together also make a triangle. Two: the student confuses the mirror set of letters with the water set. Both are cured by drill, not by cleverness. Fix the two and you will not lose a mark in this chapter again.

ASKED IN RAS

Asked in RAS Pre 2023 and 2024 - the rectangle with both diagonals (answer 8 triangles) and the 3 x 3 grid of squares (answer 14). RAS Pre 2024 also carried the rectangle with both diagonals plus one mid-line (12 triangles), the mirror-image letter question and a two-position dice question. This chapter repeats its own standard figures year after year.

Counting triangles - count in layers, never at random

There is exactly one reliable way to count triangles, and it is not to stare at the figure. You count in layers. First count the triangles made of a single smallest region. Then the triangles made of two of those regions joined. Then three. Then four. Then finally check the whole figure. Write the running total in the margin as you go – 4 + 4 = 8, not a single number pulled out of the air.

The layered counting method – use it on every figure

Mark every point where lines cross → Count the SMALLEST regions that are triangles → Count triangles made of 2 smallest regions → Count triangles made of 3 regions → Count triangles made of 4 or more regions → Check the whole outer figure → Add the layers

Standard figures – learn these numbers by heart

The figure described	Triangles	Squares
Rectangle or square + both diagonals	8	1 if it is a square
Rectangle + both diagonals + ONE mid-line	12	–
Square + both diagonals + BOTH mid-lines	16	5
Square + ONE diagonal + both mid-lines	6	5
Triangle + all three medians	16	–
Triangle with the three midpoints joined	5	–
Equilateral triangle, each side cut into 2	5	–
Equilateral triangle, each side cut into 3	13	–
Equilateral triangle, each side cut into 4	27	–

WORKED EXAMPLE 1 (RAS Pre 2023, 2024). A rectangle ABCD is drawn and both of its diagonals AC and BD are drawn. How many triangles are there in the figure in all?

- The two diagonals cross at the centre O – that is the only new point inside.
- Layer 1, smallest regions: AOB, BOC, COD, DOA = 4 triangles.
- Layer 2, two regions joined: AOB + BOC gives ABC; similarly BCD, CDA, DAB = 4.

- Layer 3, three regions joined: no triangle closes - 0.
- Layer 4, all four regions: that is the rectangle itself, not a triangle - 0.
- Total = 4 + 4 = 8. Answer: 8.

WORKED EXAMPLE 2 (RAS Pre 2024). A rectangle is divided by both its diagonals and also by the straight line joining the midpoints of its two longer sides. How many triangles are formed in all?

- The mid-line also passes through the centre O, so it cuts the top region and the bottom region into two each.
- Layer 1, smallest regions: 2 at the top + 2 at the bottom + 1 on the left + 1 on the right = 6.
- Layer 2, two regions joined: the top pair rejoins as AOB, the bottom pair as COD = 2.
- Layer 3: nothing closes - 0.
- Layer 4, half-rectangles: ABC, ABD, BCD, ACD = 4.
- Total = 6 + 2 + 4 = 12. Answer: 12.

TRAP

Do not stop at the smallest regions. In Example 1 the smallest count is 4 and 4 is sitting there in the options waiting for you. The examiner puts the half-answer in the choices on purpose. Always ask: can two of these join into a bigger triangle? Can four?

Cevians, medians and the equilateral grid

When all the extra lines come out of ONE vertex of a triangle and land on the opposite side, you do not need to count at all. Every triangle in that figure is fixed by choosing two of the points on the opposite side and joining them to the apex. So count the points on that side and use the combination formula.

With n lines drawn from the vertex, the opposite side carries n + 2 points and the answer is C(n+2, 2).

Two shortcut families you can answer in five seconds

Configuration	Rule	Values
n lines from ONE vertex to the opposite side	C(n+2, 2)	n=1: 3, n=2: 6, n=3: 10, n=4: 15
Equilateral triangle, sides cut into n parts, lines parallel to all three sides	up-triangles + down-triangles	n=2: 5, n=3: 13, n=4: 27
Triangle with all three medians	fixed figure	16
Triangle with the three midpoints joined	4 small + 1 whole	5

WORKED EXAMPLE 3. In a triangle all three medians are drawn. How many triangles are there in the resulting figure?

- The three medians meet at the centroid G and cut the triangle into 6 smallest triangles of equal area.
- Layer 1: those 6.
- Layer 2, made of two small ones: ABG, BCG, CAG = 3.
- Layer 3, made of three small ones - each median splits the triangle in half: ABD, ADC, BCE, BEA, CAF, CFB = 6.
- Layer 6, the whole triangle = 1.
- Total = 6 + 3 + 6 + 1 = 16. Answer: 16.

Counting squares and rectangles in a grid

A grid question is pure formula. For squares you add the squares of the numbers - a 3 x 3 grid holds 9 + 4 + 1 squares because a 1-unit square can sit in 9 places, a 2-unit square in 4 places and a 3-unit square in 1. For rectangles you choose lines, not cells: any two of the vertical lines and any two of the horizontal lines bound exactly one rectangle.

Grid formulas and the numbers worth memorising

Asked	Formula	Standard values
Squares of all sizes in an n x n grid	1^2 + 2^2 + ... + n^2 = n(n+1)(2n+1)/6	2x2: 5, 3x3: 14, 4x4: 30, 5x5: 55, 8x8: 204
Rectangles of all sizes in an m x n grid	C(m+1, 2) x C(n+1, 2)	3x2 grid: 6 x 3 = 18
Rectangles on a chessboard	C(9,2) x C(9,2)	36 x 36 = 1296
Rectangles that are NOT squares	total rectangles minus squares	Chessboard: 1296 - 204 = 1092

WORKED EXAMPLE 4. (a) RAS Pre 2023 - a square is divided into 9 equal smaller squares by two horizontal and two vertical lines. How many squares of all sizes are there? (b) How many rectangles of all sizes are there on a standard 8 x 8 chessboard, and how many on a 3 x 2 arrangement of small rectangles?

- (a) Squares of side 1 unit: 3 x 3 = 9.
- (a) Squares of side 2 units: the top-left corner has 2 x 2 = 4 possible positions, so 4.
- (a) Squares of side 3 units: the whole square = 1. Total = 9 + 4 + 1 = 14.
- (a) Formula check: n(n+1)(2n+1)/6 = 3 x 4 x 7 / 6 = 14. Answer: 14.
- (b) A rectangle is fixed by 2 vertical lines and 2 horizontal lines.
- (b) A 3 x 2 arrangement has 4 vertical and 3 horizontal lines: C(4,2) x C(3,2) = 6 x 3 = 18.
- (b) A chessboard has 9 vertical and 9 horizontal lines: C(9,2) x C(9,2) = 36 x 36 = 1296.

YAAD RAKHO

Do not confuse the two chessboard numbers. 204 is SQUARES, 1296 is RECTANGLES. RPSC puts both in the options of the same question. Squares add squares; rectangles choose lines.

Mirror images – the vertical-axis rule

A mirror stands vertically beside the object, so the reflection swaps left and right and leaves top and bottom alone. This is called lateral inversion. Two consequences follow, and both get asked. First, a single letter survives unchanged only if it has a vertical axis of symmetry. Second, in a word the ORDER of the letters always reverses – REASONING becomes GNINOSAER – whether or not the individual letters are symmetric.

The two invariant sets – learn them as blocks, not one by one

Reflection	Unchanged capitals	Also note
Mirror (vertical axis)	A H I M O T U V W X Y (11 letters)	Digits unchanged: 0, 8
Water (horizontal axis)	B C D E H I K O X (9 letters)	Digits unchanged: 0, 8
Unchanged in BOTH	H I O X	Plus digits 0 and 8
Word rule (mirror)	Reverse the letter order first, then mirror each letter	A word reads the same only if it is a palindrome of symmetric letters, e.g. TOOT

WORKED EXAMPLE 5 (RAS Pre 2024). (a) A clock seen in a mirror shows the time as 3:40. What is the correct time? (b) A wall clock actually shows 4:50 –

what will its mirror image show? (c) In the word MOUNTAIN, how many letters remain unchanged in their mirror image?

- (a) Rule: true time and mirror time always add up to 12:00, so subtract from 11:60.
- (a) 11:60 minus 3:40 - minutes 60 - 40 = 20, hours 11 - 3 = 8. Correct time = 8:20.
- (b) The rule works both ways: image of 4:50 = 11:60 - 4:50 = 7:10.
- (b) Check: 4:50 + 7:10 = 12:00. Correct.
- (c) Mirror-invariant set is A H I M O T U V W X Y.
- (c) In M-O-U-N-T-A-I-N the letters M, O, U, T, A, I qualify; the two N's do not. Answer: 6.

ASKED IN RAS

Asked in RAS Pre 2023 and 2024 - 'which capital letter looks exactly the same in a vertical mirror' (answer T, from the set A H I M O T U V W X Y), and RAS Pre 2024 - the mirror clock showing 3:40 (answer 8:20). Both are one-line answers if the set and the 11:60 rule are on your fingertips.

Water images - the horizontal-axis rule

A water image is what you would see reflected in still water below the object. The axis is horizontal, so top and bottom swap and left and right stay put. Therefore a letter survives only if it has a horizontal axis of symmetry - and that is a completely different set from the mirror set. In a word the letter order does NOT reverse in a water image; each letter simply flips upside down in place.

Mirror image vs water image - the single most confused pair in this chapter

Mirror image	Water image
Reflection about a VERTICAL axis	Reflection about a HORIZONTAL axis
Left and right swap; top and bottom stay	Top and bottom swap; left and right stay
Letter ORDER in a word reverses	Letter order stays the same
Survivors: A H I M O T U V W X Y	Survivors: B C D E H I K O X
Clock: 11:60 minus the given time	Clock: 18:30 minus the given time
Mirror stands beside the object	Water lies below the object

WORKED EXAMPLE 6. (a) In the word CHOICE, how many letters remain unchanged in their water image? (b) Out of the capital letters A, B, H, M, O and X, how many remain unchanged in their water image?

- Water-invariant set: B, C, D, E, H, I, K, O, X.
- (a) Take C-H-O-I-C-E letter by letter: C yes, H yes, O yes, I yes, C yes, E yes.
- (a) All 6 letters survive, and since the order does not reverse the water image still reads CHOICE. Answer: 6.
- (b) From A, B, H, M, O, X: B yes, H yes, O yes, X yes = 4.
- (b) A and M are rejected - they are mirror-symmetric, not water-symmetric. Answer: 4.

TRAP

Never use the 11:60 rule for a water clock. Mirror clock = 11:60 minus the time; water clock = 18:30 minus the time (subtract 12 hours if the answer crosses 12:00). And do not carry A and M into the water set - they are the classic wrong ticks.

Paper folding and punching

Fold a sheet in half and you have two layers. Fold again and you have four. Every fold doubles the layers, so after n folds there are 2 to the power n layers, and a single punch goes through all of them. The whole question type reduces to one line of arithmetic: holes = number of punches x 2 to the power n.

Folds to layers to holes

Count the folds, n → Layers = 2^n → One punch pierces every layer → Holes from one punch = 2^n → k punches = k x 2^n holes

WORKED EXAMPLE 7. (a) A square sheet is folded exactly in half, then in half once more, and a single hole is punched at a point not on any fold. How many holes on unfolding? (b) The same sheet, two folds, but two separate holes punched. (c) Three folds, one punch.

- (a) Two folds, so layers = 2^2 = 4. One punch pierces all 4 layers, giving 4 holes.
- (b) Layers still 4; each punch makes 4 holes, so 2 x 4 = 8 holes.
- (c) Three folds, layers = 2^3 = 8, so one punch gives 8 holes.
- General: holes = k x 2^n, where k is the number of punches and n the number of folds.

TRAP

The question says 'at a point not lying on any fold' for a reason. If the punch is made ON a fold line, the layers meet there and you get FEWER holes than 2^n - typically half. Read that clause before you multiply.

Dice and painted cubes

A cube has six faces. Any one face touches exactly four others and is opposite the fifth - that single sentence solves most dice questions. On a standard dice the opposite faces add up to 7, so the pairs are 1-6, 2-5 and 3-4. When the question gives you two positions of the same dice instead, you work out the cycle of the side faces.

Dice rules - which one applies to which question

What the question gives	Rule to apply	Result
'Standard dice' or 'ordinary dice'	Opposite faces sum to 7	Pairs 1-6, 2-5, 3-4
One face named as adjacent to four others	A face touches 4 and is opposite the 5th	The unnamed sixth number is the opposite
Two views with the SAME face on top	The other four lie in a cycle around it; opposites are two steps apart	Two pairs of opposites at once
Two views with ONE common face in different positions	Faces that never appear together in a single view are opposite	One pair of opposites

WORKED EXAMPLE 8 (RAS Pre 2024). Two positions of the same dice. Position I: top face 1, front face 2, right face 3. Position II: top face 1, front face 4, right face 5. Which number lies opposite the face bearing 2?

- 1 is on top in BOTH views, so 2, 3, 4 and 5 are the four side faces and 6 must be at the bottom, opposite 1.
- Read each view clockwise from the front: view I gives the consecutive pair 2 then 3.
- View II gives the consecutive pair 4 then 5.
- The only cycle of the four side numbers containing 2-3 and 4-5 in that order is 2, 3, 4, 5 and back to 2.

- In a four-cycle the opposite faces are two steps apart: 2 opposite 4, and 3 opposite 5.
- Answer: 4 lies opposite 2.

Painted cube cut into n^3 smaller cubes – all six faces painted

Faces painted	Count	Where they sit
Exactly 3	8 always	The 8 corners, whatever n is
Exactly 2	$12(n - 2)$	Along the 12 edges, corners excluded
Exactly 1	$6(n - 2)^2$	The inside of each of the 6 faces
None	$(n - 2)^3$	The hidden inner block
At least 1	$n^3 - (n - 2)^3$	Everything except the inner block
Only two OPPOSITE faces painted	exactly one face: $2n^2$; two faces: 0	Top layer and bottom layer only

WORKED EXAMPLE 9. A cube painted on all six faces is cut into 64 identical smaller cubes. How many have exactly two faces painted, how many exactly one, and how many none at all?

- $64 = 4^3$, so $n = 4$ and $n - 2 = 2$.
- Three faces painted = 8 (the corners, always 8).
- Two faces painted = $12(n - 2) = 12 \times 2 = 24$.
- One face painted = $6(n - 2)^2 = 6 \times 4 = 24$.
- No face painted = $(n - 2)^3 = 2^3 = 8$.
- Check the identity: $8 + 24 + 24 + 8 = 64$. The four counts must always add back to n^3.

YAAD RAKHO

Always run the addition check - 8 + 12(n-2) + 6(n-2)^2 + (n-2)^3 must equal n^3. It takes four seconds and catches every arithmetic slip. And remember the corner count is 8 for a 27-cube, a 64-cube or a 1000-cube alike; a cube has 8 corners, full stop.

Polygons, embedded figures and odd-one-out

The tail end of this chapter is three small types. Polygon questions are pure formula. Embedded-figure questions ask which of four complex figures hides a given simple shape - the method is to count the line segments and closed regions of the target shape and then look for exactly that many in each option, matching piece by piece rather than eyeballing. Odd-one-out among shapes usually turns on a definition, most often the parallel-sides definition of a quadrilateral.

Polygon formulas that get asked directly

Asked	Formula	Example
Number of diagonals	n(n - 3)/2	Octagon: 8 x 5 / 2 = 20
Sum of interior angles	(n - 2) x 180 degrees	Hexagon: 4 x 180 = 720
Each exterior angle (regular)	360/n	Decagon: 36 degrees
Sides from the interior angle	n = 360/(180 - A)	A = 144: n = 360/36 = 10
Each interior angle (regular)	(n - 2) x 180 / n	Pentagon: 108 degrees

WORKED EXAMPLE 10. (a) Each interior angle of a regular polygon measures 144 degrees - how many sides? (b) How many diagonals does a regular octagon have? (c) Three of these are alike and one is different: rhombus, square, trapezium, rectangle. Which is the odd one out?

- (a) Exterior angle = 180 - 144 = 36 degrees; number of sides = 360/36 = 10.
- (a) Same thing in one step: n = 360/(180 - A) = 360/36 = 10 sides.
- (b) Diagonals = n(n - 3)/2 with n = 8, so 8 x 5 / 2 = 20 diagonals.
- (c) A rhombus, a square and a rectangle are all parallelograms - both pairs of opposite sides parallel.
- (c) A trapezium has only ONE pair of parallel sides, so the trapezium is the odd one out.

REVISION RECAP

1. Count triangles in layers - smallest regions, then 2 joined, then 3, then 4, then the whole figure.
2. Rectangle or square with both diagonals = 8 triangles; with one mid-line added = 12.
3. Square with both diagonals and both mid-lines = 16 triangles; with one diagonal and both mid-lines = 6.
4. Triangle with all three medians = 16 triangles; with the three midpoints joined = 5.
5. n lines from one vertex to the opposite side give C(n+2, 2) triangles - n=2 gives 6, n=3 gives 10.
6. Equilateral triangle with sides cut into n parts: 5 triangles for n=2, 13 for n=3, 27 for n=4.
7. Squares in an n x n grid = n(n+1)(2n+1)/6 - remember 14, 30 and 204.
8. Rectangles in an m x n grid = C(m+1,2) x C(n+1,2); chessboard = 36 x 36 = 1296, of which 204 are squares.
9. Mirror = vertical axis; the survivors are A H I M O T U V W X Y and the letter order reverses.
10. Water = horizontal axis; the survivors are B C D E H I K O X and the order does not reverse.
11. H, I, O, X and the digits 0 and 8 survive both reflections.
12. Mirror clock: true time = 11:60 minus the image; the two times always add to 12:00.
13. Water clock: 18:30 minus the given time, dropping 12 hours if the answer overshoots.
14. Paper folding: n folds give 2^n layers, so k punches give k x 2^n holes - fewer if the punch sits on a fold.
15. A cube face touches four faces and is opposite the fifth; on a standard dice opposites add to 7.

16. Two dice views with the same top: the four side numbers form a cycle and opposites are two steps apart.
17. Painted cube: 3 faces = 8, 2 faces = $12(n-2)$, 1 face = $6(n-2)^2$, none = $(n-2)^3$, and the four add to n^3.
18. Only two opposite faces painted: exactly one face painted = $2n^2$, and no cube gets two painted faces.
19. Polygons: diagonals $n(n-3)/2$, interior sum $(n-2) \times 180$, sides from interior angle $A = 360/(180 - A)$.
20. In shape classification the trapezium is normally the odd one out - only one pair of parallel sides.

RM09

Numeracy I: Ratio, Proportion, Partnership, Percentage, Profit & Loss, Interest

3-4 questions a paper, the largest single block in the reasoning section - the bank carries 70 questions for this chapter. Interest is the biggest sub-topic: the CI-minus-SI difference was asked in RAS Pre 2018, 2021, 2023 AND 2024. Milk-and-water mixtures came in 2016 and again in 2024.

This is the arithmetic block of the reasoning paper - the numbers you did in school, asked in one line with four options. Nothing here is hard; everything here is fast if you carry the shortcut formula and slow if you set up equations. The difference between compound and simple interest alone has come in four consecutive papers.

Numeracy I

CI minus SI (the star)

- 2 years: P(R/100)^2
- 3 years: P x R^2 x (300+R)/10^6
- Reverse: P = D x (100/R)^2
- Reverse: R = 100 x sqrt(D/P)

Interest

- SI = PRT/100
- A = P(1 + R/100)^n
- Half-yearly: rate R/2, periods 2n
- CI doubling in t years = 4x in 2t

Ratio & proportion

- Share = (own part / total parts) x amount
- Combine a:b and b:c by matching b
- Mean proportional = sqrt(ab)
- Third proportional = b^2/a

Mixture & alligation

- cheaper : dearer = (d - m) : (m - c)
- Water is taken at zero cost
- Replacement: x(1 - y/x)^n
- Equal quantities: add the fractions

Percentage

- Successive: a + b + ab/100
- x% up then x% down = x^2/100 fall
- Price up x% -> cut use by 100x/(100+x)%
- Growth P(1+r/100)^n, fall P(1-r/100)^n

Profit, loss & discount

- Percentages are always on the COST price
- SP = MP(100 - d)/100
- Two discounts: d1 + d2 - d1d2/100
- Same SP, +x% and -x% = x^2/100 loss

Partnership

- Profit ratio = capital x time
- Working partner's salary comes out first
- Withdrawal changes the time, not the ratio of capital
- Joining late = fewer months

How this block is asked, and the two habits that save the marks

Every question in this chapter is a one-liner with four options, and every one of them has a shortcut. The examiner is not testing algebra; he is testing whether you carry the formula. So the whole preparation is: know which family the question belongs to, write the one formula for that family, put the numbers in. Setting up 'let the number be x' costs you thirty seconds you do not have.

- Ratio, proportion and the division of an amount into shares.
- Mixture and alligation - the milk-and-water family.
- Partnership - profit split by capital multiplied by time.
- Percentage, successive change, population growth and depreciation.
- Profit, loss, marked price and discount.
- Simple interest, compound interest and instalments.
- The difference between CI and SI - a family of its own, and the most repeated one.

Percentage to fraction - learn these and half the arithmetic disappears

Percentage	Fraction	Percentage / Fraction
50%	1/2	20% = 1/5
33 1/3%	1/3	16 2/3% = 1/6
25%	1/4	12 1/2% = 1/8
10%	1/10	6 1/4% = 1/16
8 1/3%	1/12	5% = 1/20
66 2/3%	2/3	37 1/2% = 3/8

YAAD RAKHO

Second habit: always ask 'percentage of WHAT'. Profit and loss percentages are on the cost price. Discount is on the marked price. A percentage rise and the matching percentage fall are on different bases - that is why they never cancel. Ninety per cent of the wrong answers in this chapter are right arithmetic on the wrong base.

Percentage and successive change

Percentage questions come in three shapes: plain conversion, successive change, and the 'A is x% more than B' comparison. The successive-change shape is the one that traps people, because two changes of a% and b% never simply add. The second change acts on the already-changed amount, so the net effect is a + b + ab/100, and you write a fall with a minus sign.

The percentage rules worth memorising

Situation	Formula	Note
a% change then b% change	a + b + ab/100	Use a minus sign for a fall
x% rise then x% fall	Net fall of x^2/100 per cent	Never zero
Price rises x%, expenditure fixed	Cut consumption by 100x/(100 + x) per cent	x = 25 gives 20%
A is x% more than B	B is 100x/(100 + x) per cent less than A	Different base, different number
A is x% less than B	B is 100x/(100 - x) per cent more than A	-
Growth for n years at r%	P(1 + r/100)^n	Depreciation: P(1 - r/100)^n

WORKED EXAMPLE 1. (a) A price is increased by 20% and the new price is then increased by a further 10% - what is the net increase? (b) A salary of Rs 25,000 is raised by 10% and the raised salary is then cut by 10% - what is the final salary? (c) If the price of a commodity rises by 25%, by what percentage must a family cut its consumption to keep the expenditure unchanged?

- (a) Net = a + b + ab/100 = 20 + 10 + (20 x 10)/100 = 30 + 2 = 32%.
- (a) Check the long way: 100 becomes 120, and 120 becomes 132. Yes, 32%.
- (b) Net = 10 - 10 - (10 x 10)/100 = -1%, a fall of 1%, i.e. Rs 250.
- (b) Final salary = 25000 - 250 = Rs 24,750. Long way: 25000 x 1.10 = 27,500, then 27500 x 0.90 = 24,750.
- (c) Required cut = 100x/(100 + x) = 100 x 25/125 = 20%.

- (c) Check: 100 units at Re 1 cost Rs 100; at Rs 1.25 the same Rs 100 buys only 80 units - a 20% cut.

TRAP

A 20% rise followed by a 20% fall is NOT 'no change'. It is a 4% fall, because the fall is computed on the bigger amount. The option 'no change' is put there for the hurried student. The rule x^2/100 covers every version of this trap - 10% and 10% give a 1% fall, 25% and 25% give a 6.25% fall.

Ratio and proportion

A ratio is a recipe for splitting. If an amount is divided in the ratio 2 : 3 : 7, count the parts (12), find the value of one part, and multiply. When the question gives you two ratios that overlap in one letter - A : B and B : C - you cannot read off A : C directly; you must first scale both ratios so that B is the same number in each. That single step is the whole difficulty of the topic.

Ratio and proportion formulas

Term	Definition	Formula
Share of a part	Divide the total in the ratio	(own part / sum of parts) x total
Proportion a : b :: c : d	Product of extremes = product of means	ad = bc
Mean proportional between a and b	The middle term of a : x :: x : b	x = square root of ab
Third proportional to a and b	a : b :: b : x	x = b^2/a
Fourth proportional to a, b, c	a : b :: c : x	x = bc/a
Combining a : b and b : c	Scale both until b matches	Then read off a : b : c

WORKED EXAMPLE 2. (a) Rs 5,900 is divided among A, B and C so that A : B = 3 : 4 and B : C = 5 : 6. What is A's share? (b) Two numbers are in the ratio 3 : 5; if 4 is added to each the ratio becomes 5 : 7 - what is the smaller number? (c) What is the mean proportional between 9 and 64, and the third proportional to 8 and 12?

- (a) B appears as 4 in the first ratio and 5 in the second - make it 20 in both.
- (a) A : B = 3 : 4 = 15 : 20, and B : C = 5 : 6 = 20 : 24. So A : B : C = 15 : 20 : 24.
- (a) Total parts = 15 + 20 + 24 = 59, so one part = 5900/59 = Rs 100. A's share = 15 x 100 = Rs 1,500.
- (b) Let the numbers be 3x and 5x: (3x + 4)/(5x + 4) = 5/7, so 21x + 28 = 25x + 20.
- (b) 4x = 8, x = 2, so the numbers are 6 and 10 - the smaller is 6.
- (c) Mean proportional = square root of (9 x 64) = square root of 576 = 24; third proportional = 12^2/8 = 18.

Mixture and alligation - the milk-and-water family

Alligation answers one question: in what ratio do two things of different value have to be mixed to reach a given average value. The rule is a cross-subtraction - cheaper to dearer is as (dearer minus mean) is to (mean minus cheaper). Water in a milk problem is simply an ingredient costing zero. Do not build equations for these; the cross takes five seconds.

Mixture rules - four shapes, four rules

Question shape	Rule	Note
Mix two prices to reach a mean price	cheaper : dearer = (dearer - mean) : (mean - cheaper)	Water is priced at zero

Question shape	Rule	Note
Draw y litres out of x and top up with water, n times	Liquid left = x(1 - y/x)^n	Never subtract y each time
Mix EQUAL quantities of two mixtures	Add the component fractions directly	4/5 and 3/5 milk give 7 : 3
Change a : b to a : c by adding the second liquid	Keep the first quantity fixed, recompute the second	Add the difference

WORKED EXAMPLE 3 (the 2016 and 2024 family). (a) In what ratio must water be mixed with milk costing Rs 32 per litre so that the mixture is worth Rs 24 per litre? (b) A vessel contains 100 litres of pure milk; 20 litres is drawn out and replaced by water, and the operation is performed once more - how much pure milk is left? (c) A milkman buys milk at Rs 40 per litre, adds water and sells the mixture at Rs 40 per litre, gaining 25% - what is the ratio of water to milk?

- (a) Cheaper = water at Rs 0, dearer = milk at Rs 32, mean = Rs 24.
- (a) water : milk = (32 - 24) : (24 - 0) = 8 : 24 = 1 : 3.
- (b) Use milk left = x(1 - y/x)^n with x = 100, y = 20, n = 2.
- (b) 100 x (1 - 20/100)^2 = 100 x (4/5)^2 = 100 x 16/25 = 64 litres of milk (and 36 litres of water).
- (c) He sells at Rs 40 with a 25% gain, so one litre of the mixture costs him 40/1.25 = Rs 32.
- (c) Alligation between water (Rs 0) and milk (Rs 40) at a mean of Rs 32: water : milk = (40 - 32) : (32 - 0) = 8 : 32 = 1 : 4.

ASKED IN RAS

Asked in RAS Pre 2016 and again in RAS Pre 2024 - the milk-and-water mixture. The 2016 form was the alligation ratio, the 2024 form the repeated-replacement one. Learn both the cross rule and the x(1 - y/x)^n formula; between them they cover every version RPSC has set.

Partnership

Partnership is one line: profit is shared in the ratio of capital multiplied by time. If everyone stays in for the same period the time cancels and the split is simply by capital. If one partner joins late, withdraws early or changes his investment, convert each partner's money into money-months and take the ratio of those. A working partner's salary or commission is taken out of the profit FIRST; only the balance is split.

Partnership - what to multiply

Situation	Ratio to use	Example
Same period for all	Capitals	12,000 : 16,000 = 3 : 4
Different periods	Capital x months	5000 x 12 : 6000 x 8 = 5 : 4
One partner joins after m months	Capital x (12 - m)	8000 x 12 : 9000 x 8 = 4 : 3
One partner withdraws at 6 months	Capital x 6 for him, x 12 for the other	3 x 6 : 2 x 12 = 3 : 4
Working partner with commission	Deduct the commission first, split the rest	10% of 20,000 taken out first

WORKED EXAMPLE 4. (a) A invests Rs 5,000 for 12 months and B invests Rs 6,000 for 8 months; the profit is Rs 18,000 - what is A's share? (b) A and B invest Rs 30,000 and Rs 20,000; A is the working partner and takes 10% of the profit for managing, the rest being split by capital. If the annual profit is Rs 20,000, what does A get in all?

- (a) Money-months: A = 5000 x 12 = 60,000; B = 6000 x 8 = 48,000.
- (a) Ratio = 60,000 : 48,000 = 5 : 4, so 9 parts in all.
- (a) A's share = 18000 x 5/9 = Rs 10,000.

- (b) A's management fee comes out first: 10% of 20,000 = Rs 2,000.
- (b) Balance = Rs 18,000, split in the capital ratio 30,000 : 20,000 = 3 : 2.
- (b) A gets 18000 x 3/5 = Rs 10,800, so A's total = 2000 + 10800 = Rs 12,800.

TRAP

Do not split the whole profit in the capital ratio when there is a working partner - the salary or commission is deducted before the split, and the answer that ignores it is always one of the options. Equally, 'withdraws his capital at the end of 6 months' changes the TIME to 6, not the capital.

Profit, loss, marked price and discount

Fix one rule in your head and this section is safe: profit per cent and loss per cent are always on the COST price, while discount is always on the MARKED price. A shopkeeper marks his goods above cost, then discounts the marked price, and the selling price that results is compared back to the cost price to get his profit. Three different bases in one question - that is the whole difficulty.

Profit and loss formulas

Asked	Formula	Note
Profit %	(SP - CP)/CP x 100	Always on CP
SP from CP	CP(100 + p)/100	For a loss use (100 - l)
CP from SP	100 x SP/(100 + p)	Never subtract the percentage from SP
SP from marked price	MP(100 - d)/100	Discount is on MP
Mark m% above cost, discount d%	Profit % = m - d - md/100	m=25, d=10 gives 12.5%

Asked	Formula	Note
Two successive discounts	d1 + d2 - d1d2/100	20% and 10% = 28%, not 30%
Same SP, one at +x% and one at -x%	Net LOSS of x^2/100 per cent	Never a break-even
CP of x articles = SP of y articles	Profit % = (x - y)/y x 100	Loss if y is bigger
False weight, sold at cost price	Gain % = error/(true value - error) x 100	800 g for 1 kg gives 25%
Profit of p% on CP, expressed on SP	100p/(100 + p) per cent of the SP	25% on cost = 20% of the selling price

The shopkeeper chain - work along it, never jump

Cost price (CP) → Mark up m% -> Marked price (MP) → Discount d% on MP -> Selling price (SP) → Compare SP with CP -> profit or loss %

WORKED EXAMPLE 5. (a) A shopkeeper buys an article for Rs 300, marks it 25% above cost and allows a discount of 10% on the marked price - what is his profit per cent? (b) A man sells two articles at Rs 1,200 each, gaining 20% on one and losing 20% on the other - what is his overall result? (c) A dishonest dealer sells at cost price but uses an 800 g weight for a kilogram - what is his gain per cent?

- (a) Marked price = 300 x 1.25 = Rs 375.
- (a) Selling price = 375 x 0.90 = Rs 337.50, so profit = Rs 37.50 on a cost of Rs 300.
- (a) Profit % = 37.5/300 x 100 = 12.5%. Shortcut: m - d - md/100 = 25 - 10 - 2.5 = 12.5%.
- (b) CP of the first = 1200 x 100/120 = Rs 1,000; CP of the second = 1200 x 100/80 = Rs 1,500.

- (b) Total CP = Rs 2,500 against a total SP of Rs 2,400 - a LOSS of Rs 100, i.e. 4% = x^2/100 with x = 20.
- (c) Gain % = error/(true value - error) x 100 = 200/800 x 100 = 25%.

TRAP

Successive discounts do NOT add. Discounts of 20% and then 10% on Rs 1,000 give 1000 x 0.80 x 0.90 = Rs 720, an effective discount of 28% - not 30%. The single equivalent discount is d1 + d2 - d1d2/100, and it is always LESS than the sum. The same warning applies to two successive profit percentages.

Simple interest

Simple interest is charged on the original principal every year, so the interest for each year is identical. That single property answers most questions: if a sum amounts to different values after different numbers of years, the difference between the two amounts is the interest for the gap in years, and dividing gives the interest for one year.

Simple interest - one formula, four rearrangements

Asked	Formula	Trick version
Interest	SI = P x R x T/100	Amount = P + SI
Principal	P = SI x 100/(R x T)	-
Rate	R = SI x 100/(P x T)	-
Time	T = SI x 100/(P x R)	-
Sum doubles itself	SI = P, so R x T = 100	R = 100/T
Sum triples itself	SI = 2P, so R x T = 200	R = 200/T
Amounts known for two times	Difference = interest for the gap	Then work back to P

WORKED EXAMPLE 6. (a) At what rate of simple interest will a sum double itself in 8 years? (b) A sum at simple interest amounts to Rs 2,400 in 3 years and to Rs 2,800 in 5 years - what is the sum? (c) At what rate will a sum become three times itself in 20 years?

- (a) To double, the interest must equal the principal, so SI = P.
- (a) P = P x R x 8/100, giving R = 100/8 = 12.5% per annum.
- (b) Interest for the 2-year gap = 2800 - 2400 = Rs 400, so interest for one year = Rs 200.
- (b) Interest for 3 years = Rs 600, so the principal = 2400 - 600 = Rs 1,800.
- (c) To triple, the interest must be twice the principal: 2P = P x R x 20/100.
- (c) R = 200/20 = 10% per annum.

Compound interest, growth, depreciation and instalments

Compound interest adds the interest back to the principal, so the base grows every year. Everything else follows from the one formula A = P(1 + R/100) raised to the power n. Population growth is the same formula; depreciation is the same formula with a minus sign. For half-yearly compounding halve the rate and double the number of periods; for quarterly, quarter the rate and take four times the periods.

Compound interest and its cousins

Situation	Formula	Example
Amount after n years	A = P(1 + R/100)^n	CI = A - P
Half-yearly compounding	Rate R/2, periods 2n	10% for 1 year becomes 5% for 2 periods
Quarterly compounding	Rate R/4, periods 4n	-

Situation	Formula	Example
Population after n years	P(1 + r/100)^n	20,000 at 10% for 2 years = 24,200
Depreciation	P(1 - r/100)^n	50,000 at 10% for 2 years = 40,500
Doubling at CI	Doubles in t, so 4 times in 2t, 8 times in 3t	Doubles in 5 years -> 4 times in 10
Rate from a known amount	(1 + R/100)^n = A/P	1.44 over 2 years gives R = 20%
Different rates in different years	Multiply the separate factors	P(1 + r1/100)(1 + r2/100)

WORKED EXAMPLE 7. (a) What is the compound interest on Rs 8,000 for 1 year at 10% per annum, compounded half-yearly? (b) If Rs 6,000 fetches a compound interest of Rs 2,640 in 2 years, what is the rate? (c) A sum doubles itself in 5 years at compound interest - in how many years does it become four times?

- (a) Half-yearly: use 5% for 2 periods, not 10% for 1 year.
- (a) Amount = 8000 x (1.05)^2 = 8000 x 1.1025 = Rs 8,820, so CI = Rs 820 (annual compounding would give only Rs 800).
- (b) Amount = 6000 + 2640 = Rs 8,640, so (1 + R/100)^2 = 8640/6000 = 1.44.
- (b) 1 + R/100 = 1.2, hence R = 20% per annum.
- (c) In 5 years P becomes 2P; in the next 5 years that 2P doubles again to 4P.
- (c) So 5 + 5 = 10 years. Becoming 2 to the power k times takes 5k years.

Instalments at compound interest – the present-value method

Call each instalment x → Discount the first year's instalment: x/(1 + r/100) → Discount the second year's: x/(1 + r/100)^2 → Add all the discounted values → Set the sum equal to the debt and solve for x

WORKED EXAMPLE 8. A debt of Rs 2,100 is to be repaid in two equal annual instalments at 10% per annum compound interest. What is each instalment?

- Each instalment is x; the first is paid after 1 year, the second after 2 years.
- Present value = x/1.1 + x/(1.1)^2 = 2100.
- Take the common denominator 1.21: x(1.1 + 1)/1.21 = 2100, i.e. x x 2.1/1.21 = 2100.
- x = 2100 x 1.21/2.1 = Rs 1,210.
- Check: 1210/1.1 = Rs 1,100 and 1210/1.21 = Rs 1,000, and 1100 + 1000 = Rs 2,100. Correct.
- Note the simpler instalment question: cash price Rs 12,000 against 4,000 down plus two yearly payments of 4,200 means 12,400 paid in all, i.e. Rs 400 extra.

The difference between compound and simple interest – four papers running

This is the single most repeated question in the whole reasoning paper – RAS Pre 2018, 2021, 2023 and 2024, four consecutive papers. Understand why the difference exists and you will never need to compute both interests. Simple interest charges the same interest every year. Compound interest, from the second year onwards, also charges interest on the previous year's interest. That extra bit IS the difference. For two years the extra bit is one year's interest on the first year's interest, which is exactly P(R/100) squared.

Simple interest vs compound interest

Simple interest	Compound interest
Interest on the original principal only	Interest on principal plus accumulated interest
Same interest every year	Interest grows every year
SI = P x R x T/100	A = P(1 + R/100)^n and CI = A - P
Equal for the first year	Equal for the first year - the gap opens in year 2
Doubles when R x T = 100	Doubles in t years, quadruples in 2t
Grows in a straight line	Grows faster and faster; CI always exceeds SI for n > 1

The difference formulas - memorise all four lines

Asked	Formula	Worked value
Difference for 2 years	D = P(R/100)^2 = P x R^2/10000	P=10,000, R=5: Rs 25
Difference for 3 years	D = P x R^2 x (300 + R)/10^6	P=16,000, R=5: Rs 122
Same, other form	D = P(R/100)^2 x (3 + R/100)	-
Find the sum from a 2-year difference	P = D x (100/R)^2	D=32, R=4: Rs 20,000
Find the rate from a 2-year difference	R = 100 x square root of (D/P)	D=144, P=10,000: 12%
Find the sum from a 3-year difference	P = D x 10^6 / (R^2 x (300 + R))	D=62, R=10: Rs 2,000

WORKED EXAMPLE 9 (the two-year type). (a) What is the difference between the compound interest and the simple interest on Rs 10,000 for 2 years at 5%

per annum? (b) The same for Rs 8,000 for 2 years at 10% per annum. (c) The same for Rs 25,000 for 2 years at 8% per annum.

- (a) Long way: SI = 10000 x 5 x 2/100 = Rs 1,000; CI = 10000 x (1.05)^2 - 10000 = 11,025 - 10,000 = Rs 1,025.
- (a) Difference = 1025 - 1000 = Rs 25. Shortcut: P(R/100)^2 = 10000 x (0.05)^2 = 10000 x 0.0025 = Rs 25.
- (b) P(R/100)^2 = 8000 x (0.1)^2 = 8000 x 0.01 = Rs 80. (Check: SI 1,600, CI 1,680.)
- (c) P(R/100)^2 = 25000 x (0.08)^2 = 25000 x 0.0064 = Rs 160. (Check: SI 4,000, CI 4,160.)
- Notice the pattern - the shortcut needs one multiplication, the long way needs four.

WORKED EXAMPLE 10 (three years, and the reverse questions). (a) Find the difference on Rs 16,000 for 3 years at 5% per annum. (b) The difference for 2 years at 4% per annum is Rs 32 - find the sum. (c) The difference on Rs 10,000 for 2 years is Rs 144 - find the rate. (d) The difference for 3 years at 10% is Rs 62 - find the sum.

- (a) D = P x R^2 x (300 + R)/10^6 = 16000 x 25 x 305/1,000,000 = Rs 122.
- (a) Check: SI = Rs 2,400 and CI = 16000 x (1.05)^3 - 16000 = 18,522 - 16,000 = Rs 2,522. Difference Rs 122.
- (b) 32 = P x (4/100)^2 = P x 0.0016, so P = 32/0.0016 = Rs 20,000.
- (c) 144 = 10000 x (R/100)^2, so (R/100)^2 = 0.0144 and R/100 = 0.12, giving R = 12%.
- (d) D = P x 100 x 310/1,000,000 = 0.031P, so 0.031P = 62 and P = Rs 2,000.

- For 10% over 3 years on Rs 10,000 the difference is Rs 310 - worth remembering as a benchmark.

ASKED IN RAS

Asked in RAS Pre 2018, 2021, 2023 and 2024 - four consecutive papers. Both the direct form (find the difference) and the reverse form (the difference is given, find the sum or the rate) have appeared. If you learn one formula from this whole chapter, learn D = P(R/100)^2 for two years and D = P x R^2 x (300 + R)/10^6 for three.

REVISION RECAP

1. Difference between CI and SI for 2 years = P(R/100)^2; for 3 years = P x R^2 x (300 + R)/10^6.
2. Reverse the same formulas: P = D x (100/R)^2, and R = 100 x the square root of D/P.
3. Benchmarks worth carrying: 10,000 at 5% for 2 years gives Rs 25; 10,000 at 10% for 3 years gives Rs 310.
4. SI = PRT/100 and it is the same every year; a sum doubles when R x T = 100 and triples when R x T = 200.
5. CI: A = P(1 + R/100)^n. Half-yearly means rate R/2 and periods 2n; quarterly means R/4 and 4n.
6. At compound interest a sum that doubles in t years becomes 4 times in 2t and 8 times in 3t.
7. Population growth is P(1 + r/100)^n and depreciation the same with a minus sign.
8. Instalments at CI: discount each instalment back to today and set the total equal to the debt.
9. Divide in a ratio by parts; to join A : B with B : C, scale both until B matches, then read A : B : C.
10. Mean proportional = square root of ab, third proportional = b^2/a, fourth proportional = bc/a.
11. Alligation: cheaper : dearer = (dearer - mean) : (mean - cheaper), with water priced at zero.
12. Repeated replacement leaves x(1 - y/x)^n of the original liquid - never subtract y each time.
13. Partnership shares profit in the ratio of capital x time; the working partner's commission comes out first.

14. Successive percentage changes combine as $a + b + ab/100$, never by simple addition.
15. An x% rise followed by an x% fall is always a net fall of $x^2/100$ per cent.
16. A price rise of x% needs a consumption cut of $100x/(100 + x)$ per cent to hold expenditure steady.
17. Profit and loss percentages sit on the cost price; discount sits on the marked price.
18. Mark up m% and discount d% gives a profit of $m - d - md/100$ per cent.
19. Two discounts d1 and d2 equal a single discount of $d1 + d2 - d1d2/100$ - always less than the sum.
20. Selling two articles at the same price with equal gain and loss percentages always loses $x^2/100$ per cent.

RM10

Numeracy II: Mensuration, Averages, Statistics, Data Interpretation, Probability

Numeracy II supplies roughly 8 to 10 of the 19-21 reasoning questions - mensuration and data interpretation are the two largest sub-blocks, with probability (dice especially) and calendar-clock giving one each almost every year. Pie-chart DI appeared in RAS Pre 2013, 2018 and 2024; dice probability in RAS Pre 2021, 2023 and 2024. Target full marks here in about ten minutes.

Reasoning and Mental Ability gives you 19 to 21 questions in every RAS Prelims paper, and it is the easiest block in the whole exam to convert at 90 per cent - because nothing here has to be remembered, only practised. This chapter alone, covering mensuration, averages, statistics, data interpretation, probability, speed, work, calendar and clock, is worth more marks than most of the Rajasthan content chapters. Notes for it teach METHOD, not facts.

Numeracy II

Mensuration 2D
- Square, rectangle, triangle
- Heron's formula
- Parallelogram, rhombus, trapezium
- Circle, semicircle, sector
- Path around a field

Mensuration 3D
- Cube and cuboid
- Cylinder
- Cone (slant height)
- Sphere and hemisphere
- Change of dimension: k, k squared, k cubed

Averages
- Sum divided by count
- Removal, inclusion, replacement
- Weighted average
- Average speed 2xy/(x+y)

Statistics
- Mean, Sigma fx / Sigma f
- Median from cumulative frequency
- Mode
- Range
- Mode = 3 Median - 2 Mean

Data interpretation
- Text tables
- Bar charts
- Pie charts: 1 degree = total/360
- Average, ratio, percentage change

Counting and chance
- nPr order matters
- nCr order does not
- Repeated letters n!/(p!q!)
- Coins, dice, cards, balls

Speed and work
- 5/18 and 18/5
- Trains: pole vs platform
- Relative speed
- One day's work 1/x
- Pipes and leaks

Calendar and clock
- Odd days
- 400 years = 0 odd days
- Month codes
- Angle = |30H - 5.5M|

How to attack Numeracy II - read this before the formulas

Reasoning and Mental Ability gives 19-21 questions in every RAS Prelims paper. This one chapter - mensuration, averages, statistics, data interpretation, probability, speed, work, calendar and clock - carries more marks than most of the Rajasthan content chapters you sweat over. And unlike them, nothing here has to be remembered from a book. Every answer is manufactured on the spot from a formula plus twenty seconds of arithmetic. That is why a trained candidate converts this block at 90 per cent while an untrained one loses eight marks to silly slips.

- Mensuration and DI are the two biggest sub-blocks - do them first in the paper.
- Probability with dice repeats most often; RAS Pre 2021, 2023 and 2024 all carried a dice question.
- Pie chart DI appeared in RAS Pre 2013, 2018 and 2024 - the data is always given in the stem.
- Calendar and clock give one question that takes 40 seconds if you know the two formulas.
- Nothing in this chapter needs a calculator. If your working needs one, your method is wrong.

The four-step attack on any numeracy question

Read the question and write down what is ASKED, in a box, before anything else.

→

Write the units. Metres or centimetres? Hours or minutes? km/h or m/s? Convert now, not later.

→

Name the formula out loud in your head and write it in symbols first, then substitute.

→

Do the arithmetic in one clean line, cancel before you multiply, and check the answer against common sense.

- Cancel first, multiply later - 22/7 with a radius of 7, 14 or 21 always cancels.
- Use pi = 22/7 whenever the radius is a multiple of 7; otherwise 3.14 is intended.
- Never average two averages, never average two speeds, never subtract two times - three of the most common traps in this chapter.
- If an option equals your answer times 2, or divided by 2, suspect a missing one-half in your formula.
- Sanity check every answer: a joint work time must be less than either worker alone; an average must lie between the smallest and largest value.

Plane figures - perimeter and area

The commonest mensuration question gives you one measurement of a flat figure and asks for another. Perimeter of a square, so what is the area? Three sides of a triangle, so what is the area? Diameter of a circle, so what is the area? There is no trick - only the formula and one clean substitution. The marks are lost at the substitution step, not at the recall step, so slow down there.

Plane figures - the full formula sheet

Figure	Perimeter / circumference	Area
Square, side a	4a (diagonal = a√2)	a², or d²/2 from the diagonal
Rectangle, l x b	2(l + b)	lb (diagonal = √(l² + b²))
Triangle, base b, height h	a + b + c	½ x base x height
Triangle, three sides known	a + b + c	Heron: √[s(s−a)(s−b)(s−c)], s = (a+b+c)/2
Equilateral triangle, side a	3a	(√3/4)a², height = (√3/2)a
Parallelogram, base b, height h	2(sum of adjacent sides)	base x height - NO one-half
Rhombus, diagonals d1, d2	4 x side	½ x d1 x d2
Trapezium, parallel sides a and b	sum of all four sides	½ x (a + b) x distance between them
Circle, radius r	2πr	πr²
Semicircle, radius r	πr + 2r	½πr²
Sector, angle θ	arc = (θ/360) x 2πr	(θ/360) x πr² = ½ x arc x r

Worked example - Heron's formula. The sides of a triangle are 13 cm, 14 cm and 15 cm. Find the area. You are given three sides and no height, so Heron is the only route. Learn this 13-14-15 triangle by heart; RPSC and every other exam reuse it because the answer comes out as a whole number.

- $s = (a + b + c)/2 = (13 + 14 + 15)/2 = 42/2 = 21$.
- $s - a = 21 - 13 = 8$; $s - b = 21 - 14 = 7$; $s - c = 21 - 15 = 6$.
- Area $= \sqrt{[s(s-a)(s-b)(s-c)]} = \sqrt{(21 \times 8 \times 7 \times 6)}$.
- Group the numbers before multiplying: $21 \times 6 = 126$ and $8 \times 7 = 56$, so the product is $126 \times 56 = 7056$.
- $\sqrt{7056} = 84$. Area $= 84\ cm^2$.
- Second example, same section: sector of radius 21 cm, angle 120 degrees, $\pi = 22/7$. Area $= (120/360) \times (22/7) \times 21 \times 21 = (1/3) \times 1386 = 462\ cm^2$. The full circle would be $1386\ cm^2$ and the ARC alone is only 44 cm - three different answers sitting in the options.

TRAP

The missing one-half. Triangle, trapezium and rhombus all carry a one-half; parallelogram does NOT. Trapezium with parallel sides 22 and 14 and distance 9: area $= ½ \times (22 + 14) \times 9 = 162\ cm^2$. Dropping the half gives 324 and using the difference $(22 - 14)$ gives 36 - both will be sitting in the options waiting for you.

Paths around a field, and what happens when you change a dimension

Two examiner favourites live here. First, a path of uniform width laid around or inside a rectangular field - you must find the area of the path itself, not of the field. Second, a percentage change question: the side goes up by 20 per cent, by how much does the area go up? Both are one-line problems once you stop drawing and start substituting.

Paths and dimension-change rules

Situation	Formula	Note
Path of width w OUTSIDE a rectangle l x b	$2w(l + b) + 4w^2$	the $+4w^2$ is the four corner squares
Path of width w INSIDE a rectangle l x b	$2w(l + b) - 4w^2$	corners are subtracted here
Two roads of width w crossing at right angles	$w(l + b) - w^2$	the $-w^2$ stops the crossing being counted twice
Every linear dimension multiplied by k	length x k, area x k^2, volume x k^3	the single most useful rule in mensuration
Two dimensions change by x% and y%	net change = x + y + xy/100	use a minus sign for a decrease
Three dimensions change by x% each	apply the two-dimension rule twice	or simply cube the multiplier

Worked example - path outside a field. A rectangular field is 50 m long and 40 m wide. A path 3 m wide is laid all around it on the OUTSIDE. Find the area of the path. Do it both ways once, so that you trust the shortcut afterwards.

- Long way. Outer rectangle = (50 + 2x3) by (40 + 2x3) = 56 x 46 = 2576 m^2.
- Inner field = 50 x 40 = 2000 m^2.
- Path = 2576 − 2000 = 576 m^2.
- Short way. $2w(l + b) + 4w^2$ = 2 x 3 x (50 + 40) + 4 x 9 = 540 + 36 = 576 m^2.
- Note the width is added TWICE to each dimension - once on each side. Adding it once gives 53 x 43 and a wrong answer.
- Forgetting the four corner squares gives 540 m^2, the classic distractor.

- Square side up 20%: area change = 20 + 20 + (20x20)/100 = 44% increase. Check: 100 to 120 means 10000 to 14400.
- Radius up 10%: area up 21%. Radius down 10%: area down 19%, not 20%.
- Side up 20% and down 20% in turn: net −4%, never zero.
- Edge of a cube up 50%: multiplier 1.5, so surface area x 2.25 (up 125%) and volume x 3.375 (up 237.5%).
- Radius of a sphere doubled: surface area x 4, volume x 8. Halved: surface x 1/4, volume x 1/8.

Solids - surface area and volume

Six solids, and RPSC has asked all of them. The question either gives a dimension and asks for volume or surface area, or gives the volume and works backwards to a dimension. Everything you need sits in one table. The only genuine piece of thinking is the cone, where the slant height l is almost never given - you compute it from r and h by Pythagoras.

Solids - volume, curved surface, total surface

Solid	Volume	Curved / lateral SA	Total SA
Cube, edge a	a^3	$4a^2$	$6a^2$ (diagonal = $a\sqrt{3}$)
Cuboid l x b x h	lbh	$2h(l + b)$	$2(lb + bh + hl)$
Cylinder, radius r, height h	$\pi r^2 h$	$2\pi rh$	$2\pi r(r + h)$
Cone, radius r, height h	$(1/3)\pi r^2 h$	πrl, where $l = \sqrt{(r^2 + h^2)}$	$\pi r(l + r)$
Sphere, radius r	$(4/3)\pi r^3$	—	$4\pi r^2$
Hemisphere (solid), radius r	$(2/3)\pi r^3$	$2\pi r^2$	$3\pi r^2$ (curved + flat face)

Worked example - the cone. A cone has radius 7 cm and vertical height 24 cm. Find its curved surface area, taking π = 22/7. Notice what is given: the

VERTICAL height, not the slant height. The formula πrl needs l, so the first line of your solution must be Pythagoras.

- $l = \sqrt{(r^2 + h^2)} = \sqrt{(7^2 + 24^2)} = \sqrt{(49 + 576)} = \sqrt{625} = 25$ cm.
- 7-24-25 is a Pythagorean triple - memorise it along with 3-4-5, 5-12-13, 8-15-17, 9-40-41.
- Curved surface area = $\pi rl = (22/7) \times 7 \times 25$. The 7 cancels the denominator straight away.
- $= 22 \times 25 = 550$ cm^2.
- Total surface area, if that had been asked, $= \pi r(l + r) = (22/7) \times 7 \times 32 = 704$ cm^2.
- Volume, if that had been asked, $= (1/3) \times (22/7) \times 49 \times 24 = 1232$ cm^3. Three different numbers - read the question word for word.

TRAP

Curved versus total versus volume. Three of the four options in a solids question are the OTHER two quantities for the same solid. A solid hemisphere of radius 7 has curved SA 308, total SA 462 and the full sphere 616 - all three will be offered. Also watch the units: cm^2 for any area, cm^3 for any volume. If the option carries the wrong unit, it is wrong however good the number looks.

Averages - including weighted average and average speed

Average is just total divided by count, but RPSC never asks it that plainly. The real questions are of four shapes: an item is removed, an item is added, an item is replaced, or two groups of different sizes are merged. In every one of them the trick is the same - stop thinking about averages and go back to the TOTAL. Total = average x number of items. Change the total, then divide again.

Average - the rules that answer the standard shapes

Situation	Rule	Worked value
Basic	Average = sum ÷ number of items	—
First n natural numbers	$(n + 1)/2$	first 10 give 5.5
First n odd numbers	n	first 10 odd numbers give 10
First n even numbers	$n + 1$	first 10 even numbers give 11
One item REMOVED	new sum = old sum − item, then ÷ $(n-1)$	—
One item ADDED, average rises by d	new item = new average + $n \times d$	—
One item REPLACED, average changes by d	new item = old item ± $n \times d$	—
Two groups merged	weighted average = $(n_1a_1 + n_2a_2)/(n_1 + n_2)$	—
Equal DISTANCES at speeds x and y	$2xy/(x + y)$, the harmonic mean	60 and 40 give 48
Equal TIMES at speeds x and y	$(x + y)/2$, the arithmetic mean	60 and 40 give 50

Worked example - weighted average. In a class of 50 students, the 20 boys score an average of 45 marks and the 30 girls an average of 55 marks. Find the average marks of the whole class. The tempting answer is the midpoint of 45 and 55, which is 50. It is wrong, and it is option (a) in every version of this question.

- Total marks of the boys = 20 x 45 = 900.
- Total marks of the girls = 30 x 55 = 1650.
- Total marks of the class = 900 + 1650 = 2550.
- Class average = 2550 ÷ 50 = 51 marks.

- Why 51 and not 50: there are more girls, and the girls score higher, so the combined average is pulled towards the girls' 55.
- Sanity check that costs one second - the answer must lie between 45 and 55, and closer to the bigger group's value.

Average speed - equal distance versus equal time

Equal DISTANCE each way	Equal TIME at each speed
'goes to B and returns', 'up and down the hill'	'drove for 2 hours at x, then 2 hours at y'
Use 2xy/(x + y) - the harmonic mean	Use (x + y)/2 - the plain arithmetic mean
60 and 40 give 48 km/h	60 and 40 give 50 km/h
Answer is always BELOW the simple midpoint	Answer is exactly the midpoint
More time is spent at the slower speed	Time is shared equally by definition

YAAD RAKHO

Yaad rakho - the replacement rule. When one person swaps for another in a group of n and the average moves by d, the whole change n x d is carried by that single swap. Eight people, average up 2.5 kg, the person leaving weighs 56 kg: the new person = 56 + (8 x 2.5) = 76 kg. Adding 2.5 only once gives 58.5 - the option that catches half the hall. Same logic for inclusion: 30 students average 14, teacher raises it by 1, so teacher = 15 + 30 = 45 years.

Mean, median, mode and range

Three averages of central tendency plus one measure of spread. Mean is the arithmetic average, median is the middle value after sorting, mode is the value that occurs most often, range is largest minus smallest. RPSC asks these from an ungrouped list of six or seven numbers, or from a small frequency table. The frequency-table version is where the marks are lost, because students average the values and ignore the frequencies.

Central tendency and dispersion

Measure	Ungrouped data	Grouped / frequency data
Mean	sum ÷ n	Σfx ÷ Σf - multiply each value by its frequency
Median, n odd	the ((n+1)/2)th value after SORTING	the value at the cumulative frequency crossing N/2
Median, n even	mean of the (n/2)th and (n/2 + 1)th values	average of the two middle observations
Mode	the most frequently occurring value	the value with the highest frequency
Empirical relation	Mode = 3 Median − 2 Mean	for a moderately skewed distribution
Range	maximum − minimum	a measure of dispersion, not of position

Worked example - mean of a frequency distribution. Value x is 10, 20, 30, 40, 50 with frequencies 3, 5, 8, 3, 1 respectively. Find the arithmetic mean. Make two columns on your rough sheet - one for f, one for fx - and total both.

- Σf = 3 + 5 + 8 + 3 + 1 = 20 observations in all.
- fx values: 10x3 = 30, 20x5 = 100, 30x8 = 240, 40x3 = 120, 50x1 = 50.
- Σfx = 30 + 100 + 240 + 120 + 50 = 540.
- Mean = Σfx ÷ Σf = 540 ÷ 20 = 27.
- Averaging 10, 20, 30, 40, 50 straight gives 30 - wrong, because it pretends every value occurred equally often.

Worked example - median from a cumulative frequency. A table gives the number of children per family in 50 families: 0 children in 6 families, 1 child in

10, 2 children in 12, 3 children in 14, 4 children in 8. Find the MEDIAN number of children. The cumulative frequency column does all the work.

- Cumulative frequencies: 6, then 16, then 28, then 42, then 50.
- N = 50, which is even, so the median is the average of the 25th and 26th observations.
- The 17th to 28th observations all lie in the '2 children' row, because the cumulative jumps from 16 to 28 there.
- So the 25th and 26th are both 2. Median = (2 + 2)/2 = 2.
- The MODE here is 3, because 14 is the highest frequency. Median 2, mode 3 - two different answers in the same table.
- Empirical relation check, on a different data set: mean 26 and median 28 give mode = 3(28) − 2(26) = 84 − 52 = 32.

Data interpretation - tables, bar charts and pie charts

A DI question hands you a small data set and asks two or three arithmetic questions about it. In RAS the set is tiny - five rows at most - and it is written out in words in the stem, so you never have to read a graph badly. The arithmetic is only average, ratio, difference and percentage. What kills candidates is misreading which row the question wants. Worked example - a pie chart. The monthly expenditure of a family totalling ₹36,000 is shown by a pie chart with these central angles: Food 120 degrees, House rent 90 degrees, Education 60 degrees, Transport 45 degrees, Miscellaneous 45 degrees. Find (a) the spend on food, (b) by how much house rent exceeds education, and (c) house rent plus transport as a percentage of the total.

- First line of every pie question: 360 degrees stands for the whole, here ₹36,000. So 1 degree = 36000 ÷ 360 = ₹100.

- (a) Food = 120 x 100 = ₹12,000. Equivalently 120/360 = one third of ₹36,000.
- (b) House rent = 90 x 100 = ₹9,000; education = 60 x 100 = ₹6,000; difference = ₹3,000. Faster: (90 − 60) x 100 = ₹3,000.
- (c) House rent + transport = 90 + 45 = 135 degrees. Percentage = (135/360) x 100 = 37.5%.
- For a percentage question you never need the rupees at all - work in degrees and divide by 360.
- Check that the angles add to 360: 120 + 90 + 60 + 45 + 45 = 360. If they do not, you have misread a row.

ASKED IN RAS

Pie-chart DI on household or budget expenditure appeared in RAS Pre 2013, 2018 and 2024. The wording changes, the method never does: convert 360 degrees to the given total, get the value of one degree, then answer. Expect two questions from the same chart.

Worked example - a table. The wheat production of a district in lakh tonnes was 24 in 2019, 30 in 2020, 27 in 2021, 36 in 2022 and 33 in 2023. Three questions get asked from a table like this: the average, a percentage change between two years, and a count of years above or below the average. A bar chart is the same data drawn as five bars, and it asks the same three questions - never redraw it, just read the heights into a column. A second, double-bar type gives boys and girls in five schools - A 240 and 160, B 180 and 220, C 300 and 200, D 150 and 150, E 210 and 190 - and asks for a ratio or for which school has the highest percentage of girls.

- Average production = (24 + 30 + 27 + 36 + 33) ÷ 5 = 150 ÷ 5 = 30 lakh tonnes.
- Percentage rise 2019 to 2020 = (30 − 24)/24 x 100 = 6/24 x 100 = 25%. Always divide by the ORIGINAL year, not the new one - dividing by 30

gives the wrong 20%.

- Years ABOVE the average of 30: 24 no, 30 is equal and therefore not above, 27 no, 36 yes, 33 yes. Answer: 2 years.
- Schools table - total boys = 240+180+300+150+210 = 1080; total girls = 160+220+200+150+190 = 920; ratio 1080 : 920 = 27 : 23 after dividing by 40.
- Highest percentage of girls: A 160/400 = 40%, B 220/400 = 55%, C 200/500 = 40%, D 150/300 = 50%, E 190/400 = 47.5%. Answer: School B.
- Note that School C has 200 girls, more than D's 150, yet a lower percentage. Largest NUMBER is not largest SHARE - that is the whole point of the question.

How to work a DI set in under two minutes

Copy the data into a clean column on your rough sheet before reading any of the questions. → Add the column and write the TOTAL at the bottom - most DI questions need it. →

Compute the average and write it beside the total; the 'above average' question is then free. → Read each question and underline whether it wants a value, a difference, a ratio or a percentage. →

For percentage change, fix the denominator first: it is always the earlier or original value. → Answer all questions from the same set together while the numbers are still in front of you.

Permutation and combination

One question, sometimes two, and it is always decided in the first three seconds by a single decision: does the ORDER matter? Arrangements, seatings, words made from letters, numbers made from digits - order matters, use permutation.

Selections, committees, teams, handshakes, chords, groups - order does not matter, use combination. Get that decision right and the arithmetic is trivial.

Permutation and combination - formulas and standard results

Thing asked	Formula	Standard value
Arrange r out of n, order matters	nPr = n!/(n−r)!	5P2 = 20
Select r out of n, order does not	nCr = n!/[r!(n−r)!]	5C2 = 10
Arrange all n distinct letters	n!	DELHI = 5! = 120
Arrange n letters with repeats	n!/(p! q! ...)	BANANA = 6!/(3!2!) = 60
Handshakes / lines / chords among n	nC2 = n(n−1)/2	12 people give 66
Diagonals of an n-sided polygon	n(n−3)/2	octagon gives 20
Useful identities	nC0 = nCn = 1; nC1 = n; nCr = nC(n−r)	10C7 = 10C3 = 120
Two independent choices	MULTIPLY the counts	never add them

Worked example one - BANANA. In how many different ways can the letters of the word BANANA be arranged? Worked example two - a committee of 3 men and 2 women is to be formed from 6 men and 5 women; in how many ways? Do both, because between them they cover the two things that go wrong: repeated letters, and forgetting to multiply.

- BANANA has 6 letters: B once, A three times, N twice.
- Arrangements = 6!/(3! x 2!) = 720/(6 x 2) = 720/12 = 60.
- Treating all six letters as distinct gives 720 - the standard wrong answer.
- Committee: choose the men first. 6C3 = (6 x 5 x 4)/(3 x 2 x 1) = 120/6 = 20 ways.
- Choose the women: 5C2 = (5 x 4)/(2 x 1) = 20/2 = 10 ways.

- The two choices are independent, so MULTIPLY: 20 x 10 = 200 ways. Adding gives 30, which is the distractor.
- Handshake check on the same logic: 12 people, each pair shakes once, so 12C2 = (12 x 11)/2 = 66. Using 12 x 11 = 132 counts every handshake from both sides.

Permutation versus combination - decide in three seconds

Permutation - ORDER matters	Combination - order does NOT matter
Arrange, seat, line up, form a word or a number	Select, choose, form a committee or a team
Prize positions: first, second, third	Three winners with no ranking
$nPr = n!/(n-r)!$	$nCr = n!/[r!(n-r)!]$
nPr is always the LARGER of the two	$nCr = nPr \div r!$
AB and BA are two different outcomes	AB and BA are the same outcome
Digits forming a number, letters forming a word	Handshakes, chords, lines through points

Probability - coins, dice, cards and balls in a bag

Probability is favourable outcomes divided by total outcomes, and in RAS the total is always one of four small numbers: 2 to the power n for coins, 36 for two dice, 52 for cards, or the ball count in a bag. So the whole question reduces to counting the favourable cases correctly. Write the sample-space size FIRST, then count. Every answer must lie between 0 and 1 - if yours does not, you have inverted a fraction.

Probability - the four standard settings

Setting	Total outcomes	Things you must know
n coins tossed	2^n	exactly r heads = $nCr \div 2^n$; 3 coins give 8 outcomes

Setting	Total outcomes	Things you must know
One die	6	primes are 2, 3 and 5 - so P(prime) = 1/2; 1 is NOT prime
Two dice	36	doublets = 6, so P = 1/6; P(sum 7) = 6/36 = 1/6, the likeliest sum
Two dice, ways to make a sum	2:1, 3:2, 4:3, 5:4, 6:5, 7:6	8:5, 9:4, 10:3, 11:2, 12:1 - it is a symmetric ladder
Pack of 52 cards	52	4 suits x 13; 26 red and 26 black; 4 aces
Face cards	12 in all	J, Q, K in each suit; red face cards = 6 only
Balls drawn together, no replacement	nC2 for two balls	P(both black from 4W and 5B) = 5C2 ÷ 9C2
Balls drawn with replacement	multiply the single probabilities	$(5/9)^2$ is NOT the same as 5C2 ÷ 9C2
Complement	P(not E) = 1 − P(E)	use it for 'at least one'

Worked example one - two dice, sum of 8. Worked example two - a bag with 4 white and 5 black balls, two drawn together, both black. These two carry the two counting errors that RPSC exploits: forgetting that dice are ordered, and forgetting that 'drawn together' means without replacement.

- Two dice: n(S) = 6 x 6 = 36, because the first die and the second die are distinguishable.
- Sum 8 comes from (2,6), (3,5), (4,4), (5,3), (6,2) - five ordered pairs.
- P(sum 8) = 5/36. Counting the unordered sets 2+6, 3+5, 4+4 as only three cases gives the wrong 1/12.
- Balls: total ways of drawing 2 from 9 = 9C2 = (9 x 8)/2 = 36.
- Ways of drawing 2 black from 5 = 5C2 = (5 x 4)/2 = 10.
- P(both black) = 10/36 = 5/18. The distractor 25/81 comes from doing $(5/9)^2$, which would be right only WITH replacement.

- Third quick one: three coins, exactly two heads. Sample space 8; favourable HHT, HTH, THH; P = 3/8. Exactly three heads would be 1/8.

ASKED IN RAS

A dice probability question appeared in RAS Pre 2021, 2023 and 2024 - three papers running. The forms used were P(prime on one die) = 1/2, P(doublet) = 1/6, P(sum 8) = 5/36 and P(sum a multiple of 3) = 12/36 = 1/3. Learn the sum ladder and all four are instant.

YAAD RAKHO

Yaad rakho - the sum ladder for two dice reads 1, 2, 3, 4, 5, 6, 5, 4, 3, 2, 1 for sums 2 through 12, and adds to 36. So for any sum question, read the count straight off the ladder and divide by 36. Sum a multiple of 3 means sums 3, 6, 9, 12 - that is 2 + 5 + 4 + 1 = 12 ways, so 1/3. For cards, remember 6 red face cards, not 12: the colour condition halves it.

Time, speed and distance - including trains and boats

Speed = distance divided by time, and everything in this sub-topic is that one relation dressed differently. The two decisions that matter are the unit conversion and the choice of distance. A train crossing a pole covers only its own length; a train crossing a platform covers its own length PLUS the platform. Get those two right and the rest is division.

Time, speed and distance - conversions and relative motion

Situation	Rule	Example value
Basic relation	Speed = Distance ÷ Time	210 km in 3 h gives 70 km/h

Situation	Rule	Example value
km/h to m/s	multiply by 5/18	54 km/h = 15 m/s; 72 km/h = 20 m/s
m/s to km/h	multiply by 18/5	20 m/s = 72 km/h
Train crossing a pole or a man	distance = length of the train	—
Train crossing a platform or bridge	distance = train length + platform length	—
Two bodies moving OPPOSITE ways	relative speed = sum of the speeds	5 and 3 give a gap of 8 km per hour
Two bodies moving the SAME way	relative speed = difference of the speeds	5 and 3 give a gap of 2 km per hour
Boat downstream / upstream	u + v and u − v	u = boat in still water, v = stream
Same distance at two speeds	distance is fixed, so time varies inversely	speeds 5 : 6 give times 6 : 5

Worked example one - a train 240 m long running at 72 km/h crosses a platform 360 m long; find the time. Worked example two - walking at 5 km/h a student reaches school 6 minutes late, walking at 6 km/h he is 4 minutes early; find the distance. The second is the harder standard type and it is worth learning as a pattern.

- Train: convert first. 72 km/h = 72 x 5/18 = 20 m/s.
- Distance covered = train + platform = 240 + 360 = 600 m.
- Time = 600 ÷ 20 = 30 seconds. Using only the 240 m gives 12 seconds, the trap answer.
- Late-and-early: the two walking times differ by 6 + 4 = 10 minutes = 1/6 hour.
- Let the distance be d km. Then d/5 − d/6 = 1/6.
- (6d − 5d)/30 = 1/6, so d/30 = 1/6, giving d = 5 km.

- Check: at 5 km/h it takes 60 minutes, at 6 km/h it takes 50 minutes - a gap of exactly 10 minutes.

- The late and the early minutes are ADDED to get the time gap, never subtracted.
- Two men walking apart at 5 and 3 km/h are 8 x 4 = 32 km apart after 4 hours; walking the same way they would be only 8 km apart.
- Direct proportion shortcut: 210 km in 3 hours means 350 km takes 3 x (350/210) = 5 hours, with no need to find the speed.
- A question in seconds and metres is a train question; a question in hours and kilometres is a journey question. Convert to match, once, at the start.
- If the answer comes out over 100 m/s you have converted the wrong way - that is faster than any train.

Time and work, pipes and cisterns

The single rule for this whole sub-topic: never work with days, always work with ONE DAY'S WORK. If A finishes a job in x days, A does 1/x of it in a day. Add the one-day works to combine workers, subtract to find one worker from a pair, and turn the total back into days at the very end by taking the reciprocal. Pipes and cisterns is the identical sum, except that an emptying pipe or a leak carries a minus sign.

Work and pipes - the standard results

Situation	Rule	Example
A alone takes x days	one day's work = 1/x	x = 12 gives 1/12
A and B together	1/x + 1/y, so time = xy/(x + y)	12 and 24 give 8 days
B alone, given A and A+B	1/y = 1/(A+B time) − 1/x	10 and 15 give 30 days

Situation	Rule	Example
Men, days, hours, work	M1 D1 H1 / W1 = M2 D2 H2 / W2	12 men x 18 days = 216 man-days
More men, same job	days fall in inverse proportion	27 men need 216/27 = 8 days
Filling pipe	rate = +1/x	—
Emptying pipe or leak	rate = −1/y	—
Filling pipe with a leak	net rate = 1/x − 1/y	10 and 15 give 1/30, i.e. 30 hours
Efficiency ratio	if A is twice as fast as B, A takes half the days	times are inverse to efficiencies

Worked example one - A and B together finish a work in 10 days and A alone in 15 days; how long does B alone take? Worked example two - a tap fills a tank in 10 hours but a leak can empty the full tank in 15 hours; with both working, how long does it take to fill? Notice the two sums are numerically identical - that is the point.

- One day's work of A and B together = 1/10. One day's work of A = 1/15.
- One day's work of B = 1/10 − 1/15 = 3/30 − 2/30 = 1/30.
- So B alone takes 30 days. Subtracting the days, 15 − 10 = 5, is the standard error and it is always an option.
- Tank: the tap fills 1/10 per hour, the leak empties 1/15 per hour.
- Net = 1/10 − 1/15 = 1/30 of the tank per hour, so the tank fills in 30 hours.
- Adding instead of subtracting gives 6 hours - impossible, since a leak must make it SLOWER than the tap's own 10 hours.
- Two pipes filling in 6 and 4 hours: 1/6 + 1/4 = 5/12 per hour, so 12/5 hours = 2 hours 24 minutes = 144 minutes.

YAAD RAKHO

Yaad rakho - the LCM method kills fractions. Take the total work as the LCM of the given days. A takes 12 days and B 24 days: LCM = 24 units of work, so A does 2 units a day and B does 1. Together 3 units a day, so 24/3 = 8 days. No fractions anywhere. Same for tanks: tap 10 h, leak 15 h, LCM 30 litres, tap +3 and leak −2, net +1 litre an hour, so 30 hours.

Calendars and clocks

Two short topics that together give a guaranteed question, and both run on one idea each. Calendar runs on ODD DAYS - the days left over after taking out complete weeks. Clock runs on the fact that the minute hand gains on the hour hand at 5.5 degrees a minute. Learn the two counting tables below and neither question can take you more than a minute.

Calendar odd days and clock angles

Item	Value	Note
Ordinary year	1 odd day	365 = 52 weeks + 1
Leap year	2 odd days	366 = 52 weeks + 2
100 years	5 odd days	24 leap + 76 ordinary years
200 / 300 / 400 years	3 / 1 / 0 odd days	the calendar repeats every 400 years
Day code from odd days	0 Sun, 1 Mon, 2 Tue, 3 Wed, 4 Thu, 5 Fri, 6 Sat	count from Sunday
Leap year test	divisible by 4; a century year must be divisible by 400	1900 was NOT a leap year, 2000 was
Month codes Jan to Dec	0, 3, 3, 6, 1, 4, 6, 2, 5, 0, 3, 5	add to the year's odd days

Item	Value	Note
Angle between the hands	|30H − 5.5M| degrees	if over 180, subtract from 360
Hand speeds	hour 0.5 deg/min, minute 6 deg/min	minute hand gains 5.5 deg/min
In 12 hours	coincide 11 times, opposite 11 times, right angles 22 times	in 24 hours, double each

Worked example one - the Constitution came into force on 26 January 1950; what day of the week was it? Worked example two - at what time between 4 and 5 o'clock do the two hands coincide? The first is the full odd-day drill, the second is the gain-of-5.5-degrees drill.

- Break 1950 into 1600 + 300 + 49 completed years, then the days of January.
- 1600 years give 0 odd days; 300 years give 1 odd day.
- The 49 years 1901 to 1949 hold 12 leap years and 37 ordinary years, so 12x2 + 37x1 = 61 days = 8 weeks + 5 odd days.
- January up to the 26th is 26 days = 3 weeks + 5 odd days.
- Total odd days = 0 + 1 + 5 + 5 = 11, and 11 − 7 = 4. Code 4 counted from Sunday is Thursday. 26 January 1950 was a Thursday.
- Clock: at 4 o'clock the hour hand stands at 120 degrees and the minute hand at 0, so a gap of 120 degrees must be closed.
- The gap closes at 6 − 0.5 = 5.5 degrees a minute, so the time is 120 ÷ 5.5 = 240/11 = 21 and 9/11 minutes past 4.

Finding the day of the week for any date

Split the year as 400s + 100s + the completed years since the last century. → Odd days: 400 years give 0, 300 give 1, 200 give 3, 100 give 5. →

For the completed years, count leap years L and ordinary years O and take 2L + O. → Add the days elapsed in the current year, month by month, up to the given date. →

Add everything and divide by 7; keep the remainder only. → Read the remainder as 0 Sunday, 1 Monday, and so on up to 6 Saturday.

TRAP

Three traps in one box. First, 15 August 2026 is a Saturday, so 15 August 2027 is a Sunday - the stretch contains February 2027, an ordinary year, so only ONE odd day is added; students add two out of habit. Second, at 3:30 the angle is 75 degrees, not 90 - by half past, the hour hand has itself crawled half way from 3 towards 4. Third, a century year is a leap year only if it divides by 400, so 1900 had 365 days.

REVISION RECAP

1. Square: area a squared, diagonal a root 2, area from diagonal d squared over 2. Rectangle: area lb, diagonal root(l squared + b squared).
2. Heron: s = (a+b+c)/2, area = root[s(s-a)(s-b)(s-c)]. The 13-14-15 triangle gives 84.
3. One-half belongs to triangle, trapezium and rhombus. Parallelogram is base x height, no half.
4. Circle 2 pi r and pi r squared; sector area = (theta/360) pi r squared and arc = (theta/360) 2 pi r.
5. Path w wide outside a rectangle = 2w(l+b) + 4w squared; inside, subtract the 4w squared.
6. Change every linear dimension by k: length x k, area x k squared, volume x k cubed. Side up 20% means area up 44%.

7. Cone: find l = root(r squared + h squared) first. CSA pi r l, TSA pi r (l+r), volume one-third pi r squared h.
8. Hemisphere total surface = 3 pi r squared, because the flat face counts. Sphere = 4 pi r squared.
9. Go back to the TOTAL in every average question: total = average x number of items.
10. Weighted average = (n1a1 + n2a2)/(n1+n2). Never average two group averages of unequal size.
11. Equal distances: average speed = 2xy/(x+y). Equal times: (x+y)/2. 60 and 40 give 48, not 50.
12. Grouped mean = Sigma fx / Sigma f. Median needs the cumulative frequency; mode is the largest frequency.
13. Mode = 3 Median - 2 Mean; Range = maximum - minimum.
14. Pie chart: 360 degrees = the whole total, so find the value of one degree first. Percentages need only the angles.
15. Percentage change divides by the ORIGINAL value, never the new one.
16. Order matters means nPr; order does not means nCr. Independent choices multiply, they never add.
17. BANANA = 6!/(3!2!) = 60; handshakes among n = nC2; diagonals of an n-gon = n(n-3)/2.
18. Two dice: 36 outcomes, sum ladder 1-2-3-4-5-6-5-4-3-2-1, doublet 1/6, sum 8 is 5/36. Six red face cards.
19. Train past a pole covers its own length; past a platform, length plus platform. 5/18 to go from km/h to m/s.
20. Work in one day's work 1/x, never in days. Leak subtracts. Odd days: ordinary 1, leap 2, 400 years 0. Clock angle = |30H - 5.5M|.